User-Centric Design Requirements for the Next Gen Utah Water Measurement Data Dashboard

Tech Services Section, Utah Division of Water Rights
September 2026 | DRAFT — Work in Progress

Executive Summary

This document describes and synthesizes requirements and key activities for six distinct user personas — each representing a group of water data users with shared roles, responsibilities, and goals centered around measurement data— and provides mockups of the desired custom dashboard for each.

1. **Donna**, the Director and Policy Maker
2. **Sarah**, the Scientist, Researcher, or Consultant
3. **Wiley**, the Water User
4. **Cody**, the Commissioner
5. **Diego,** the Data Manager, QA/QC Staff
6. **Fatima**, the Field Engineer

These requirements are grounded in user-centric design (UCD) research, including staff interviews, a focus group, and a survey that reflect direct input from Division staff, commissioners, and other stakeholders who work with measurement data daily. The document was developed over the past two years, compiled from several sources and may use a better formatting or more concise narrative. It is meant to be a foundational reference and justification to design decisions for internal staff and external contractors as the Division develops the Next Gen Water Measurement Data Dashboard.

This information also directly informs the interface design and feature development for measurement data managed by the HydroServer database and its generic web interface.[1] The document provides links to Figma mockups for each persona, which illustrate the user's journey in navigating the interface and accomplishing their essential tasks. Together, these

[1] HydroServer Database: https://hydroserver.waterrights.utah.gov/

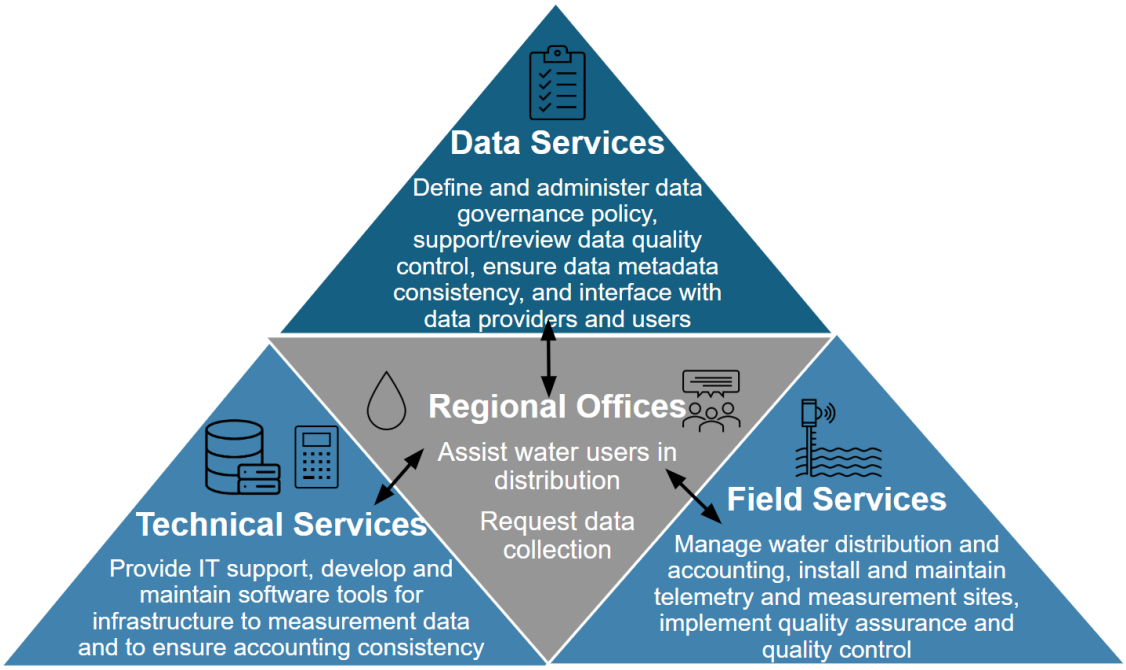
persona-driven requirements are intended to guide the Division toward a modernized, standardized, and user-friendly dashboard that meets the needs of internal staff and the public alike.

Strategic Background & Legislative Alignment

The Utah Division of Water Rights (DWRi) is a state government agency and a division of the Utah Department of Natural Resources (DNR). According to the Utah Code, the DWRi is designated as the water rights authority of the state of Utah, and is vested with powers required to perform necessary responsibilities. The State Engineer is the director of the division, and is responsible for the general administrative supervision of the state waters and the measurement, appropriation, apportionment and distribution of those waters. The Office of the State Engineer was created in 1897 and is one of the oldest state government entities.[2]

Charged with the administration of the laws governing water rights in Utah, the DWRi is the institutional memory of water data for the state. Division functions are carried out within six Sections: 1) Application & Records; 2) Adjudications; 3) Dam Safety & Stream Alterations; 4) Field Services; 5) Technical Services; and 6) Data Services. The division also has seven regional offices throughout the state located in the cities of Logan, Salt Lake City (two regional offices), Vernal, Price, Richfield and Cedar City.

[2] What is the Division of Water Rights: https://waterrights.utah.gov/wrinfo/



Why this initiative exists

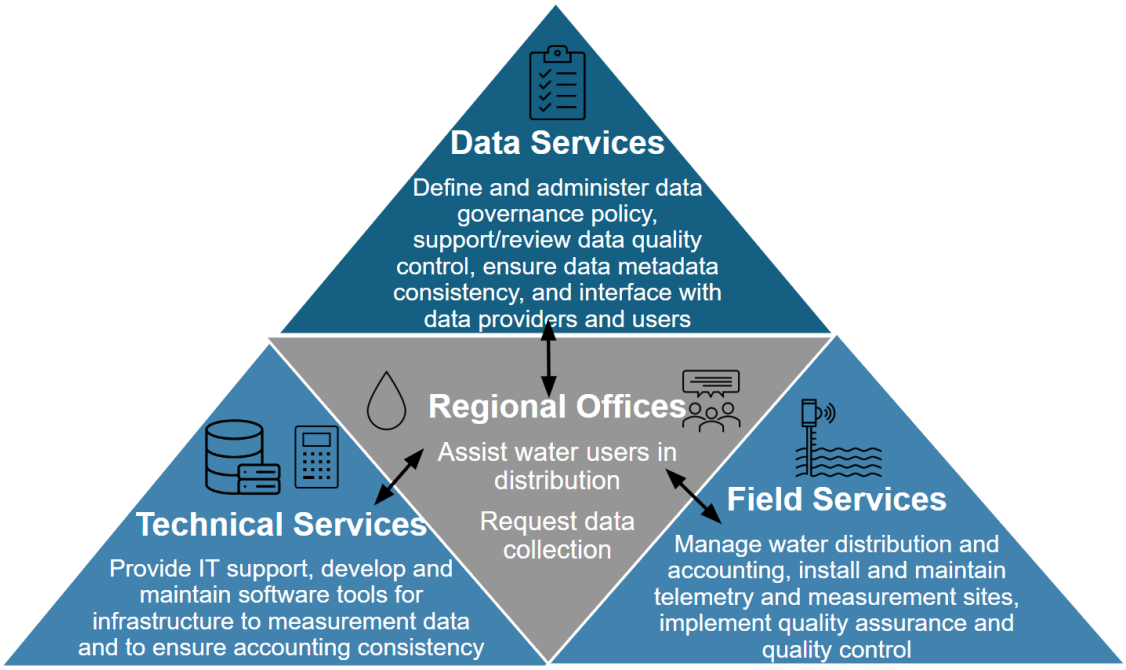
- In response to the 2023 Legislative Audit,[3] the DWRi is overhauling its existing water distribution data system into a new, modernized one.
- The new system is meant to better support the DWRi's daily operations in administering the appropriation, measurement, apportionment, and distribution of the state's surface water and groundwater.

Where it fits in the Division's Action Plan

- This effort falls under the Division's Magnificent 7 pillar Action Plan[4], specifically:
  - **Chapter 3**: "Water Right Data Enhancements"
  - **Action Item No. 1**: "Improve Measurement Data Infrastructure"
- The Action Plan states its goal directly: "Because of the growing need for water rights data by multiple agencies, our goal is to create an action plan that will guide daily operations and improve data sharing."

Background on DWRi's data role

[3] Legislative Performance Audit: https://le.utah.gov/interim/2024/pdf/00001105.pdf
[4] Division of Water Rights Action Plan 2024-2025: https://waterrights.utah.gov/action-plan/DWRi-Action-Plan-2024.pdf



- For decades, DWRi has collected and managed detailed data on water use across the state.
- This data is used extensively for water distribution decisions and is accessible on the Division's website.
- DWRi recognizes its existing data systems need to be more understandable and accessible to meet its statutory mandate.

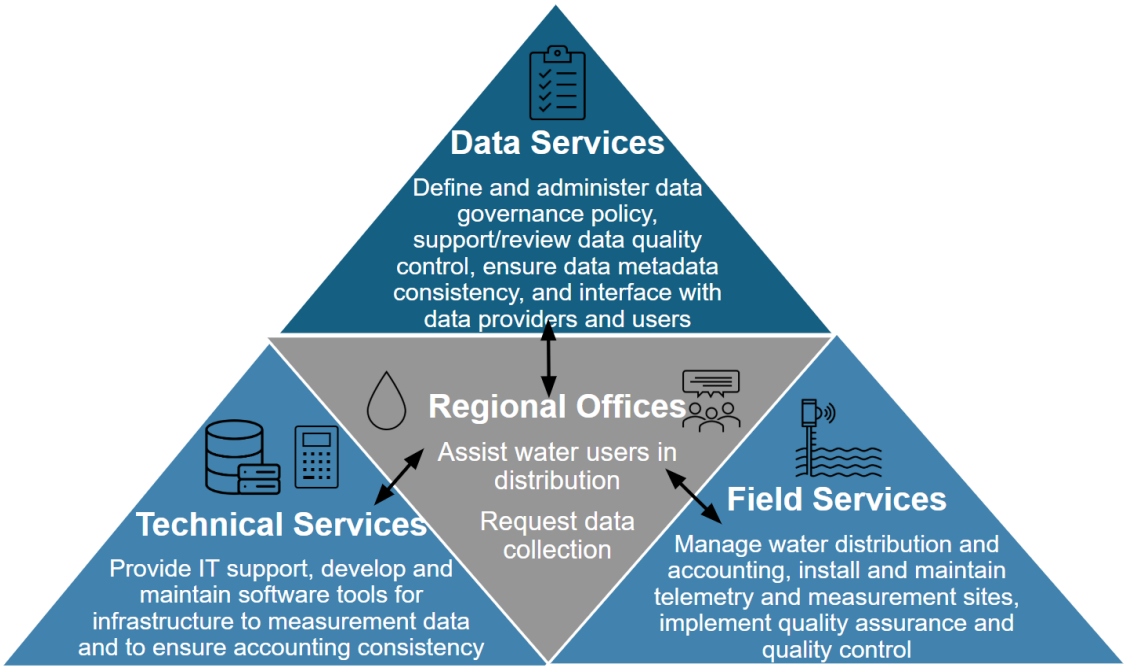
The legal basis for DWRi's authority

- Water in Utah is "the property of the public" (Utah Code 73-1-1).
- The Division is an office of public record (Utah Code 73-2-11).
- The DWRi director "shall be responsible for the general administrative supervision of the waters of the state and the measurement, appropriation, apportionment, and distribution of those waters" (Utah Code 73-2-1(3)(a)).

The problem the audit identified

- DWRi currently makes a variety of water data available online.
- However, this information is not generally well documented, consistent, or easily understandable — a gap highlighted directly by the 2023 Legislative Audit.[5]

[5] Legislative Performance Audit: https://le.utah.gov/interim/2024/pdf/00001105.pdf



- This finding lines up with the Legislative Auditor General's High-Risk List, which names Utah's water use and water infrastructure needs among the state's twelve highest-risk areas.[6]

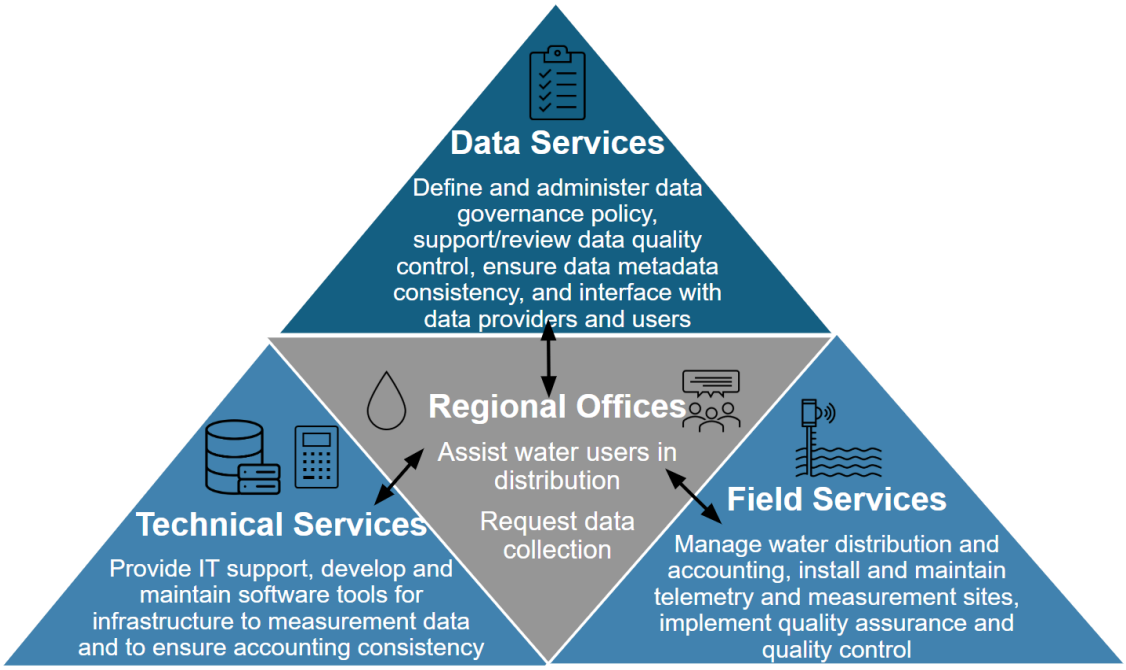
Modernization Goal: To develop a new custom web tools ecosystem that supports the standardized management of the State of Utah water diversion and water use data and provide streamlined access that informs the operation and planning of water resources. This goal addressed the Audit's main recommendation: "DWRi should prioritize excellence in data management to ensure state water data can be used effectively for a wide scope of management needs."

User-Centered Design (UCD) Methodology

The dashboard development process puts customers first by adopting a **User-Centered Design (UCD)** approach. Rather than letting software logic and developer efficiency dictate design decisions, UCD focuses on user needs first and engineers the technical solution around them.

Architects vs. Engineers: Traditional software development often lets engineers dictate the design based on ease of coding. Under UCD, the users act as the architects defining

[6] High-Risk List: Identifying and Mitigating Critical Vulnerabilities in Utah - 2023. https://le.utah.gov/interim/2023/pdf/00004965.pdf



the vision, while developers serve as the civil engineers tasked with making that visual, functional structure a reality.

Personas & Use Cases

To guide feature development throughout the project lifecycle, DWRi staff established **six user personas**. These personas group water data users in Utah into themes based on shared roles, background, and data needs, ensuring development directly supports real-world use cases.

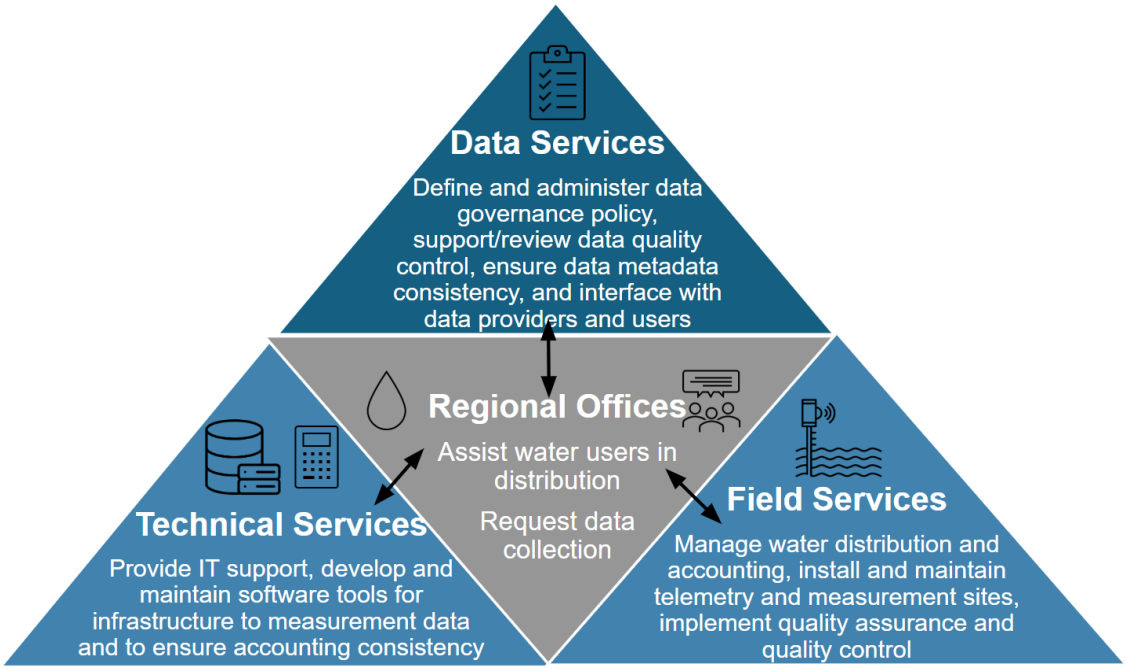
Division Personas & Data Governance

Three of the six personas represent internal staff across four primary DWRi sections (Figure 1):

- Technical Services
- Field Services
- Data Services
- Regional Offices

These roles align directly with the Measurement Data Governance Plan[7] , which defines responsibilities across the data lifecycle:

[7] Water Measurement Data Governance Plan for the Utah Division of Water Rights: https://docs.google.com/document/d/12dwWXs9Hj_TktB-jTwO0w9ezVMdwseMU9CPRW7e3TCQ/edit?usp=sharing



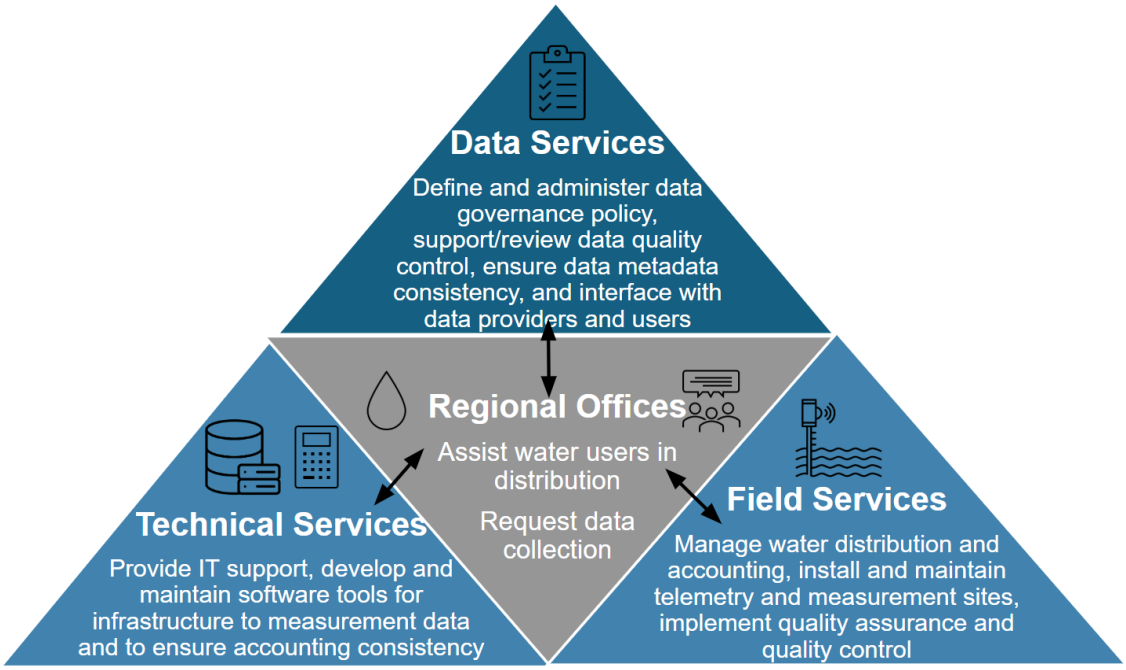
Quality Assurance (QA) — Pre-Collection: Putting proactive processes in place to prevent errors. Examples include calibrating instruments, training staff, and establishing automated validation rules before data is recorded.

Quality Control (QC) — Post-Collection: Reviewing and evaluating recorded data to identify errors, anomalies, or missing values. Examples include applying corrections, interpolation, and adjustments so final datasets are accurate and analysis-ready.

Figure 1: Unique and complementary staff roles pertaining to measurement data across the Technical Services, Field Services Data Services Sections and the Regional Offices

Each of the six user groups requires unique user experience optimized for their specific tasks, level of access, and data interaction frequency. The personas are given fictional names to humanize the user experience.

7. **Donna**, the Director and Policy Maker
8. **Sarah**, the Scientist, Researcher, or Consultant
9. **Wiley**, the Water User
10. **Cody**, the Commissioner
11. **Diego,** the Data Manager, QA/QC Staff
12. **Fatima**, the Field Engineer



These requirements are from a user-centric approach independent of or agnostic to the underlying design. The requirements are a synthesis of collective input from Water Rights staff and other stakeholders as described in an earlier foundational document collated by David Jones in 2023 entitled "User Stories: As a [persona], I [want to], [so that]." The described requirements here will likely evolve over time but likely not radically.

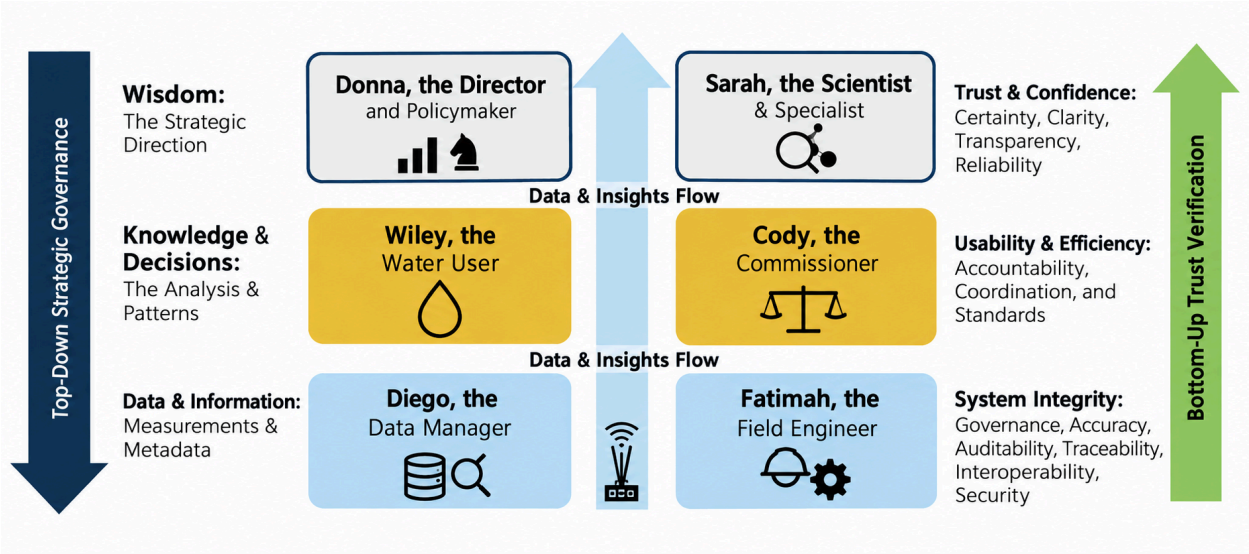
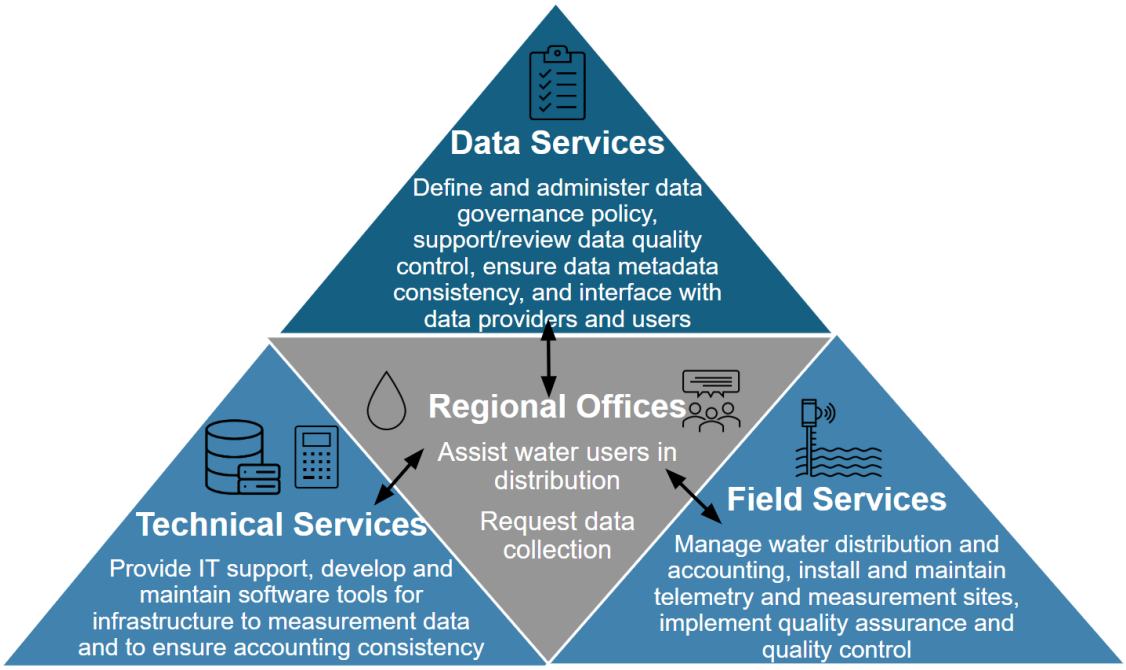


Figure 2: The six personas and their key roles or needs motivating and guiding the design of Next-Gen Water Use and Distribution Data System. Water users and the commissioners are at the heart of the system, supported by DWRi staff to provide operational assistance and provide transparency and insights to the leadership. The graph is generated by Gemini.



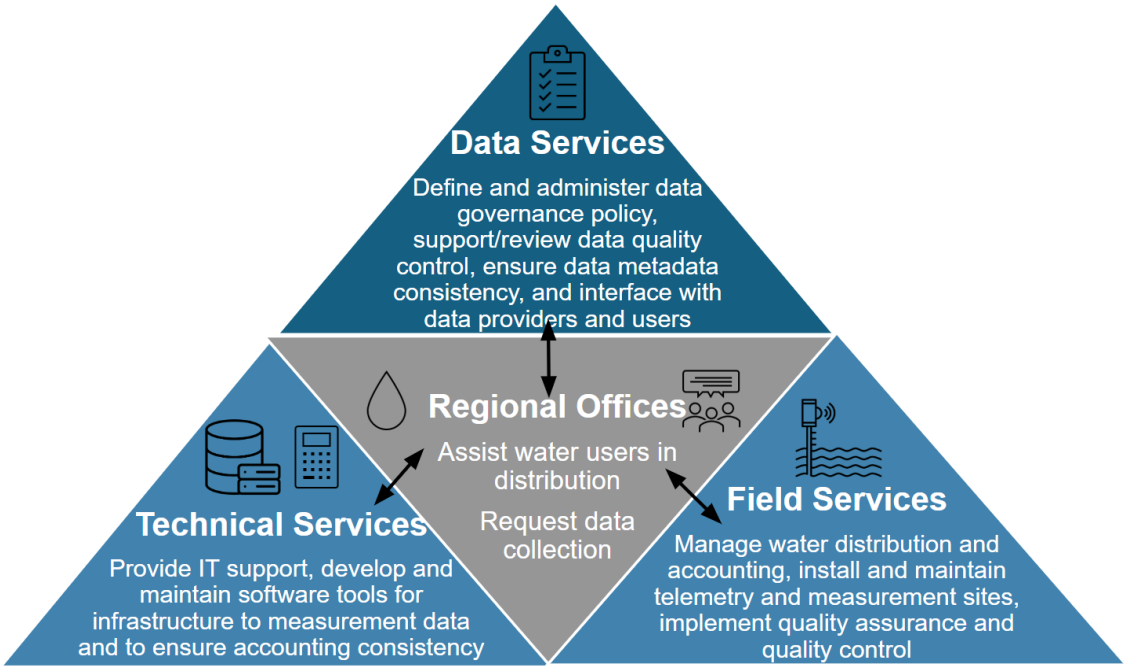
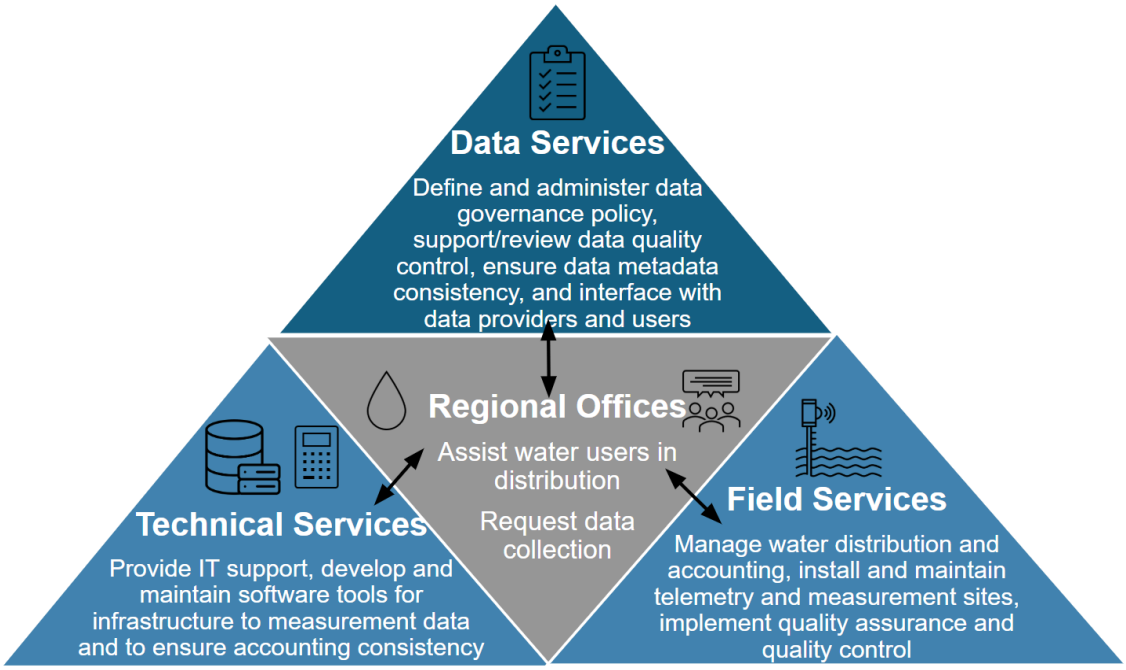

Data Services
Define and administer data governance policy, support/review data quality control, ensure data metadata consistency, and interface with data providers and users
Regional Offices
Assist water users in distribution
Request data collection
Technical Services
Provide IT support, develop and maintain software tools for infrastructure to measurement data and to ensure accounting consistency
Field Services
Manage water distribution and accounting, install and maintain telemetry and measurement sites, implement quality assurance and quality control

Table 1: The six personas and how they address requirements mentioned in the Legislative Audit and the Best Practices reports.

Persona	Key Role	Directions from the Audit and Best Practices Reports
1. Donna	**Director**	“High-level summaries and charts”, "Shaping vision with stakeholders", "Weigh competing interests", "Transparency and Accountability","Report to the Utah Legislature"
2. Sarah	**Scientist**	"Publicly accessed and used", "FAIR (Findable, Accessible, Interoperable, Reusable)", "Research integrity", "Coordination among agencies”
3. Wiley	**Water User**	"Operational flexibility", "Equitable access", "Protect water users"
4. Cody	**Commissioner**	"Distribution accountability", "Real-time measurement", "Enforced lines of responsibility"
5. Diego	**Data Manager**	"Data integrity", "Data map and data dictionary", "Eliminate Data Silos"
6. Fatima	**Field Engineer**	"Accurate and complete" data, "Meter and monitor", "Innovation... make processes more efficient"

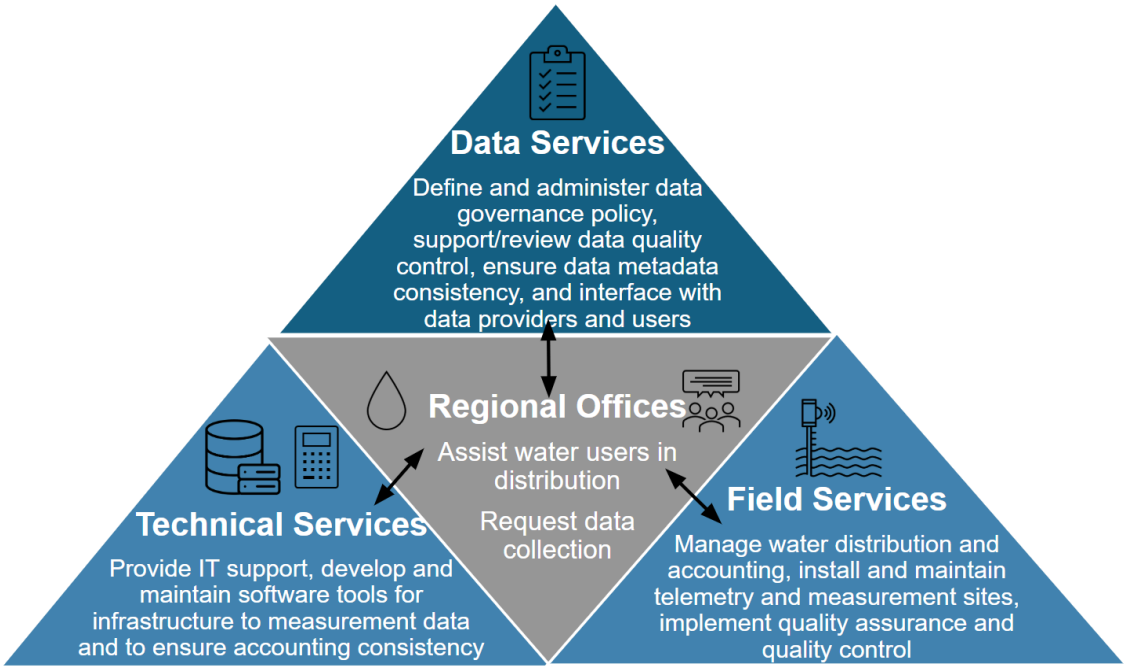
The New Distribution Data Dashboard Survey[8] was fielded in March 2026 by the Division of Water Rights' Data Services Section, developed in response to the 2023 Legislative Audit as part of an effort to overhaul the Division's water distribution data

[8] New Distribution Data Dashboard Survey https://docs.google.com/forms/d/e/1FAIpQLSdrtuC9iGd3XhbwGp7PxsdGEENWjQ_qvkpwIelS55AIpWc2eg/viewform?usp=sharing&ouid=109773658733321178469



system. The survey drew 33 responses across three staff groups[9]: Commissioners or Deputies, Distribution Engineers, and Regional Office Engineers. Commissioners or Deputies made up the large majority — 21 of the 33 respondents (nearly two-thirds) — with Distribution Engineers accounting for 7, Regional Office Engineers for 4, and one additional response from Measurement & Telemetry staff.

[9] Survey results https://docs.google.com/spreadsheets/d/1LupUwRyWvIvDxD2lONkH8t7_Ge0EbamlUNO6yLHWM_E/edit?usp=sharing



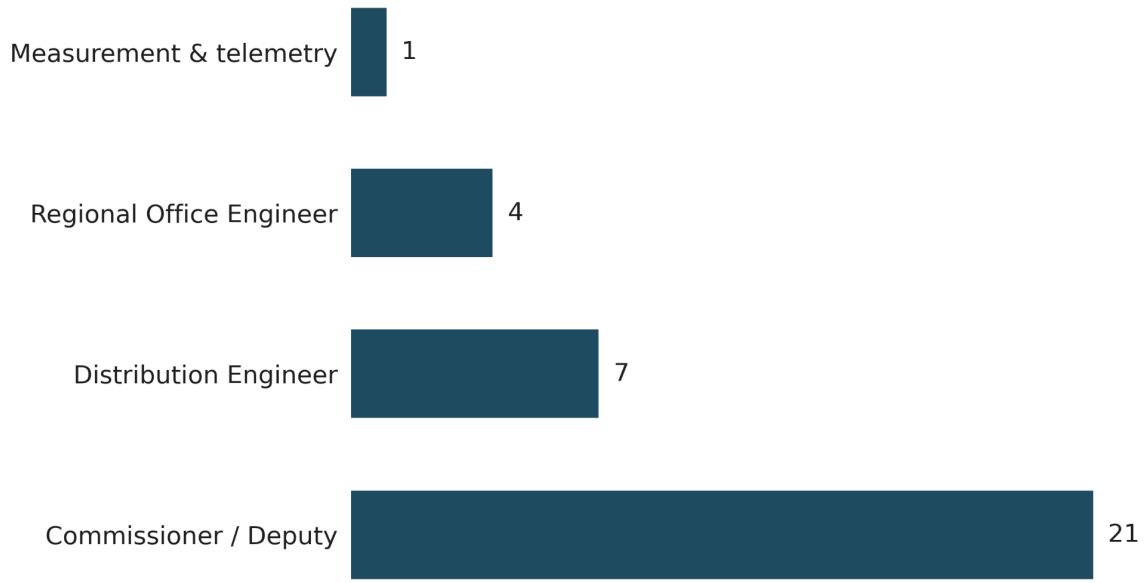
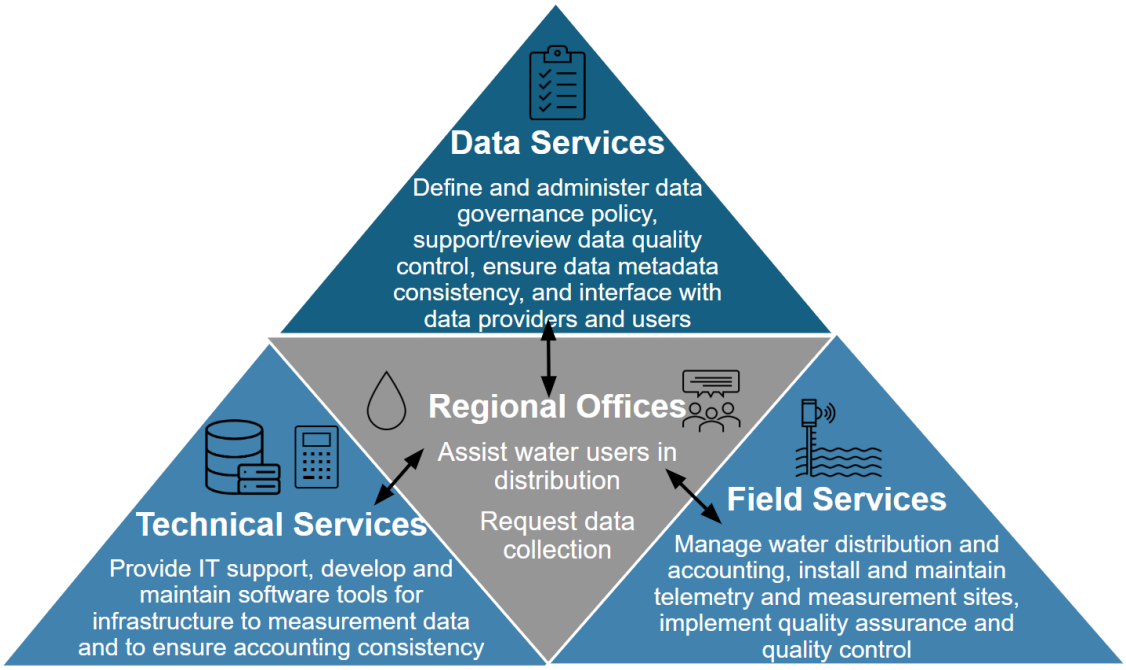


Figure 3: Survey respondents by staff role (n=33)

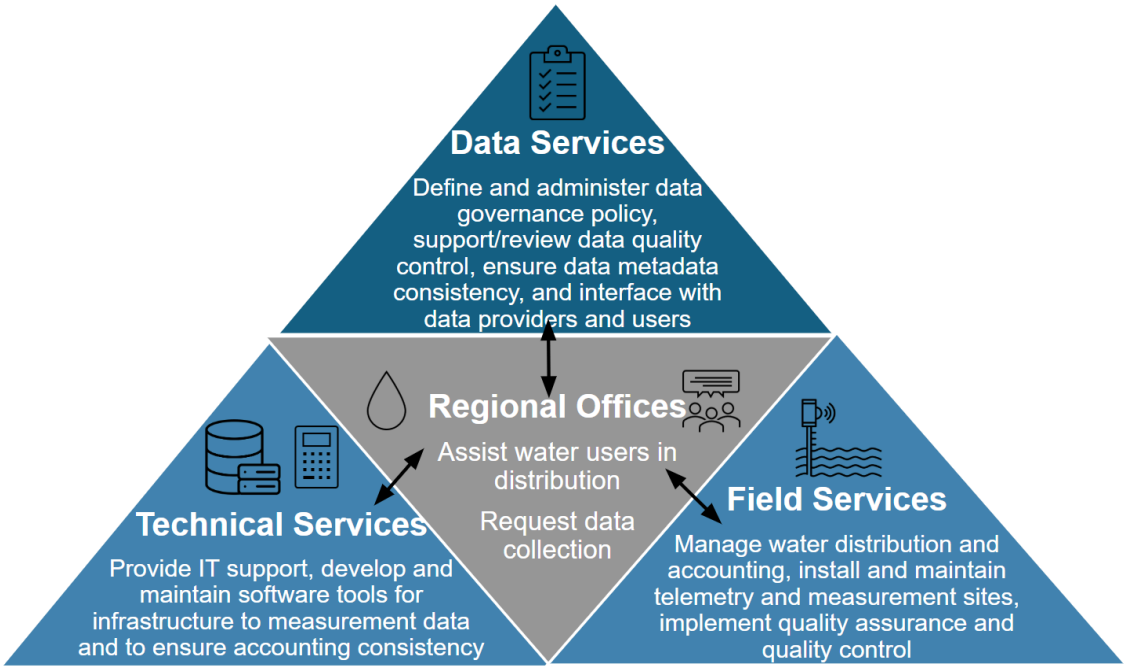
This desired dashboard will be the primary interface for different users to log in and get various types of access to water distribution and use data. So ideally, the user can complete their journey to get accomplished as smoothly as possible without feeling they need to use different tools or logins. Having one universal account for all sorts of modular tools would be super ideal.



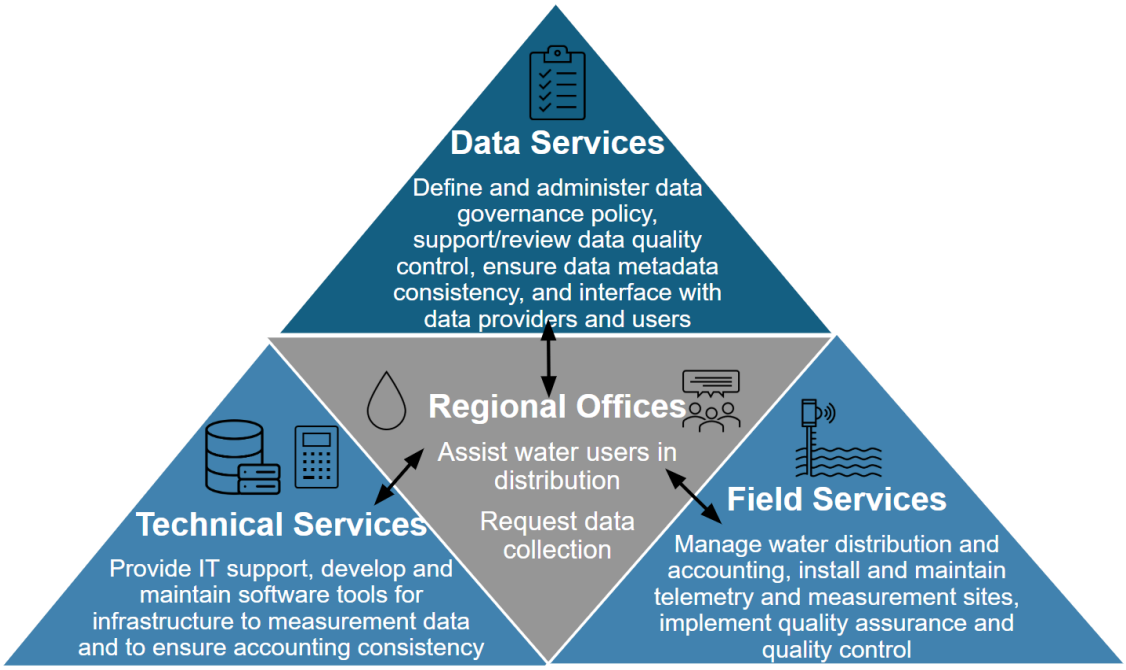
Both public and internal users here interact with an online tool for different purposes without knowing anything about the backend database(s) and APIs. The online tool should be portable and can be viewed on regular screens in Chrome, Edge, Safari, and Firefox browsers.

The interface will be publicly accessible through a web browser and will provide access, filtering, and analytics of water data across the state. It will also offer further functionalities for users with accounts, especially internal users. The following are the most basic functionalities available to all users, including those not signed in:

- **View Map:** View a map of all water distribution sites in Utah.
- **Change basemaps:** Switch map backgrounds (e.g., topographical, satellite).
- Site Filtering: Filter sites by one or more of the following controlled vocabulary metadata that describes the site or its measurement data. Users can build filters manually through selectors, or describe what they need in plain language and let AI translate that into the equivalent filter set — for example, typing "active reservoir sites in the Bear River system with telemetry" or "streamflow gages in Utah with data since 2020" should return the same result as selecting those fields by hand. The AI-assisted path is meant to sit alongside manual filtering, not replace it — users can always drop into the manual controls to fine-tune or verify what the natural-language query selected.
    1. Distribution System
    2. Distribution System Groups
    3. Distribution System subgroups



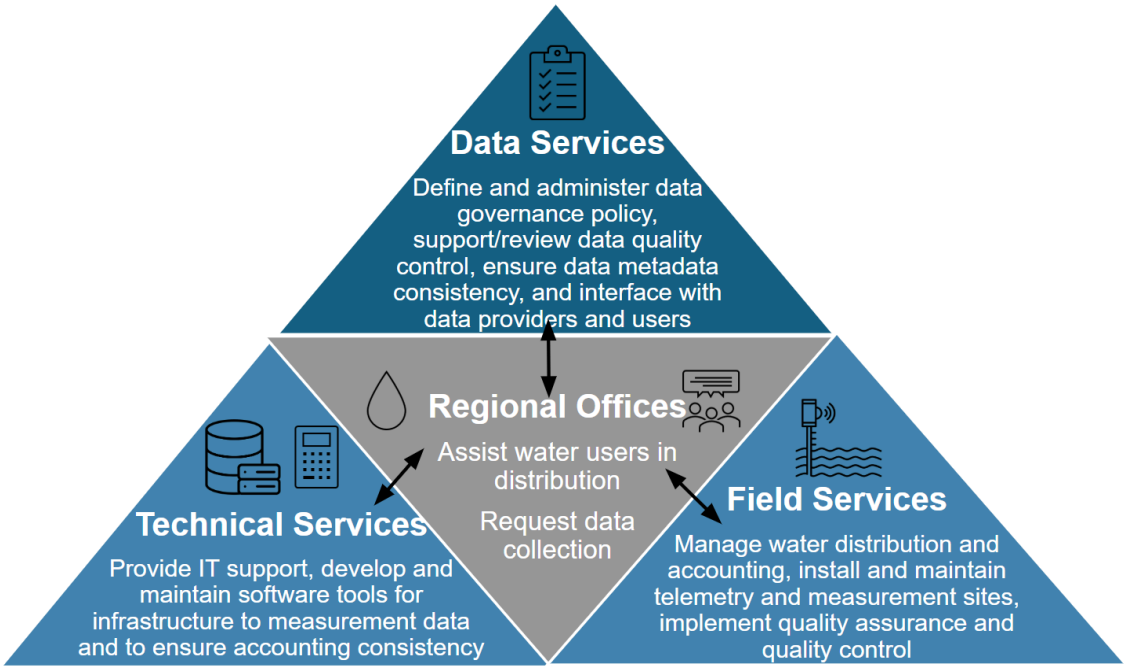
4. Site type (conveyance, stream, reservoir, reservoir release)
5. Telemetry availability (yes, no, unknown)
6. Data provider name (e.g., USGS, DWRi, CUWCD)
7. Observed property (Reservoir storage, flow, gage height)
8. Control Structure (culvert ditch/canal flume - Parshall Flume, Ramp Flume Gate, Headgate)
9. Material (concrete, metal concrete, rock/masonry, wood earthen)
10. Control Structure Size (0.75 ft, 1 ft, 1.5 ft, 1.7 ft)
11. Water source type (e.g., SW, GW)
12. Measuring device type (e.g., logger/recorder, Campbell CR series sensor, acoustic doppler sensor, magnetic sensor, pressure sensor, radar sensor, ultrasonic staff gage)
13. Measurement method type (instrument deployment, manual reading)
14. Observation Type (Observed, Computed - add up two measurements, Calculated - model result, Estimated - professional, Unevaluated)
15. Available data date range (min and max dates for each site)
16. Site Status (i.e., active, inactive, proposed)
17. State (ID, UT, WY, CO)
18. Time step (e.g., monthly, daily, hourly)
19. Aggregation Statistic (Average, Cumulative)
20. Method Type (Instrument deployment, manual reading, site visit)



21. Others?

Ask in plain language: A natural-language search bar lets any user — signed in or not — ask a direct question and get the relevant sites, chart, or table back. Under the hood, the AI maps the request onto the same 21 metadata fields used by the manual filters above — it isn't a separate search index, just a different entry point into the same structured data. That mapping is what makes the answer trustworthy: every result the AI returns can be traced back to the specific field values it matched, and a user can always drop into the manual filters to see or adjust exactly what was selected. This matters most for general users, who know what they're looking for in plain terms but shouldn't need to learn a controlled vocabulary to find it. For example:

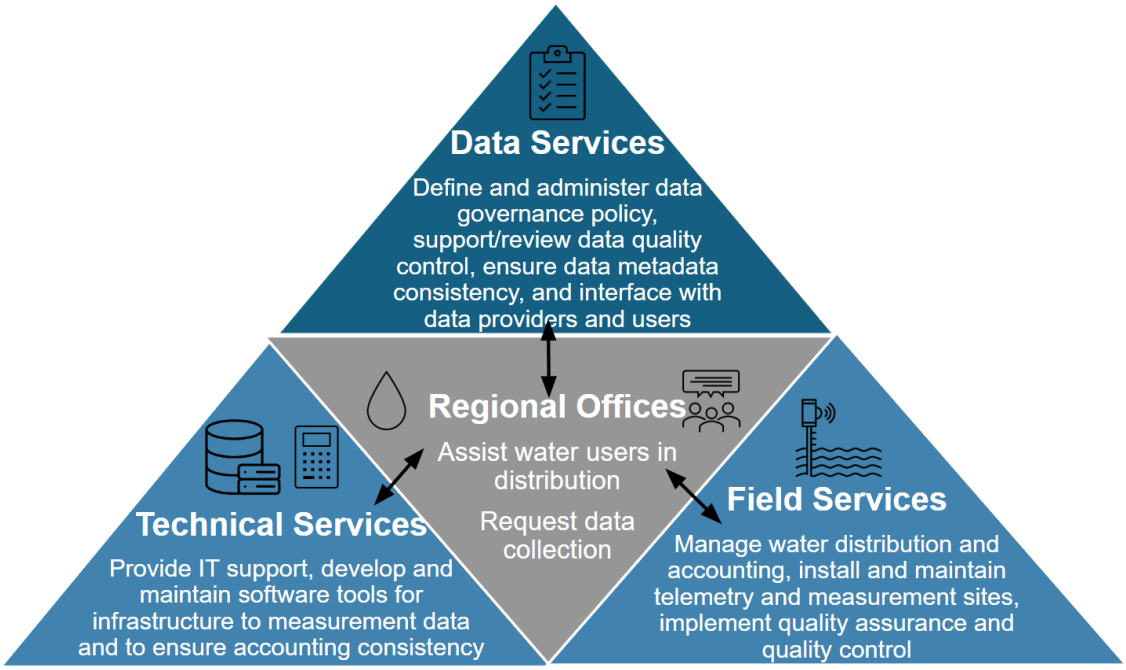
- "Active reservoir sites in Utah with telemetry" → Site Status = active, Site type = reservoir, State = UT, Telemetry availability = yes
- "USGS streamflow gages reporting hourly since 2020" → Data provider name = USGS, Observed property = flow, Time step = hourly, Available data date range = since 2020
- "Concrete Parshall flumes in the Bear River system" → Distribution System = Bear River, Control Structure = Parshall Flume, Material = concrete



- "Groundwater sites with a pressure sensor that are still just proposed" → Water source type = GW, Measuring device type = pressure sensor, Site Status = proposed
- Show stations in my system with flow declining fast over the past 3 days
- What's the current release flowrate from Sevier Bridge Reservoir?

Each persona (except the public one) will have an account (probably connected to an API key under the hood). After they log in, they can see the sites with permission to edit. They also could toggle to a different tab and view and filter other sites in the State. While the tool will be public-facing with no login required, individual users who want to make edits or view their custom set up, they need to have an account to save their preferences.

- **Note**: Many of these metadata filters are not directly supported in the underlying HydroServer architecture. Supporting the ability to filter sites by these criteria in the dashboard directly influences how the data must be managed and structured in the database to ensure performant queries.
- Quick Site View: Display a small pop-up card when clicking on a site; this card will show the station and site name, the last reading, and the reading value, and include a hyperlink to the site landing page for detailed data access.



- Analytics: Display dynamic analytics (e.g., pie charts of water distribution flow, measuring device type). While this feature is not critical to operational needs, its dynamic and interactive capability offers tremendous value and insights to all users by summarizing the data in many different dimensions (see the Donna, the Director persona for the value proposition). Relevant from the Commissioner Survey: When you are deciding how to distribute water, which information do you need to see together? “The average usage on all the system.”

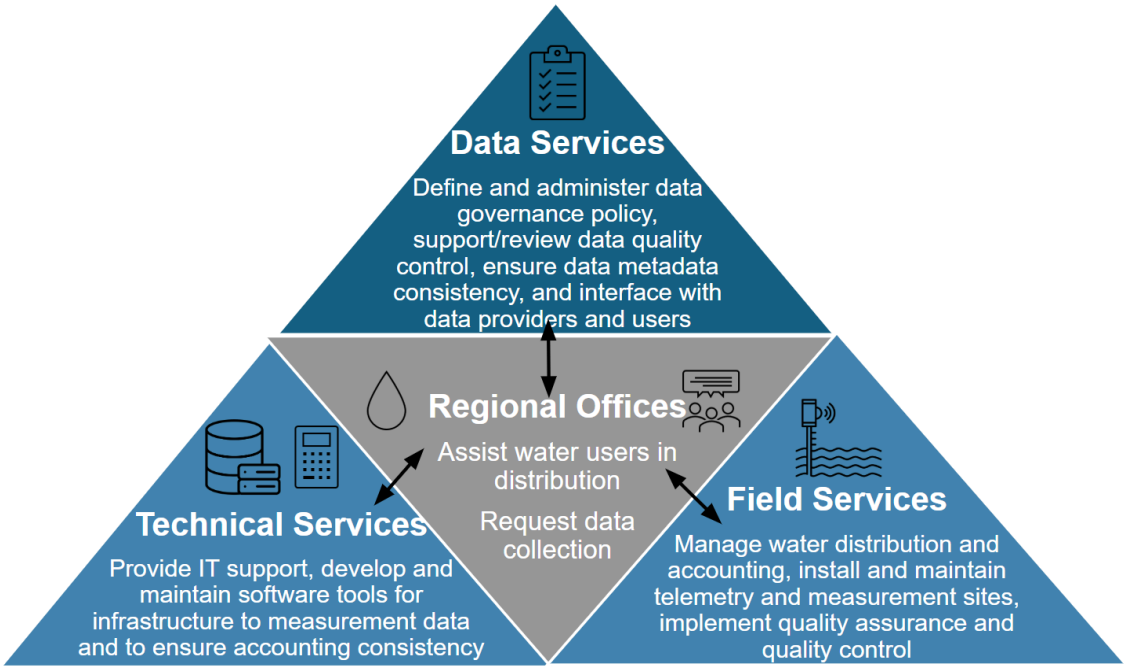
The six personas, their Figma mockups, and detailed permissions/features are outlined below:

1. Diego, The Data Manager (Data Steward/Administrator)

Diego is a Data Steward in the Data Services Section. He is the Data Governance implementation expert tasked with implementing and enforcing data policies, managing and reviewing data quality efforts, and serving as the subject matter expert for data both internally and to the public (see the Appendix).

Task Category 1: Data System Authority

Admins are tasked with ensuring the Division maintains a consistent and authoritative database with controlled metadata for all surface and groundwater data. Admins control and distribute edit



access to the database. This structure is consistent with the Division's Technology Modernization Roadmap, which recommends a modernized, services-oriented measurement data system (HydroServer) with clearly documented metadata and access control.[10]

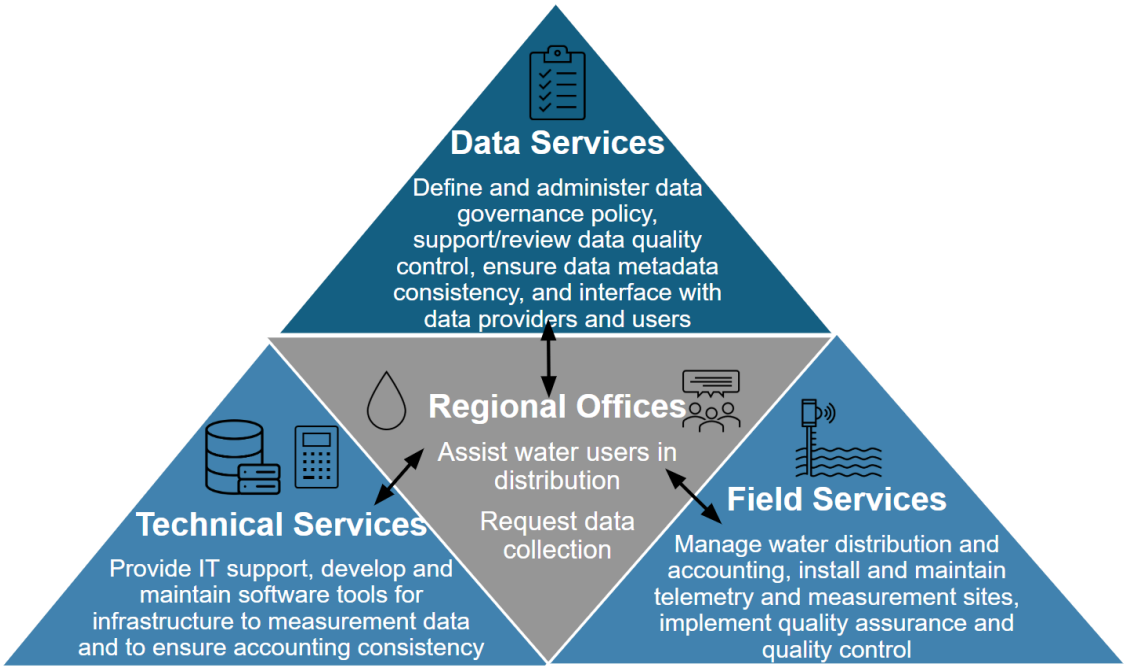
- Access Structure: Admins have the exact same access roles for all sites (universal administrative access). This setup is designed to create redundancy and flexibility, simplifying onboarding/offboarding and eliminating the complexities of site ownership transfer.
- Historical Tracking: When an Admin leaves, their history and name remain in the system to preserve an auditable trail.

Task Category 2: Dashboard Responsibilities

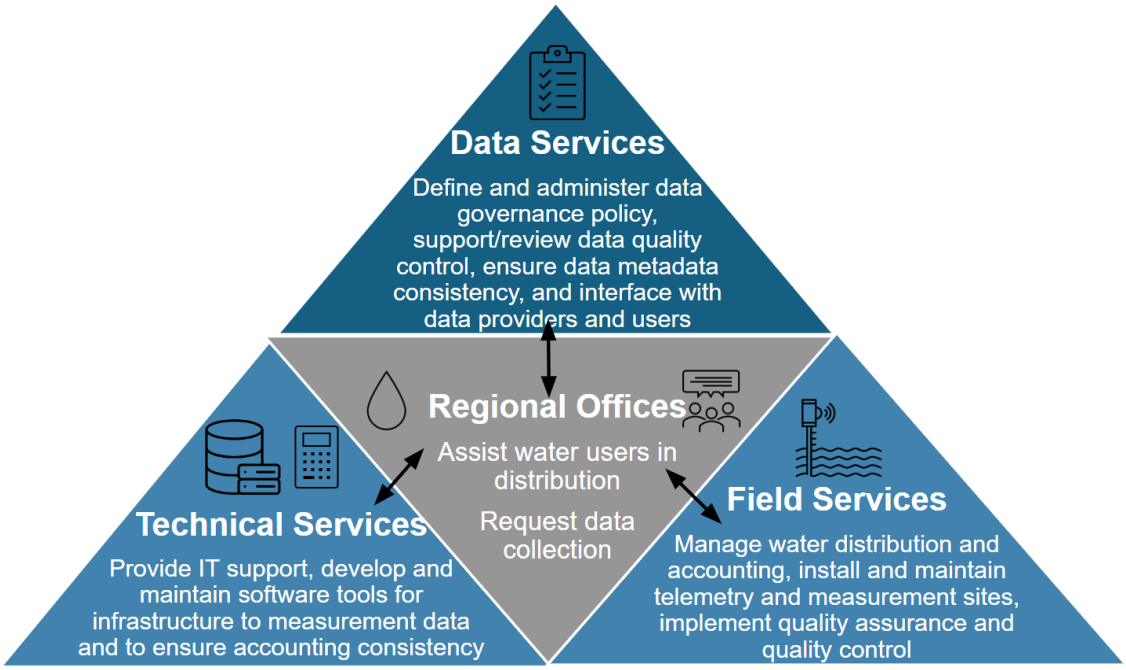
Diego needs to use the dashboard to accomplish the following high-level administrative and quality control tasks:

- User and Permission Management: Manage users and assign permissions (e.g., Public User, Admin, Commissioner, Engineer). Specifically, he assigns Distribution Engineers or sites to the Distribution Engineer and Commissioner roles.

[10] Horsburgh, J.S., Slaugh, D., Lippold, K. Utah Division of Water Rights Technology Modernization Roadmap. Utah Water Research Laboratory, Utah State University, September 2024.



- Metadata Control: Has the sole responsibility to create or delete distribution systems and to edit universal metadata (sensors, observed properties, processing levels, units, result qualifiers, etc.) to maintain consistency across the entire database.
- Data Orchestration Setup: Set up new data providers in the orchestration system and add new measurement sites to the orchestrator for data ingestion.
- System Health Monitoring (Overall Connections Health): Monitor the data streaming health to ensure the orchestrator is operational. Subscribe to a daily digest notification — for example, delivered at 8 a.m. on weekdays, with the Monday digest also summarizing Saturday and Sunday — that lists which Orchestration System data connections need attention: whether an entire connection is down, or how many individual tasks within a connection have failed. Each listed connection links directly to a filtered view of its tasks in the Orchestration System, with the tasks needing attention sorted to the top by default.
    - Example: “The Central Utah connection needs attention (5 tasks failed)” or “The USGS connection is completely down.”
    - Desired notification: Get alerts when a new user profile is created and when a new observation site is added.
- Data Sanity Check (QA/QC): Visualize, summarize, and inspect data as a quality assurance check. This includes viewing plots of discharge, water level, and battery voltage;

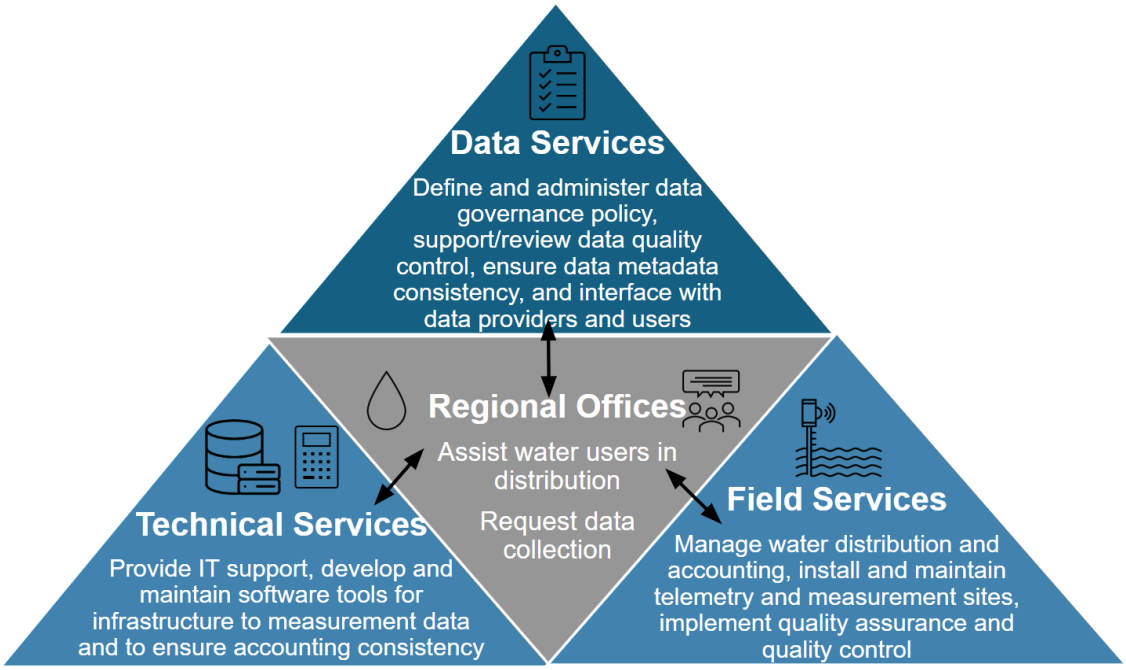


viewing site photos and location; and using tools like the tea-cup diagram for any "favorite" site.

- Automated Data Quality Monitoring (QC Checks): Configure quality-control thresholds for individual datastreams (daily and sub-daily), then subscribe to a notification digest — weekly, daily, twice daily, or hourly — that lists only the datastreams with QC violations, sortable by violation type. Subscriptions can be filtered by distribution system (e.g., Sevier River) and set to all datastreams or a selected subset.
    - Supported rule types: automated range checks (including battery-voltage checks), rate-of-change checks, persistence checks, missing-data checks, and zero-value checks.
    - Each monitoring task records who configured it and when it was last updated, runs independently of other tasks so one failure does not affect others, and can be tested in a development or test environment before being promoted to production.
    - (Source: HydroServer Automated Data Quality Monitoring Functional Specifications)

Task Category 3: Analytics, Reporting, and Customer Support

Diego is also expected to accomplish the following tasks:

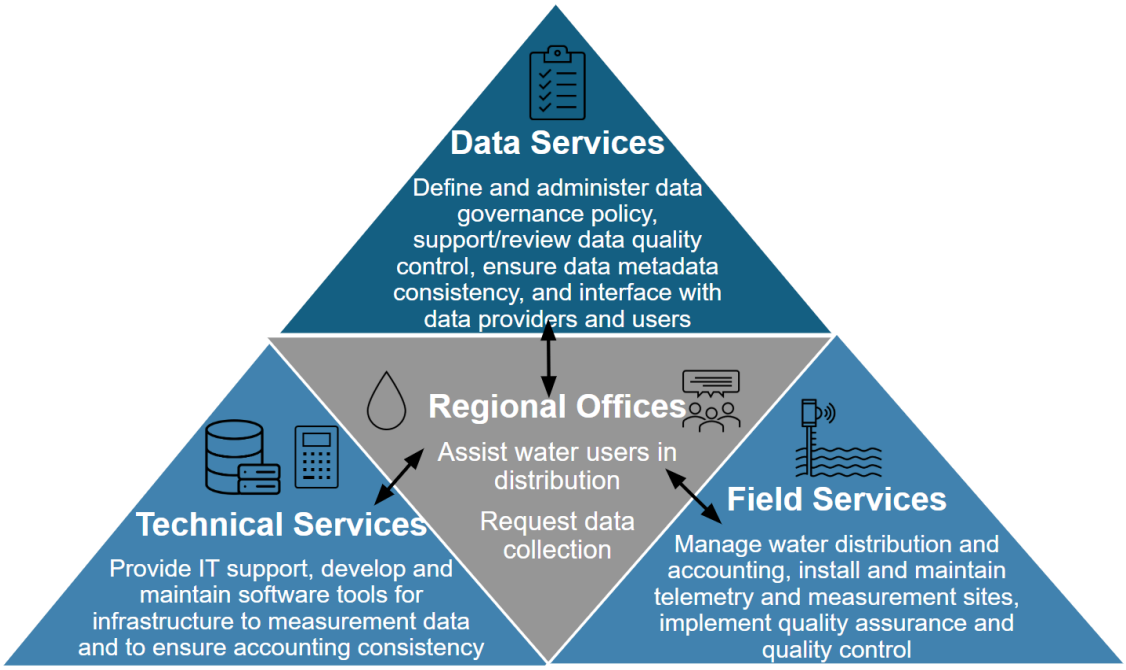


- Analytics and Reporting: Track and report analytics (number of users, visitors, and data download requests) for the dashboard and the APIs per day, month, and year using external tools like Google Analytics and Microsoft Clarity.
- Customer Support and Prioritization: Provide timely customer support. This involves tracking and prioritizing software bugs to be fixed and advocating for additional features based on user requests. Diego wants the dashboard to generate unique URLs that encode the selected parameters and zoom level so they can replicate it when shared by a user. This URL custom encoding is important to let Diego and users share the exact dashboard setup that could be problematic.

Future Work (not part of this planned Dashboard but relevant to this persona)

Water Use Program Data Manager

- All the same permissions as the water user, except cannot edit boundary areas.
- Access to “Manage Systems” dashboard
    - Notifications tab - notification when comment is added to water reporting system and notification when report is submitted for review.
    - Water Supplier tab - displays all boundary areas in the category selected; public, secondary, industrial, management.

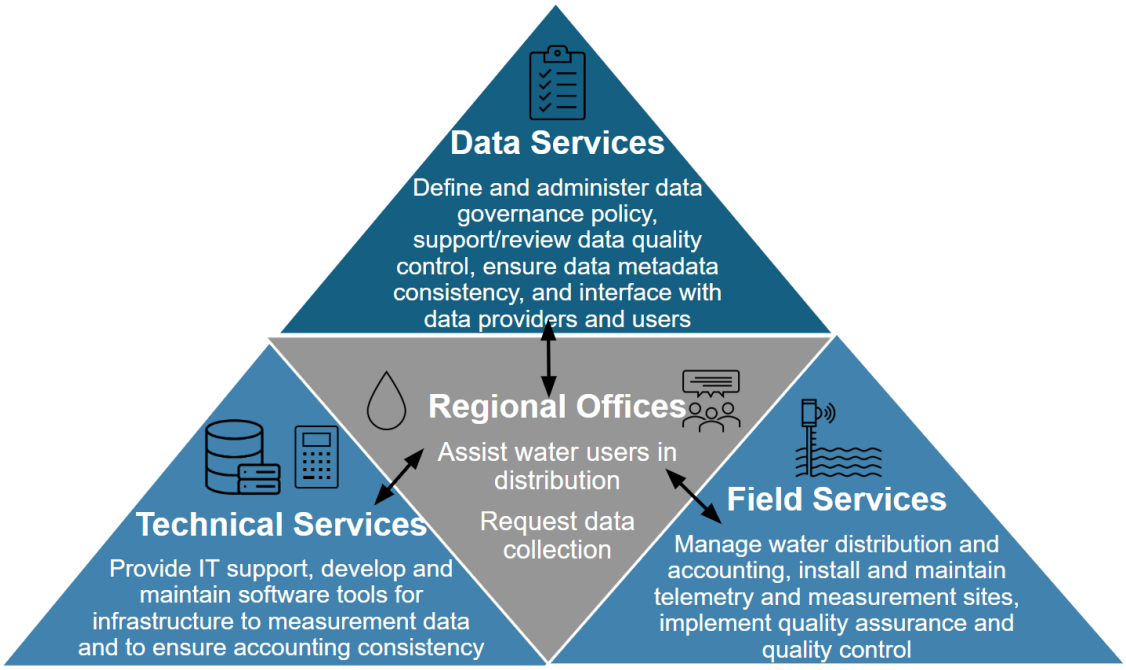


- Ability to filter between: completed, in progress, submitted for review, nothing reported.

Foundational User Stories (2023)

The following needs, drawn from the 2023 foundational user-stories document, describe Data Manager / Data Governance responsibilities that inform Diego's requirements above:

- **Jim:** As a data manager, Jim needs a way to indicate whether data has been reviewed and is of good quality, and wants a clear, consistent process and set of definitions for doing so.
- **Jared:** Jared wants distribution accounting information maintained in a standardized, transparent way, following double-entry accounting principles, so that once a transaction is "finalized" it cannot be deleted or modified.
- **Jared:** Jared wants historical use groups preserved so that water right research is more efficient.
- **Leah:** Leah needs the commissioner's decision data to be in the database in real time and fully QA'ed.
- **Automation engineers and commissioners:** need to be informed automatically of discrepancies in the data.



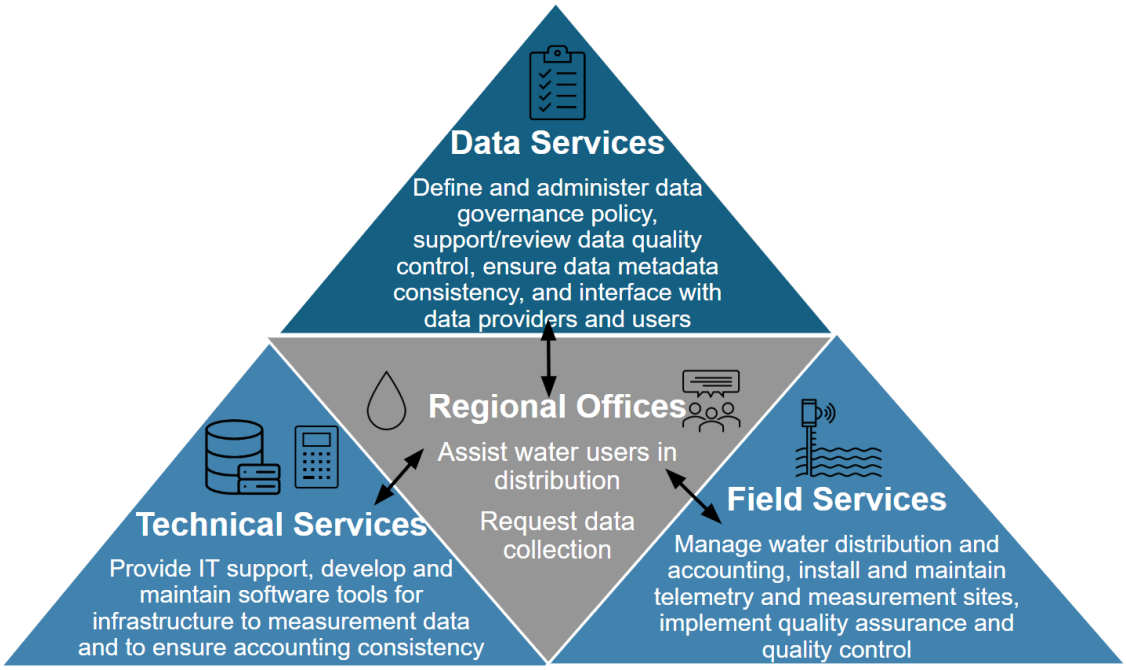
2. Cody, the Commissioner, and their Deputies

Cody is a Water Commissioner (Commissioner) appointed by the State Engineer. He is responsible for physically distributing water according to its legally established rights within a specific distribution system. They are tasked with overseeing head gates, measuring water flow, accounting for storage, and preparing annual reports on water measurements and issues. Their work ensures the orderly and legal allocation of available water to all water rights holders based on the priority of their rights.

Core Responsibilities & Data Needs

Cody requires timely access to accurate daily water flow or volume measurements. He needs a simple way to review and, if needed, correct values measured by automated devices. He also needs to report or enter manual site measurements into the database.

Cody plays a critical role in Data Governance as he is the closest to the measurement sites and most knowledgeable of the distribution infrastructure. He is directly responsible for reviewing and verifying the accuracy, completeness, and consistency of data to identify and correct errors. At the end of the year, he reports his distribution decisions and data in an annual report.



Cody is often a water user or rancher who is not tech-savvy and requires a simple user interface and experience. The dashboard must be user-friendly for the majority of users (at least 60% of the 55 Commissioners and Deputies, or about 30 of them). The Distribution Engineer will assist the remaining Commissioners. The Technology Modernization Roadmap similarly calls for a dedicated, standardized data-entry tool for River Commissioners along with a training program to build their technical skills.[11]

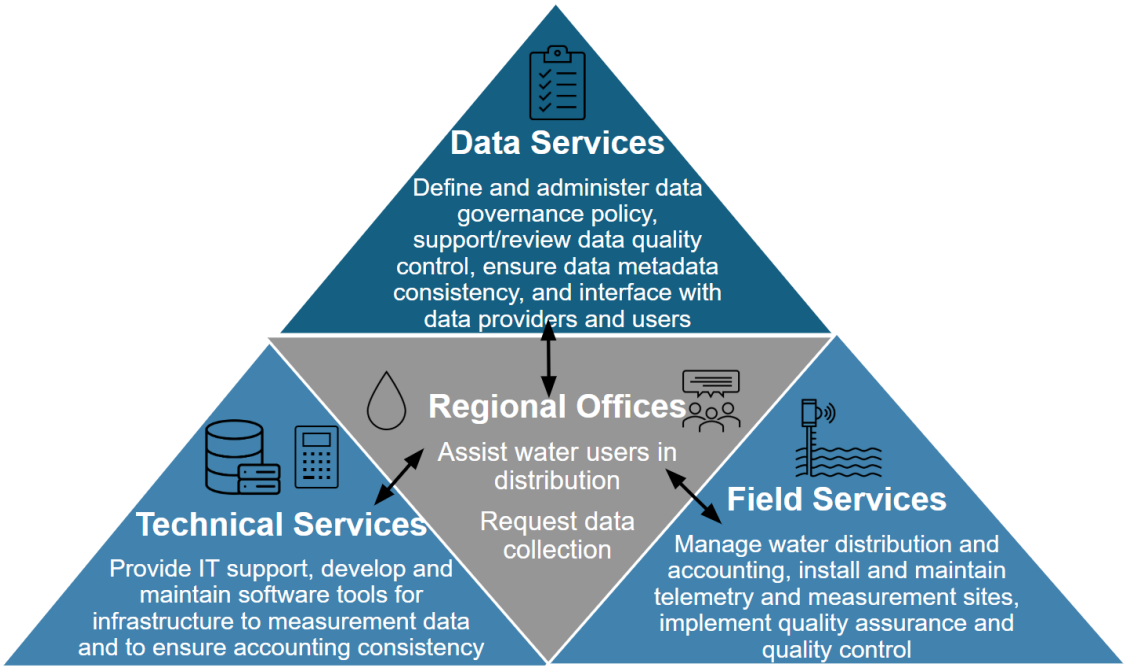
> SURVEY INSIGHT · **Cody checks in constantly, and does it himself.** 55% of respondents access distribution data daily or more often during irrigation season (27% multiple times a day, plus daily and several-times-a-week checkers). When asked how they fill data gaps, respondents overwhelmingly named judgment and legwork — interpolation, site visits, and field notes — not a tool, confirming Cody's role as the frontline data steward.

"Site visit, deputy data, interpolation of data, and others." — Kent S Wilkerson, Commissioner, Weber and Ogden

"As I go check stations, I take written notes of changes or problems to correct the data within the model." — Porter Monroe, Commissioner, Lower Sevier

"If I don't have accurate data to rely on, I estimate usage with users' input." — David Jessop, Commissioner, Virgin River

[11] Horsburgh, J.S., Slaugh, D., Lippold, K. Utah Division of Water Rights Technology Modernization Roadmap. Utah Water Research Laboratory, Utah State University, September 2024.



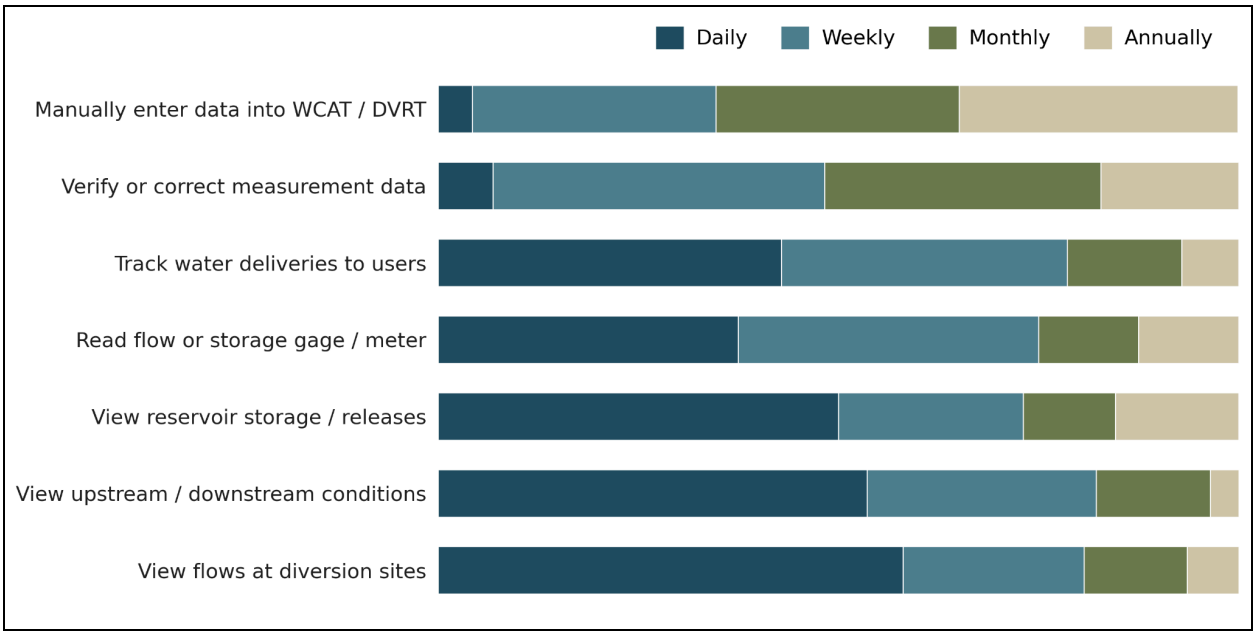
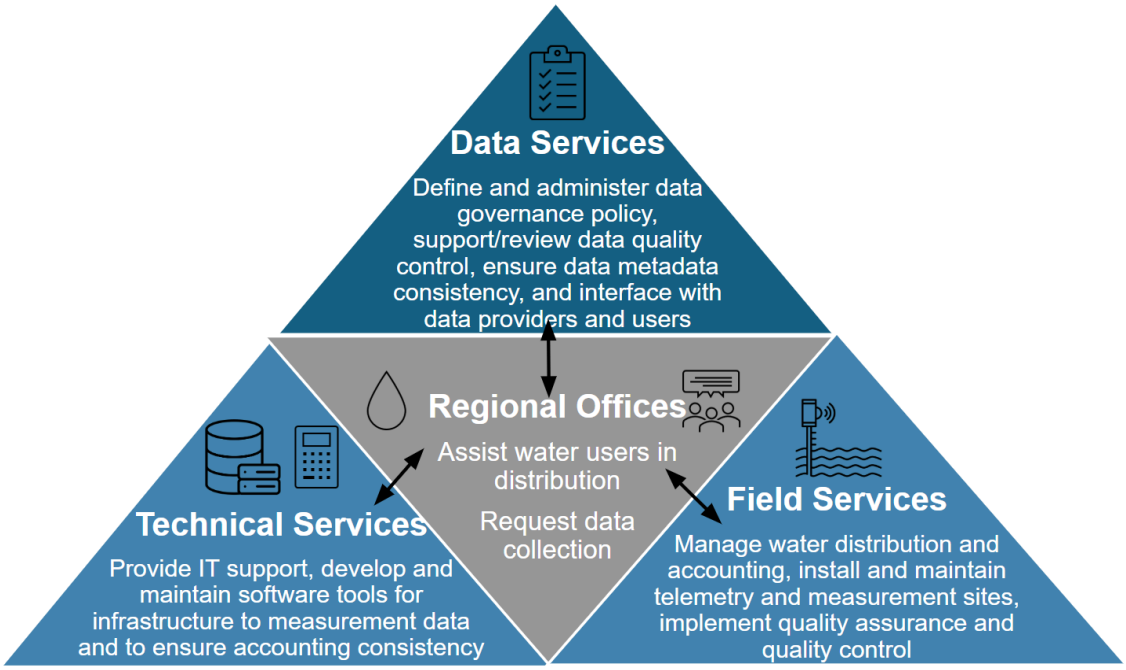


Figure 4: How often survey respondents perform each distribution task, from daily viewing tasks to less-frequent manual entry and verification.



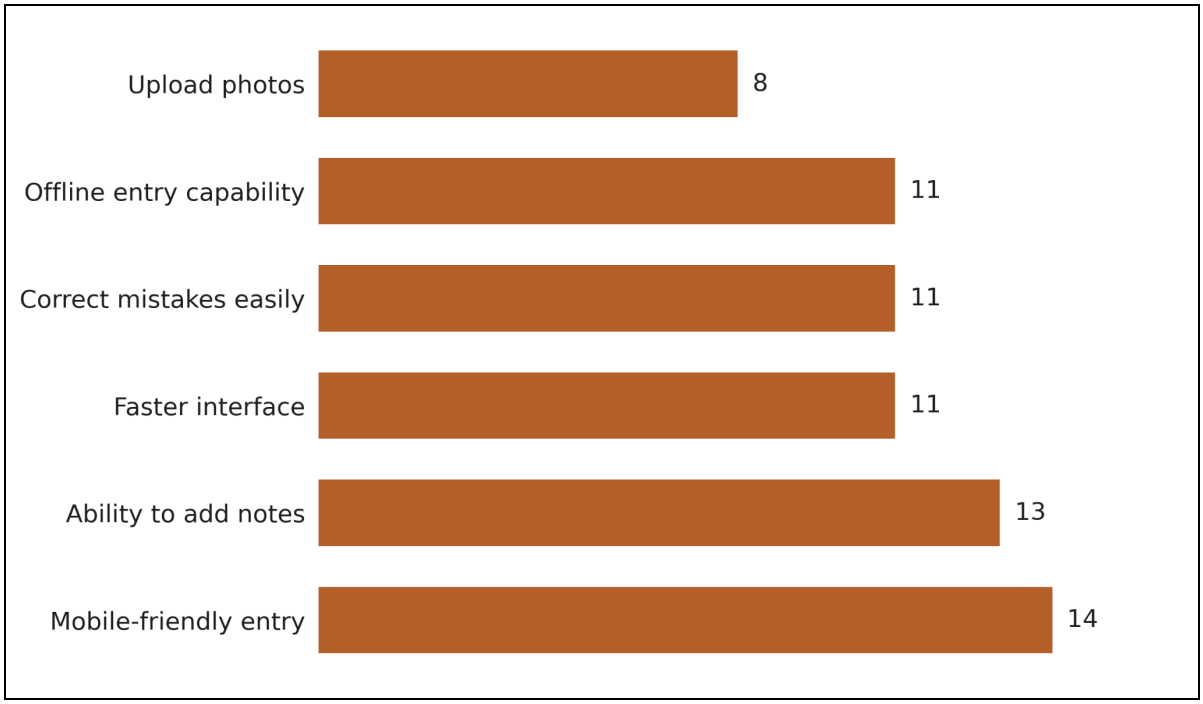
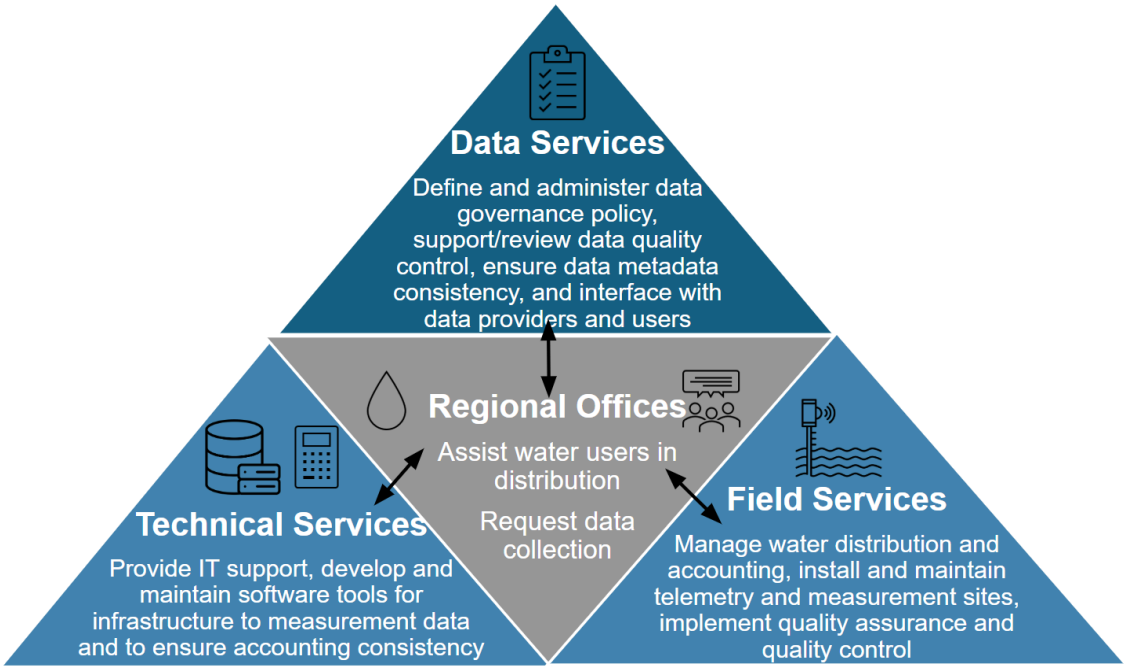


Figure 5: Top requested improvements to manual data entry (survey Q7).



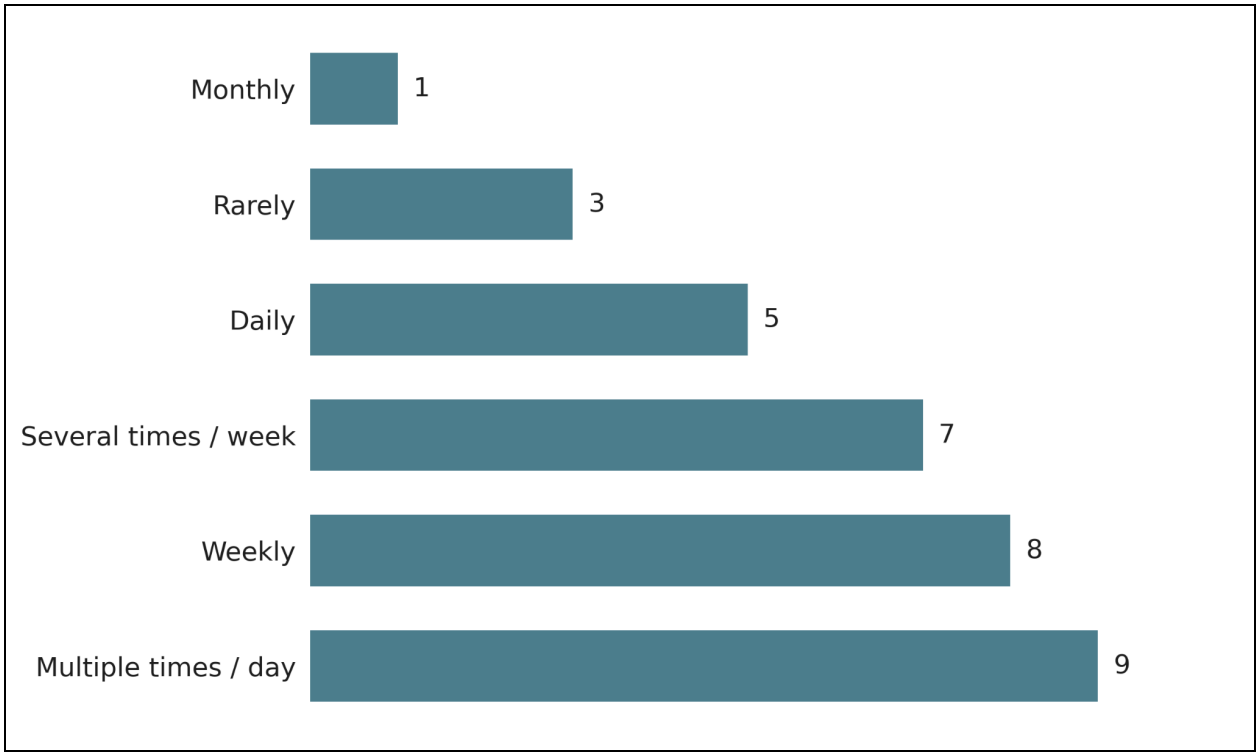
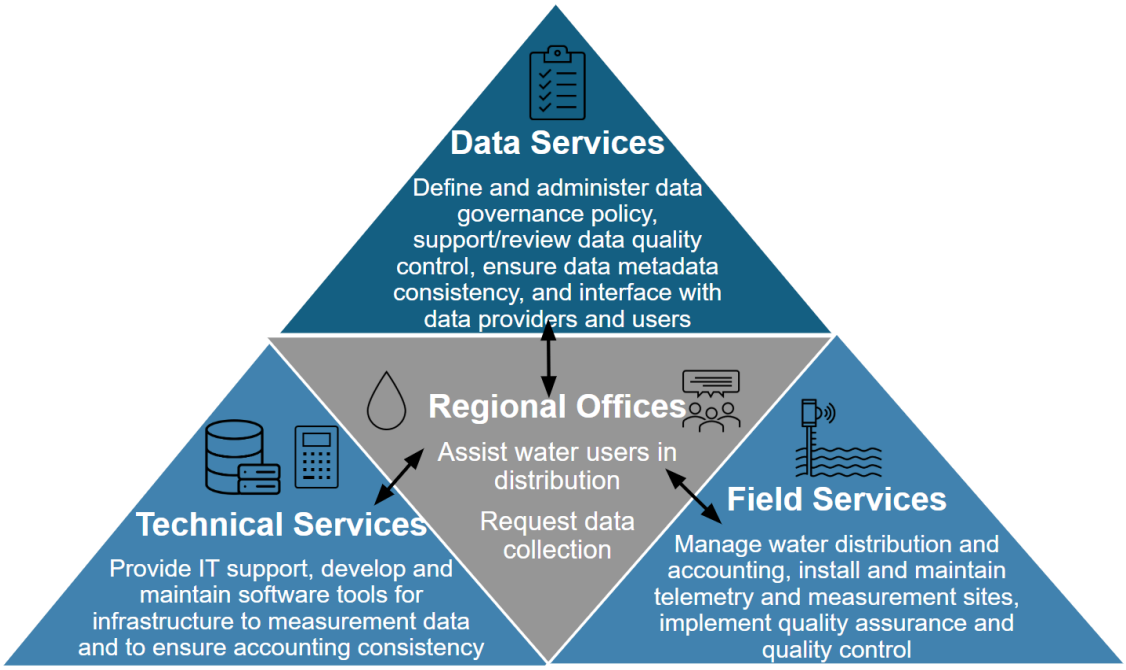


Figure 6: Frequency of accessing distribution data during irrigation season (survey Q4)

Where does the frustration concentrate? Here are specific fixes & desired improvements



01 — Entry is slow and click-heavy
"Have to click and enter each entry at a time." — Michael Nay, Distribution Engineer, Richfield/Cedar "A lot of clicks to get to the daily average table." — Cash Stallings, Regional Office Engineer, Southeastern

02 — No single map view of everything
"Unable to visit one screen (map) and see all flow values at the same time." — Tim Heyrend, Distribution Engineer, Vernal/Hobble Creek/Spanish Fork

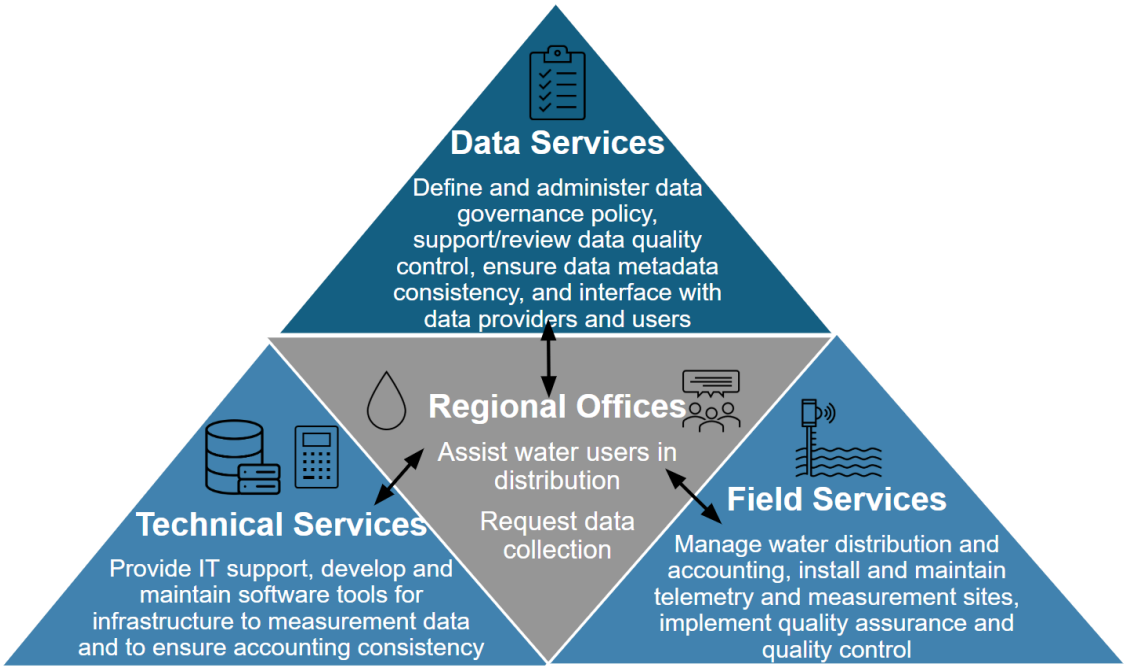
03 — Can't copy-paste from existing spreadsheets
"The inability to copy and paste from my Excel workbook." — Michael Lasswell, Commissioner, Logan River

04 — Login and passwords get in the way
"The water commissioners don't log in often, and when they do, it's a headache for them." — Caleb McCurry, Distribution Engineer, Little Cottonwood Creek

05 — Real-time and scraped data feel disconnected
"Only the data 'we' collect shows up on these pages — I have to go to all the different sites to see the real-time data." — David Horsley, Distribution Engineer, Huntington/Price/San Pitch/Sevier

Suggested System Fixes

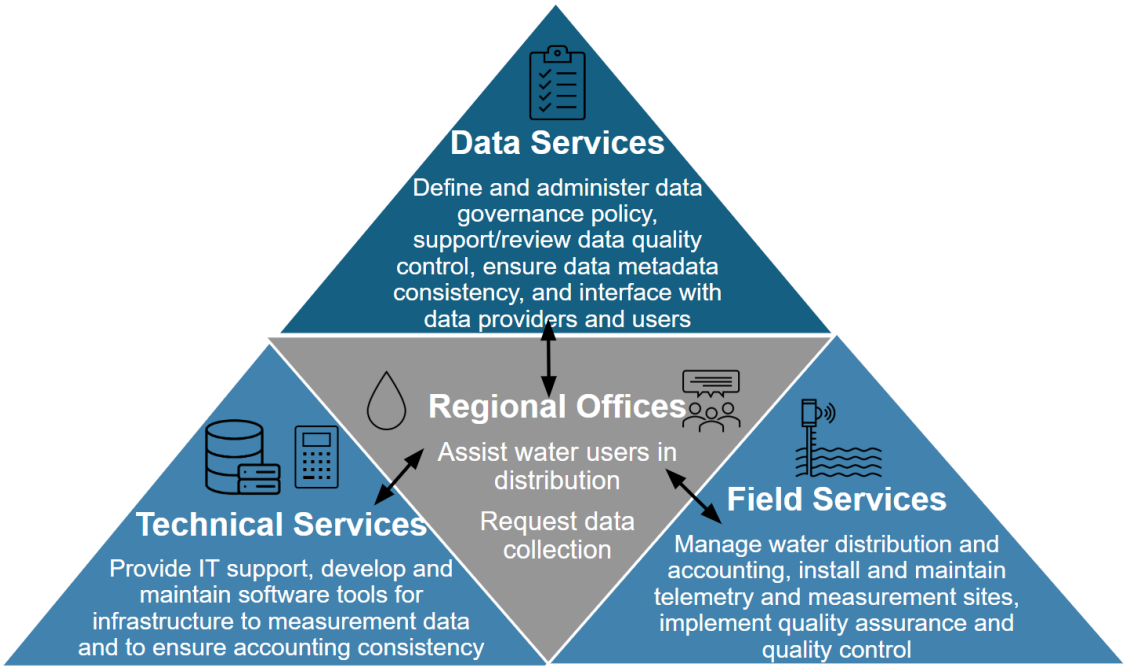
- Less complex log-in process (Kent S Wilkerson, Commissioner, Weber and Ogden)
- Change the units to the data available, to avoid manual calculation to daily averages or other units (David Jessop, Commissioner, Virgin River)
- Formatting should allow editing data charts to fit into the (physical) annual report (Jay Leavitt, Commissioner, Santa Clara River)
- The inability to copy & paste from an Excel workbook (Michael Lasswell, Commissioner, Logan River)



- Able to pull data without manual entry (Scott Bergendorf, Commissioner, Provo River and Hobble Creek)

Desired Improvements

- Recent history — shown the way USGS sites display it (Kent S Wilkerson, Commissioner, Weber and Ogden)
- Upstream and downstream flows — for a system, shown together (Brayden Rindlisbach, Commissioner, Upper Jordan)
- "I would like to see the monthly average as well as the daily and weekly averages." — Porter Monroe
- A schematic — showing river flow data
- Real-time reservoir elevations and stream flow rates (Clinton Hansen, Commissioner, Little Bear River)



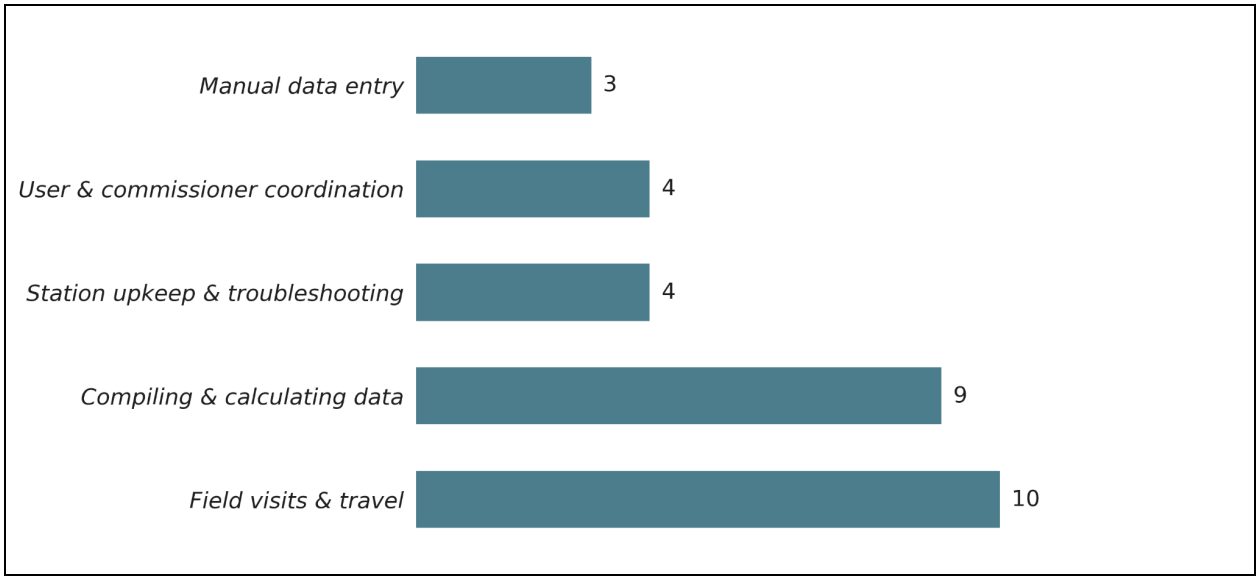


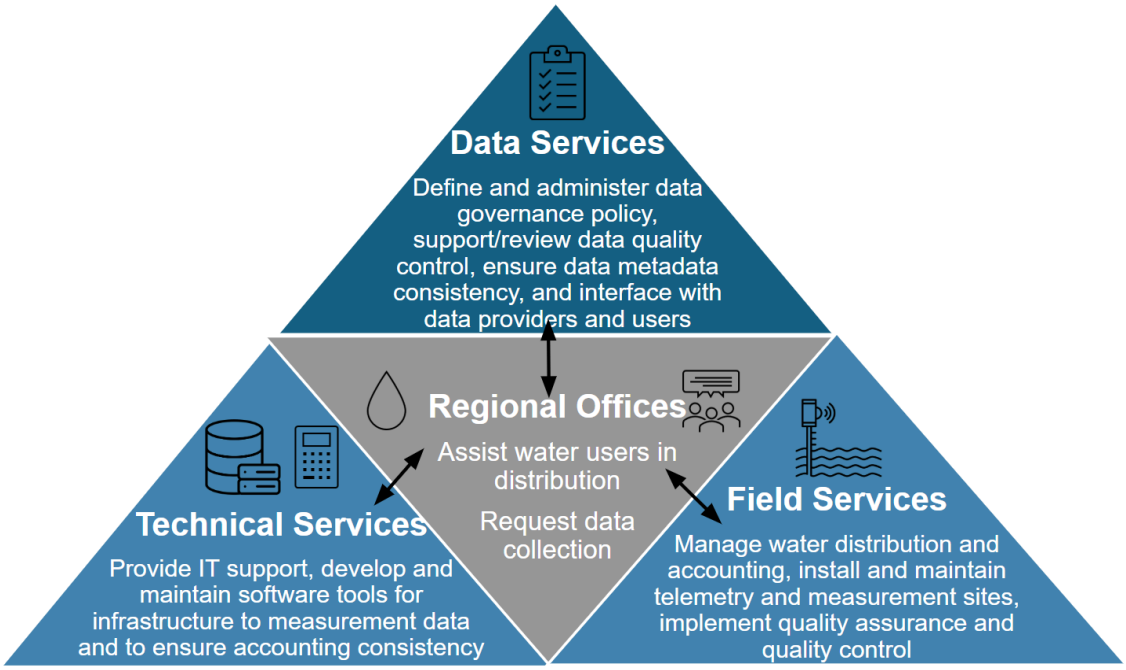
Figure 7: Where commissioners and engineers spend most of their time (survey Q9)

The Pattern Across The Survey

The biggest time costs are physical — driving to gauges, reading meters, fixing stations — not the software itself. But when respondents do sit down to work in the system, entry friction (clicks, no copy-paste, no mobile or offline mode) is what they name first when asked what to fix.

Major Time Drains

- Compiling and verifying data
- Fixing stations



- Waiting for data from irrigation companies to finish reports on time
- 3–4 surveys of the system per week, bi-weekly data compilation and reporting
- Manual data entry into DVRT
- Crunching the numbers of the daily data

Key Dashboard Tasks

Cody needs to use the dashboard to accomplish the following:

- **View and Access:** Access all assigned sites within their specific distribution system (not the entire State).

  Survey evidence: viewing tasks are the most frequent thing Cody does — 58% view flows at diversion sites daily, 54% view upstream/downstream conditions daily, and 50% check reservoir storage daily.

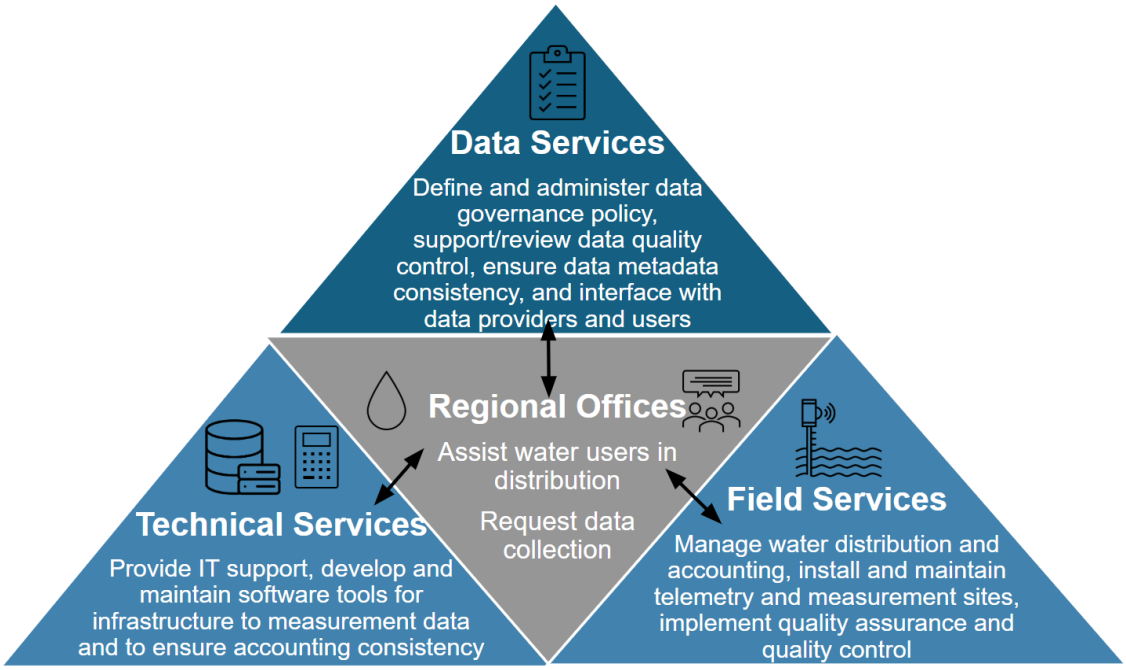
- **Visualize Data:** View plots of key metrics (discharge, water level), site location, photos, tea-cup diagrams, and year-over-year comparisons.

  Survey evidence: the top unprompted requests were recent history shown the way USGS sites display it, upstream and downstream flows shown together, a schematic of river flow, and real-time reservoir elevations and stream flow rates — all visualization asks.

- **Track Flow:** Identify the total water volume delivered to a user for a specified period (past day, week, month, or since the start of the irrigation season).

  Survey evidence: 43% of respondents track deliveries to users daily and another 36% weekly — this is a near-continuous task during the season, not an occasional lookup.

- **Enter Data:** Record readings from sites without automated equipment using a simple interface.



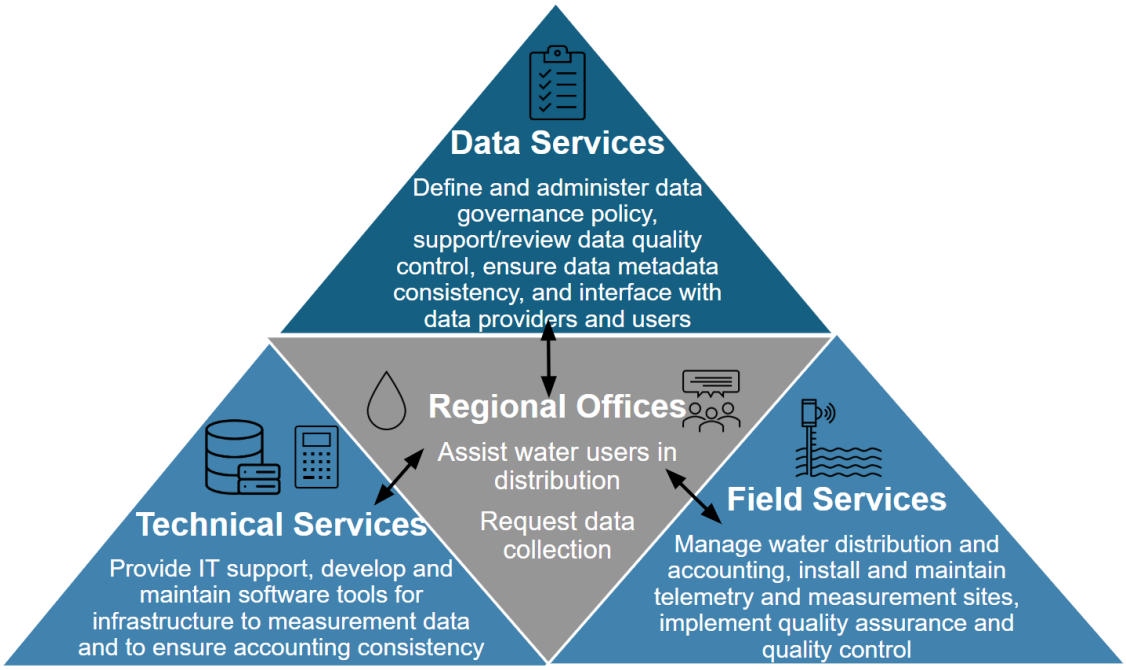
Survey evidence: this is the one task that runs opposite every viewing task — among those who manually enter data into WCAT/DVRT at all, most do it only monthly or annually, and 9 of 32 respondents never do it. It is also the task generating the most complaints (see below).

- **Edit/Verify Data:** Correct provisional time-series data for assigned sites, elevating the processing level (e.g., to “reviewed”). Daily, weekly, monthly, yearly.

  Survey evidence: verification clusters around weekly (41%) and monthly (34%), rather than daily — the interface should support both daily and a batch review workflow.

How data gaps are estimated or filled?

- ➢ Site visit, deputy data, interpolation of data, and others
- ➢ Manual verification
- ➢ Interpolation
- ➢ If I don't have accurate data to rely on, I estimate usage data with users' input
- ➢ Fill in using observations
- ➢ Through analysis
- ➢ Interpolation in the state recording system for commissioners
- ➢ Base estimates on past and current data
- ➢ Use previous day's data until a better estimate can be determined
- ➢ Checking water flow
- ➢ Visual recording
- ➢ As I go check stations, I take written notes of changes or problems to correct the data within the model
- ➢ Linear interpolation
- ➢ Excel
- ➢ Call users when changes are seen



- **Manage Site Metadata:** Change site details (name, location, elevation, min/max capacities, distribution system, photos). Cody wants to see the changes automatically logged with a time stamp and the user's name to maintain historical records.

  Survey evidence: respondents specifically asked to be able to upload photos

- **Report:** at the end of the year, generate an annual report summarizing daily flow and volume data for all the sites in his distribution system.

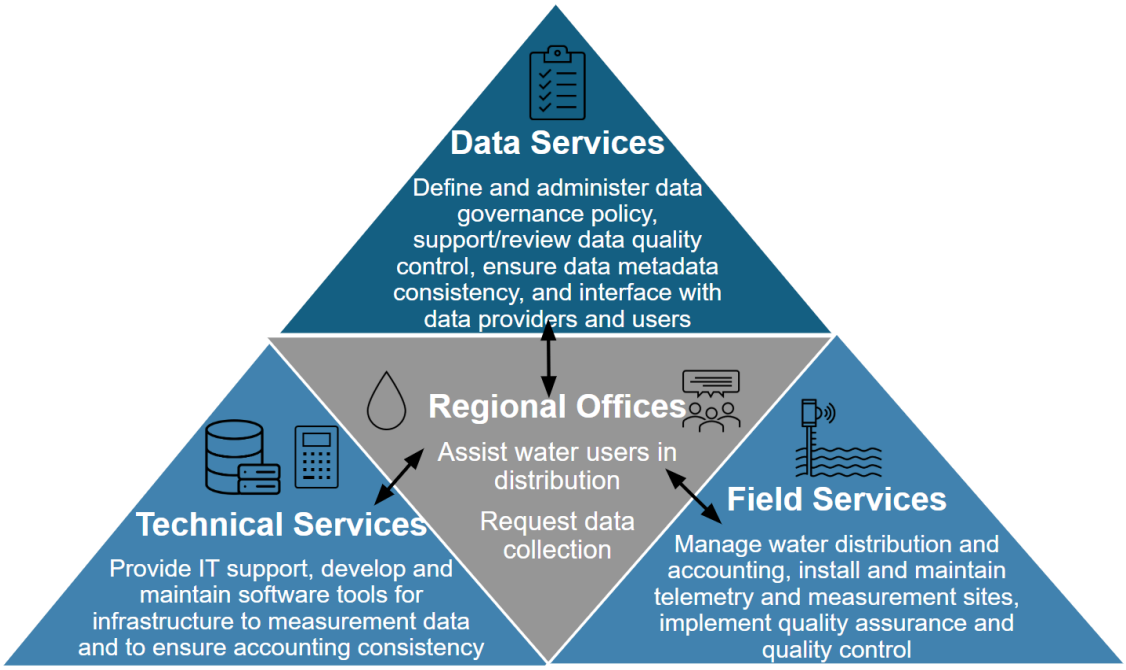
  Survey evidence: “compiling and calculating data” was the second-largest time drain named in the survey (9 mentions, behind only field travel), with respondents describing “3–4 surveys of the system per week, bi-weekly data compilation and reporting” and “crunching the numbers of the daily data.” One respondent also asked that chart formatting be editable enough to fit into the physical annual report — the export path matters as much as the data itself.

- **Set up Notifications:** Set up alerts for new sites added to their system and flags for existing sites (e.g., No Signal, Reading outside min/max flow rate).

Mobile Data Access and Entry (Nice to Have)

This capability is highly valuable as it solves the core issue of data latency and digitization friction inherent in the current workflow (waiting until year-end to input handwritten notes). Draft mockups are available here: Smartphone Accessible Mockup.

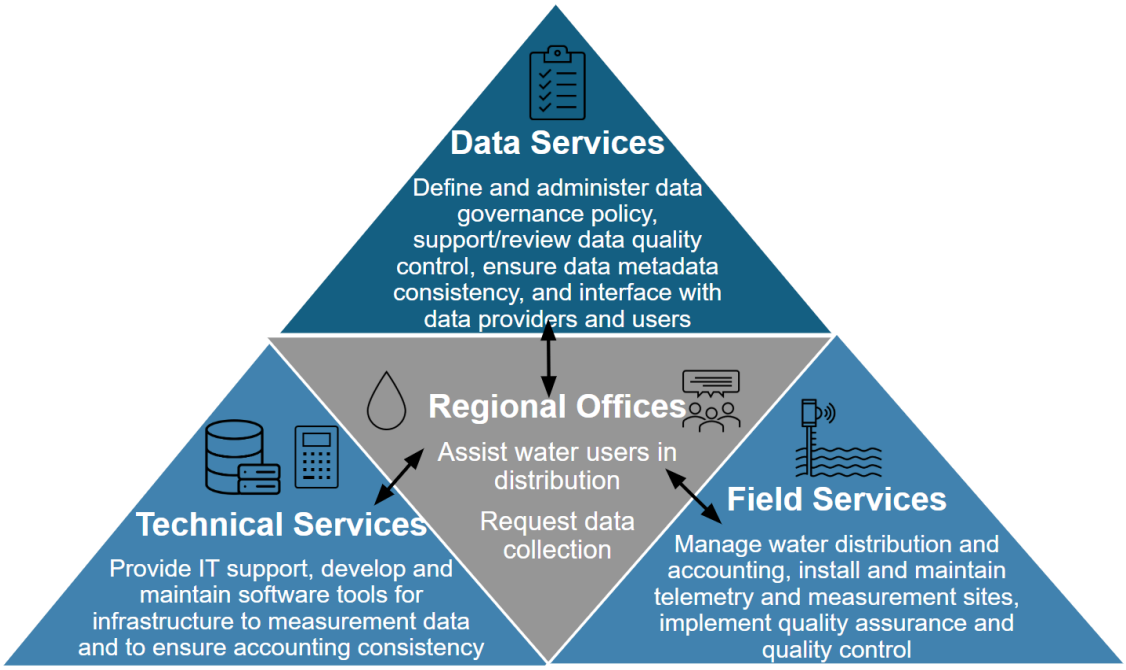
Why a Mobile App?



- **Timeliness:** Allows Cody to perform quicker reviews of provisional data and enter his field measurements immediately at the site, which drastically improves data freshness and reduces the risk of error from deferred entry. The app will mostly replace the notebook.
- **Mobility:** Since Cody is “in the field” and closest to the measurement sites, a smartphone or tablet application is the most logical tool for him to interact with the system.
- **Data Integrity:** Entering data while standing at the site ensures higher accuracy compared to entering readings weeks or months later from a notebook.
- **User Experience (UX):** A dedicated app is essential to cater to the user profile: Cody is not tech-savvy and needs a simple, intuitive interface that works well on a small screen.

> SURVEY INSIGHT · **The survey makes the strongest possible case for this feature.** Field visits and travel were, by a clear margin, the single biggest time drain respondents named — more mentions than any other task. And when asked what would improve manual entry, the top four requests were all field-first: mobile-friendly entry (14 of 33), the ability to add notes (13), the ability to correct mistakes easily (11), and offline entry capability (11), with photo upload close behind (8). Cody isn't asking for a nicer desktop screen — he's asking to never have to sit down at one.

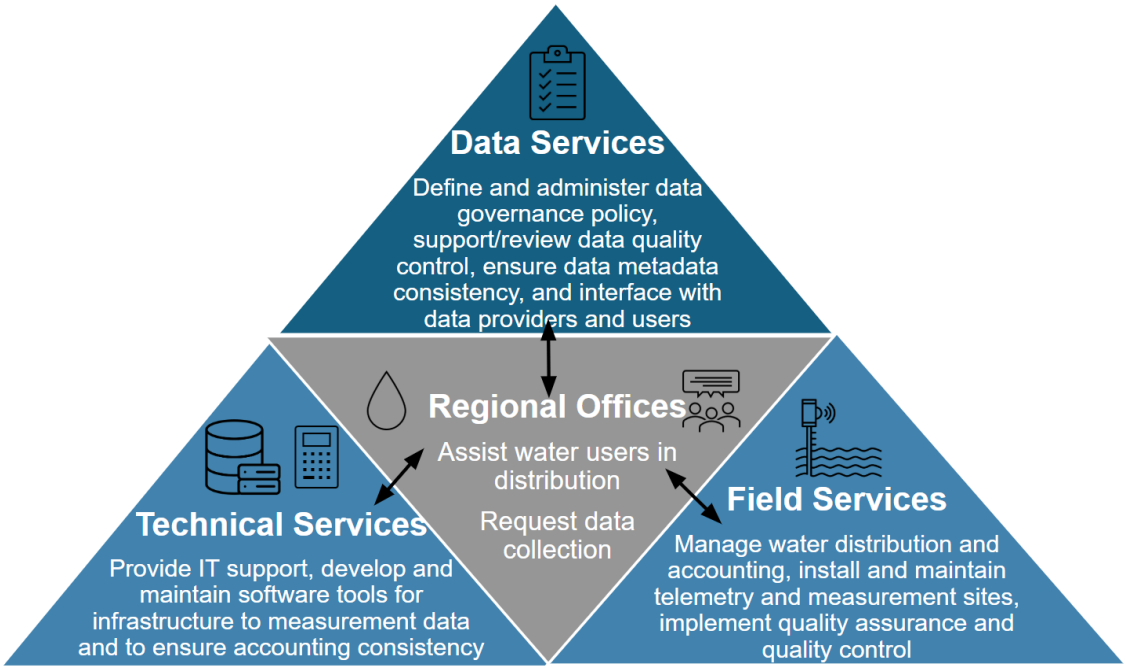
“Have to click and enter each entry at a time.”
“The inability to copy and paste from my Excel workbook.”
“The water commissioners don't log in often, and when they do, it's a headache for them.”
These three complaints — slow entry, no copy-paste from the spreadsheets Cody already keeps, and login friction for occasional users — describe exactly the desktop workflow the mobile app is meant to replace. A field-first design doesn't just add convenience; it removes the reasons Cody currently avoids entering data promptly.



Foundational User Stories (2023)

These needs, drawn from the 2023 foundational user-stories document, describe Commissioner and distribution-accounting requirements that inform Cody's requirements above:

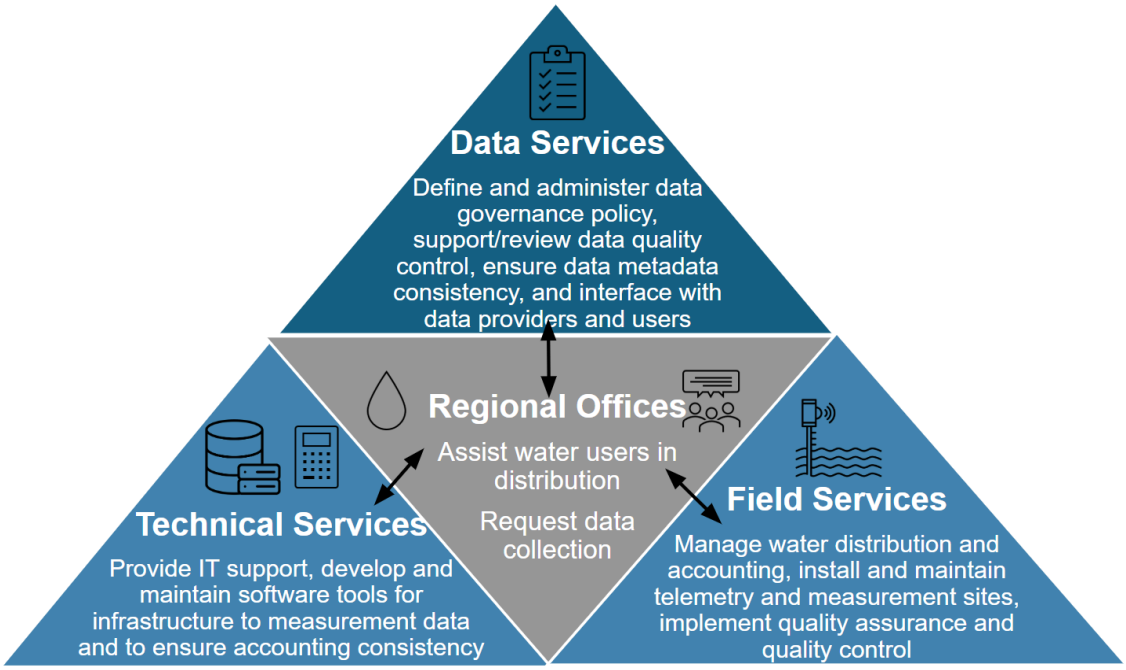
- **Leah:** Leah (former employee) wants a record of provisional data showing the information she had at the time of distribution, so she can defend her actions.
- **The public and storage-account commissioners:** want to see final (not provisional) data, so they have the best available information for their purposes.
- **Commissioners:** need to batch-print a report of calculated values (apportionments and estimates) to complete their annual report.
- **Groundwater distribution commissioners:** need to evaluate whether supplemental water sources are being used in accordance with existing rights.
- Users ought to be able to see — maybe with a graph — if someone is over-using water.
- Distribution accounting needs to support flexible accounting periods, such as an Oct–15 water year.



3. **Fatima, the Field Engineer, Automation, Regional Engineer, or Modeler**[12]

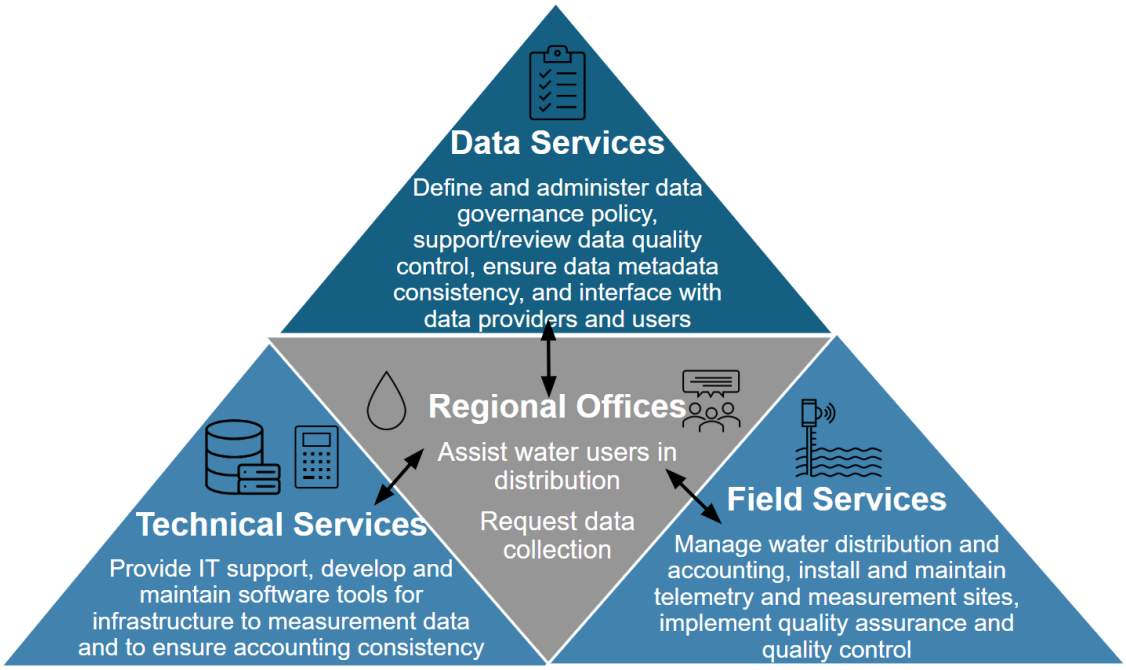
Across her various titles, Fatima consistently performs engineering tasks essential for managing Utah's water resources. A common thread in her work is the focus on water distribution, measurement, technical compliance, and support for commissioners. She is consistently involved in managing water distribution—applying technical expertise to ensure the system operates legally and equitably—and providing support to field personnel. This support includes training water commissioners and evaluating their reporting. Fatima is

[12] "Discuss Automation and Field Services New Data System Design Requirements" Focus Group meeting on Jul 29, 2025. (16 invited attendees: Caleb McCurry, Neil Hooker, Susan Odekirk, Ian Stockdill, Jacob Olsen, Everett Taylor, David Horsley, Tim Heyrend, Preston Benko, Cole Panter, Alexander Heilig, Adel Abdallah, Michael Nay, David J. Jones, Blake Bingham, and James Reese).



regularly involved in improving program effectiveness, whether by developing and improving water distribution program policy or assisting with water right enforcement activities as needed. Measurement oversight is a key function, though the physical fieldwork for this task is mainly a responsibility of the Automation Engineer.

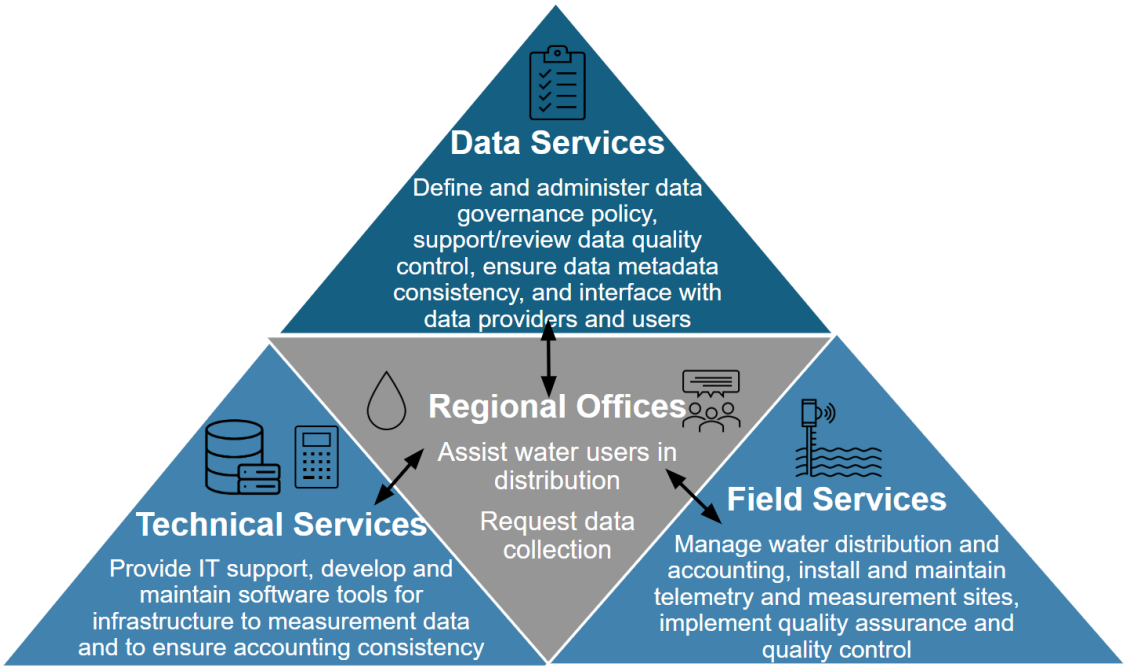
As a distribution/automation engineer, Fatima sits between the commissioners and water users on one side and the data services/QC team on the other. She wants the system to surface insights and analytics (e.g., which stations are telemetered, which sites are proposed/active/offline, trends in data quality issues at a given site) that ultimately feed into distribution accounting—the core purpose the system must



support—rather than being an end in themselves.

Table 2: Summary of Key Fatima, the Field Engineer Roles and Responsibilities related to Water Data Management

Role	Responsibilities
Distribution Engineer	Primarily focuses on investigating, documenting, evaluating, and overseeing water distribution, ensuring compliance, training Commissioners and evaluating their reporting.
Automation Engineer	Is responsible for installing, maintaining, and inspecting measurement devices on water diversions throughout the State. They follow QA protocols to prevent or reduce the chance of errors in measuring data before collection. They also focus on analyzing, summarizing, and reviewing the accuracy of measured data using engineering judgment to support equitable distribution.
Regional Engineer	Assists water users in distribution outside established Distribution Systems; assists in the management and supervision of established distribution systems within the region; operates river modeling programs, and assists with water right enforcement.
Water Rights Engineer - Modeling	Collaborates with staff to create Water Commissioner Accounting Tool (WCAT) models that account for water right distribution. They also assist in understanding these accounting models, and work with data/automation teams to determine new measurement locations to meet modeling needs.



SURVEY INSIGHT · Engineers See the Same Friction from the Field Side

Distribution Engineers and Regional Office Engineers made up 11 of the 33 respondents to the New Distribution Data Dashboard Survey. Their open-ended answers echo Fatima's QA/QC, audit-trail, and site-management priorities above: This lines up with the Division's own Hydroinformatics and Technology Gap Analysis, which found that DWRi lacks a traceable audit trail for data modifications and sufficient tools for data quality review.[13]

"There needs to be a "history" or "log" of changes to the data." — Jacob Olsen, Distribution Engineer, Logan/Weber/Ogden/Provo/Clear Creek

"Clumsiness of correcting data." — Ian Stockdill, Distribution Engineer, Upper Bear

"I want to be able to reload data to all the diversions (if there is new underlying data) by clicking one button." — Skyler Buck, Regional Office Engineer, Northern Region

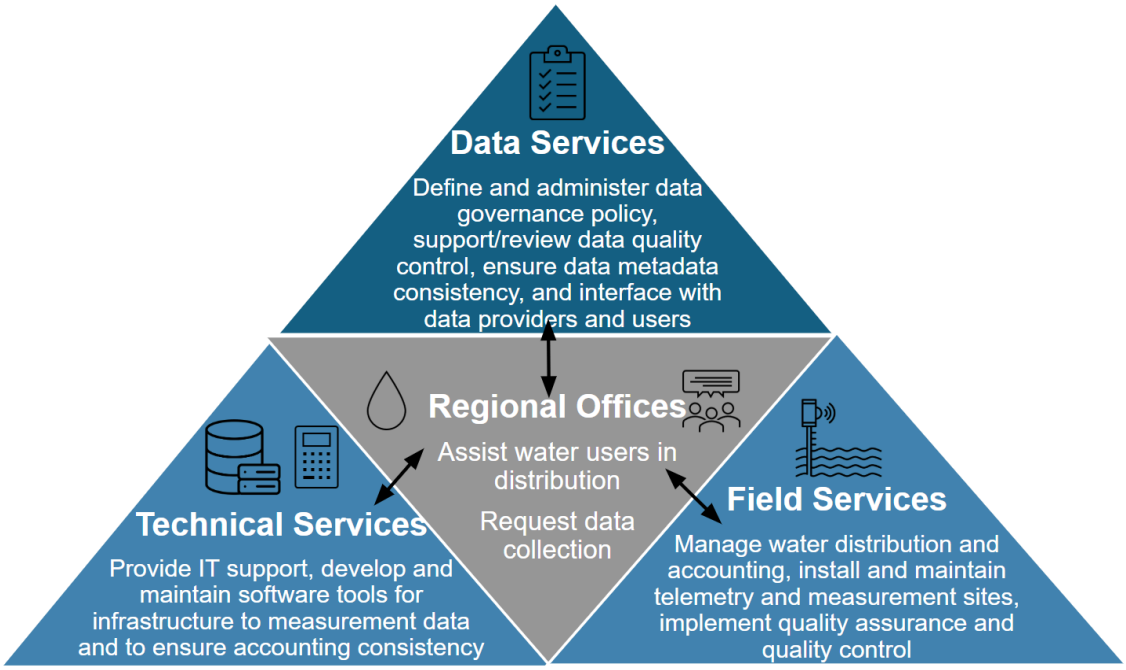
"It would help if flags would show up on the map if not in water right." — David Horsley, Distribution Engineer, Huntington/Price/San Pitch/Sevier

"Maybe add the ability to copy and paste data from a google spreadsheet." — Caleb McCurry, Distribution Engineer, Little Cottonwood Creek

Key Dashboard Tasks

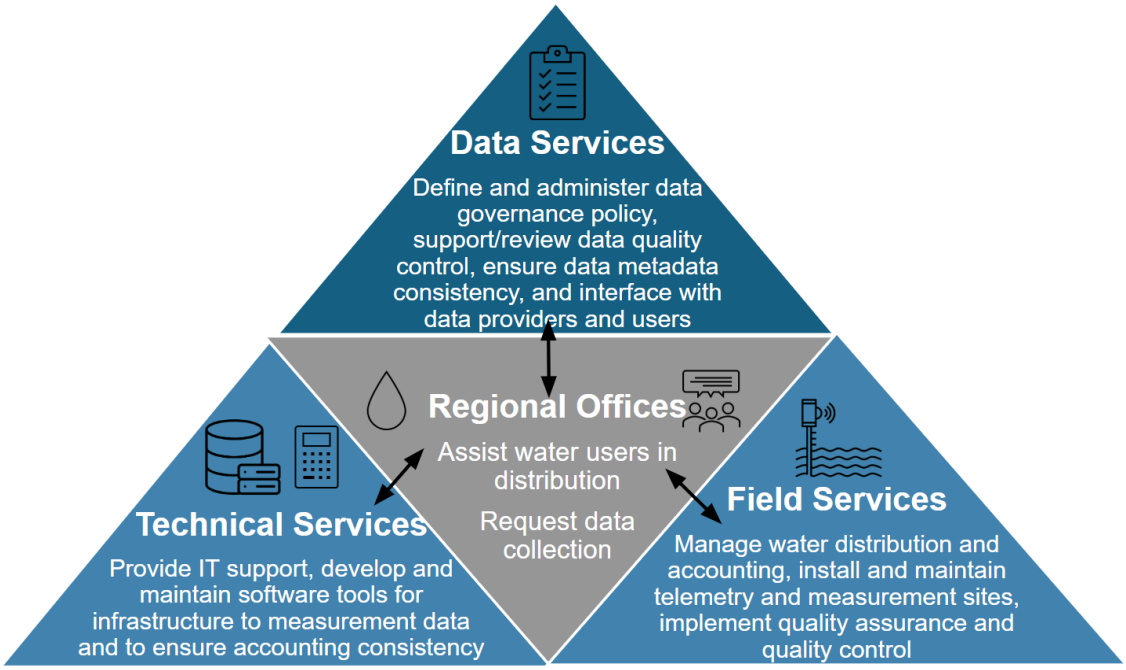
Fatima needs to use the dashboard to streamline her activities across the different roles to accomplish the following:

[13] Utah Division of Water Rights Hydroinformatics and Technology Gap Analysis. Utah Water Research Laboratory, Utah State University, 2024.



Task Category 1: New Site/Device Setup & Editing

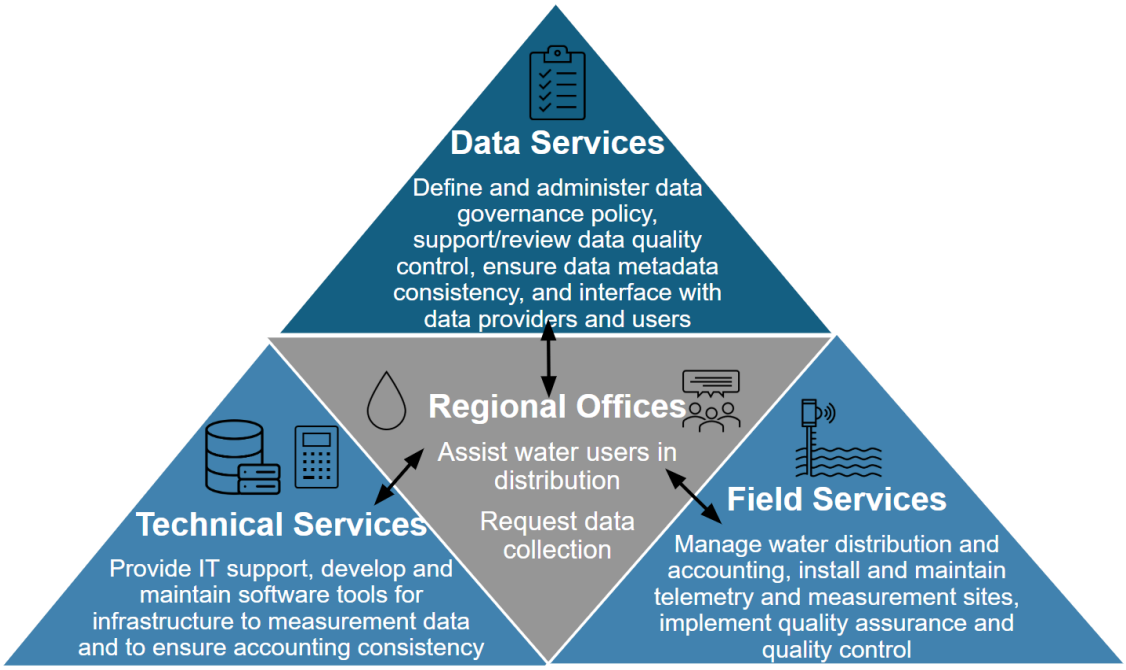
- Add a new site: once a new measurement station is installed, she needs to add a new site by entering site information (site name, site type, min/max capacities, site location, elevation, distribution system, add site photos).
- Edit Sites: She needs the ability to edit the same fields as in the Add Site function and deleting/adding photos.
- Add a site to groups or sub-groups.
- Set up and edit a Data Stream: To set up a data stream, the user will choose observed properties and sensors from a drop-down menu. She can edit time series data through the Quality Control tools. Enter measurement readings.
- Create another Data Stream: She must create a new data stream for a site when the measuring device changes to a different setup (e.g., changing the size of a Parshall flume or switching from a flume to a meter).
- Set threshold capacity: Set a sensor's minimum and maximum capacity for QA/QC purposes. Any data reported outside this allowable range must be flagged and color-coded to quickly identify sites needing repair.
- Orchestration Setup: Add the site to an existing orchestrator and input its parameters (e.g., scraping frequency). Should this be Diego's sole responsibility?
- **Bulk data entry:** Enter or paste large batches of historical or catch-up readings (e.g., a full month or year at once, not limited to ~15 rows at a time) using a data-entry tool



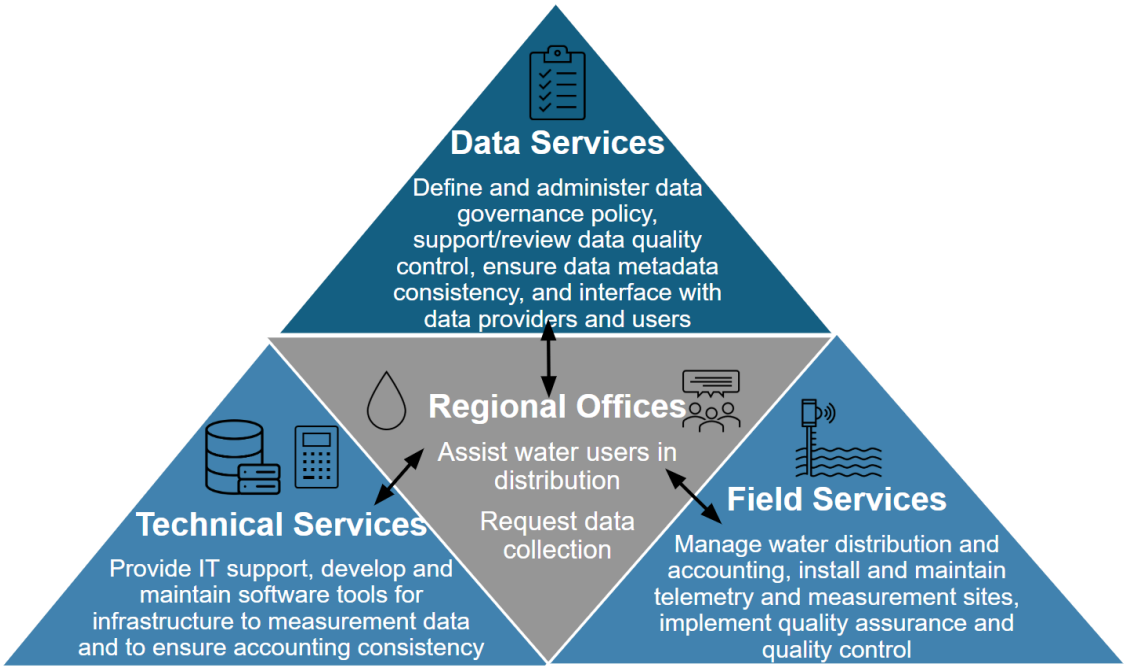
modeled on the existing WCAT workflow, so commissioners' and engineers' habits of entering data in bulk are preserved in the new system. (Requested by Ian Stockdill)

Task Category 2: Daily Operations – Data Integrity & QA/QC

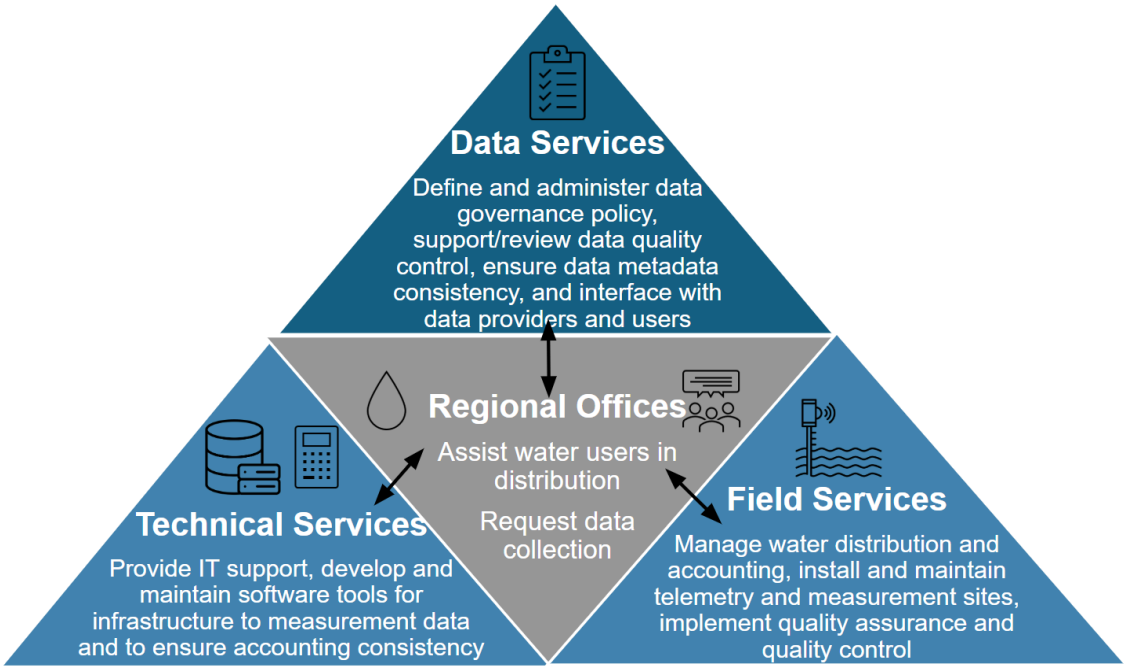
- Filter sites and visualize data (see the last real-time data per site, view plots of discharge, water level, battery voltage, for any favorite site; view photos and location; tea-cup diagram, year over year comparison).
- Additional filter under "Site Data" to sort distribution stations by the installation date range.
- Advanced Data Visualization: she needs the ability to view plots of discharge, water level, **battery voltage** (important enough to expose to engineers even where it may be hidden from commissioners), view photos and location, and use tools like the tea-cup diagram and year-over-year comparison. (Requested by Caleb McCurry and David Horsley, Focus Group meeting)
- **Water type/source context:** See what kind of water a given measurement or site represents (e.g., storage water vs. natural flow, a combined flow, or a site serving multiple water users/rights/shares) so the number is meaningful on its own, with a path toward linking sites to the water right network as that integration matures. (Requested by Susan Odekirk)



- Document Known Data Issues: Record known issues with site data (historical or ongoing) to provide context for potentially bad data.
- **Layered data & version history:** View and work with distinct data layers — an un-editable raw layer (automation engineers may delete, but not edit, erroneous raw readings, e.g., when a sensor was known to be out of water) and a continuously editable "best available data" layer that remains open indefinitely (no year-end lock) so late-discovered issues can still be corrected. (Discussed by Jacob Olsen, David Horsley, Susan Odekirk, and Everett Taylor)
- **Change tracking with qualifiers/flags:** Tag edits with a defined set of qualifier keywords (e.g., "interpolate," "ice-affected") that can be combined, plus an optional free-text justification field, so it's clear why a value was changed. (Requested by Jacob Olsen)
- **Point-level audit trail:** Click any data point (or a highlighted/flagged stretch of the time series) to see its full edit history — who changed it, when, and why — including cases where a later edit overwrote an earlier one. (Discussed by Adel Abdallah and Jacob Olsen, Focus Group meeting)
- **Trend view of recurring issues:** See, both graphically (e.g., color-coded change markers on a time series) and in table form, which sites are chronically flagged or repeatedly corrected, to help prioritize attention. (Requested by Jacob Olsen)

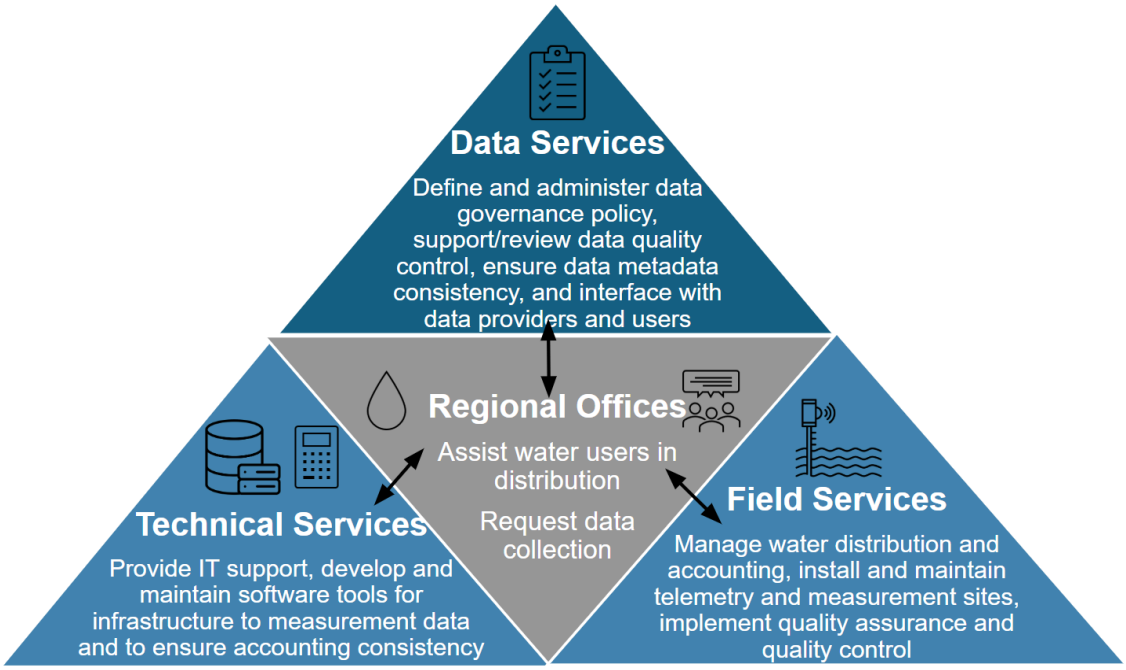


- **Point-in-time reconstruction (future/aspirational):** Ability to see what the "best available" value was as of a specific past date — useful for reconstructing the data as it stood when a distribution decision was actually made — flagged by the team as valuable but more technically complex to implement. (Requested by David Horsley and Jacob Olsen)
- Log Data Shifts: Log when shifts are applied to a site (e.g., to correct a rating curve) separate from general site comments.
- **Rating curves/regressions:** Track the history of rating-curve regressions and shifts used to convert a raw measurement (e.g., head) into a reported value (e.g., flow), since these are currently maintained informally (e.g., in personal spreadsheets or overwritten datalogger programs) and prior regressions are otherwise lost — noted as a gap the core system (HydroServer) does not currently address and needs further scoping. (Raised by Susan Odekirk and Ian Stockdill)
- Audit Trail: She wants to see the history of what changes were made to the raw or preliminary data and who made those changes.
- **Reporting:** Generate an annual report or a multi-station report (PDF) for a selected site and year. (Demonstrated by Alexander Heilig)



Task Category 3: Field Operations (Quality Assurance)

- Site visit documentation: document her site visit and report maintenance actions (e.g., sensor fixed). The history of all site visits and maintenance must be kept as part of the site's metadata which informs the quality control actions and accuracy of the data.
- **Voice-to-text visit notes:** Record an audio note describing a site visit in the field (rather than writing everything down in real time) with the option for AI transcription/summarization into a concise record — intended to let her stay focused on the visit (e.g., listening to a landowner) instead of transcribing on the fly. (Proposed by Adel Abdallah, supported by Susan Odekirk)
- **Offline capture:** Field data, photos, audio, and visit reports should be storable on the device while offline and sync automatically once connectivity is restored. (Discussed by Alexander Heilig)
- Asset management: record used device make/model and site photos to support the annual inventory requirement.
- **Multiple, categorized site photos:** Capture more than one photo per site (equipment, flume, box, general site view, etc.), with a clear distinction between a photo of the site

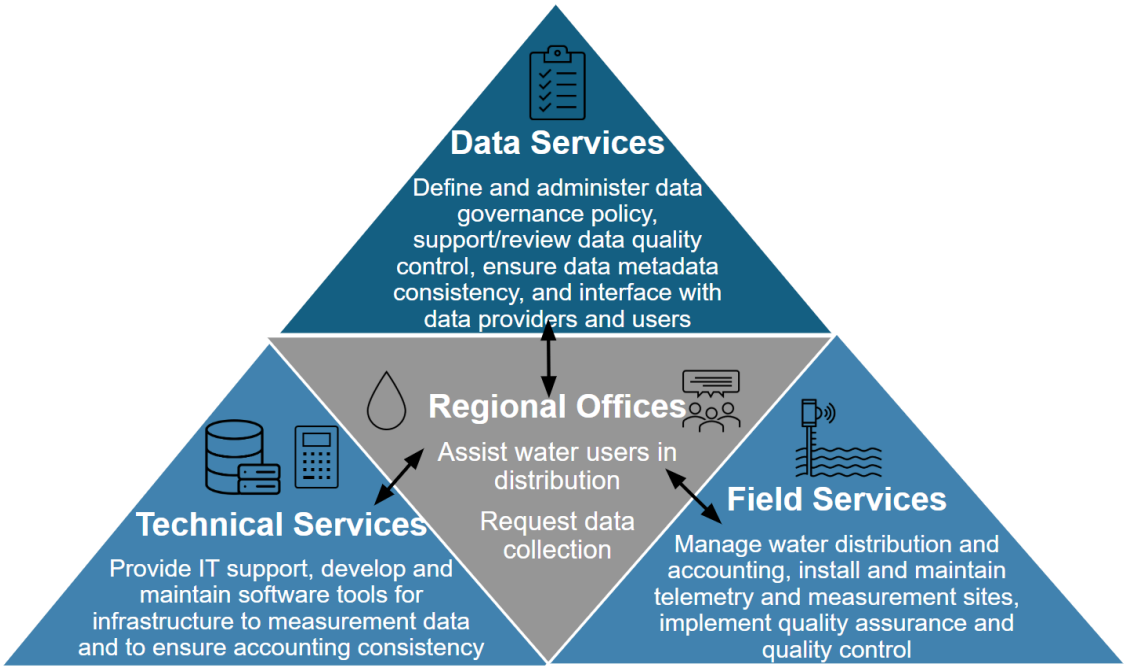


itself versus a photo/scan of a specific asset (e.g., datalogger barcode). (Requested by David Horsley and Caleb McCurry)

- **Public vs. internal photos/notes:** Separate photos and information intended for public-facing use (e.g., a general site photo) from internal-only content (e.g., wiring details, gate codes, key numbers) so internal material isn't inadvertently exposed. (Raised by Ian Stockdill and Susan Odekirk)
- Inspection history: She wants to know when the device for a site was last inspected or created and by whom.
- Prioritize Maintenance: Filter sites based on an attribute indicating whether they collect data year-round or only during the irrigation season. This helps the automation engineer prioritize maintenance schedules.
- **Visit scheduling (possible future feature):** See a schedule/cadence for revisiting each station if the division adopts a formal site-visit rotation. (Shown by Alexander Heilig)

Task Category 4: Notification (Alerts) Management

- Set up and receive notifications (alerts) when a datastream on one of her sites has a problem (ranging from no data transmission to out-of-range signals).



- System-wide updates: View notifications if assigned sites were edited and if a site was added/flagged within her assigned system (with the ability to filter these three types of notifications).

Mockups of Personas' Journey for the Next-Gen Utah Water Use and Distribution Data System: FEEDBACK

Alexander Heilig and Adel Abdallah — Data Services Section, Utah Division of Water Rights — August 2025 — DRAFT and work in progress

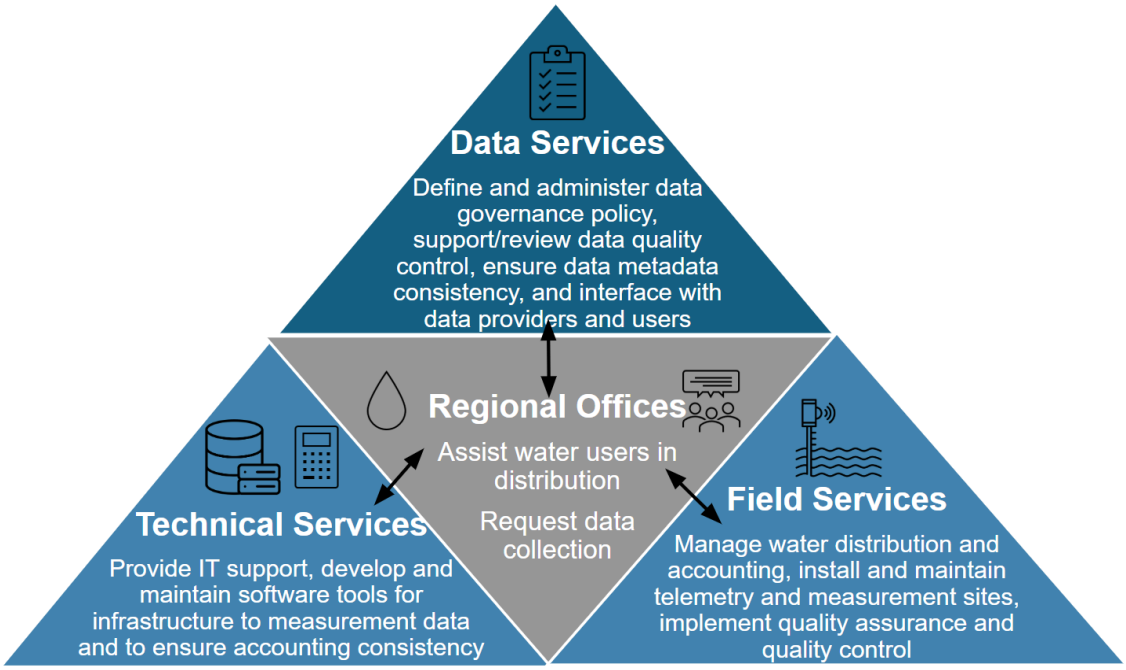
7/29/2025 — Jacob Olsen: Dory the Distribution Engineer

> Under "Manage Sites" it would be nice to have the ability to sort the sites alphabetically or by other metadata.

8/4/2025 — Cole Panter: Field Services - Smartphone

Suggest having a checkbox to "not show on log in" again but having a tour to start I think is useful.

8/4/2025 — Cole Panter: Field Services - Smartphone



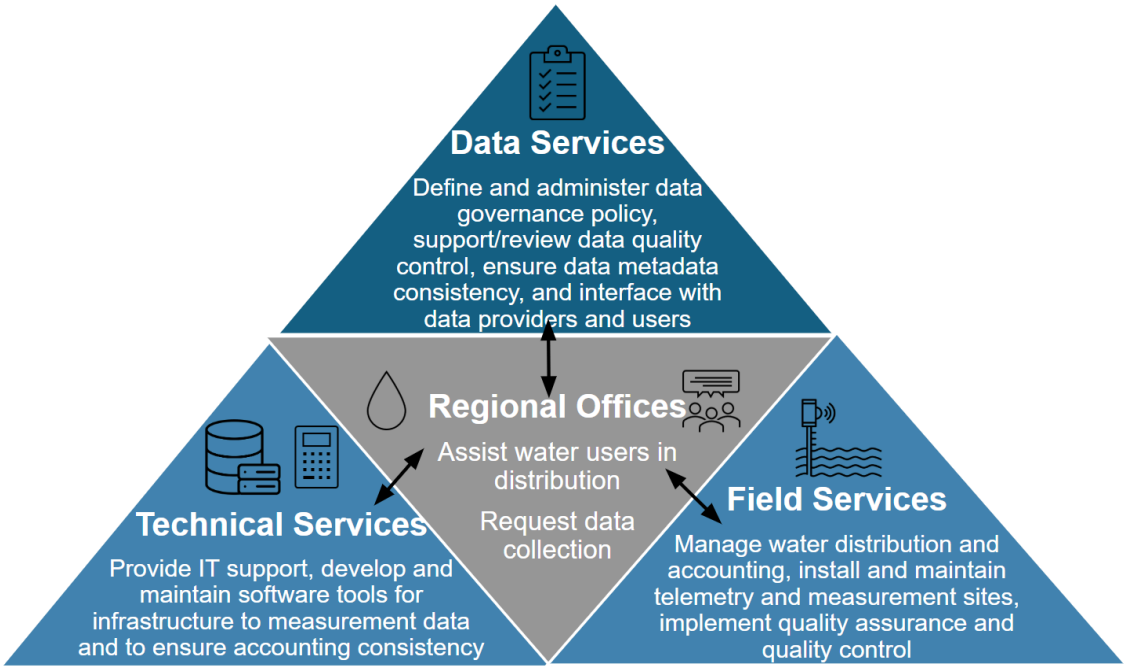
The Phone App looks very beneficial. I think the overall structure looks great. My suggestions / wish list would be:

- Site home page shows the flow and front facing photo of the site with general comments (options to edit). A list of options would then include: Assets, Water Measurement, Addresses, Past Site Visits, Report Site Visit.
- Assets Page: Scan asset and remove asset option. If this could be linked to a Google Drive doc on the G:Drive that would be amazing.
- Water Measurement: Report measurement and access historical measurements.
- Addresses: Inputs for Cell IP Address and Port, Radio Addresses, Pakbus Number, Pakbus security, etc. Linked to G:Drive as well.
- Report Site Visits: Visit Type, Pictures, Comments. Pictures uploaded to Site photo file on G:Drive.

8/4/2025 — Cole Panter: Dory the Distribution Engineer

The browser site looks great. I really like the concept. Some initial comments:

- Are notifications standard for each site or can they be customized by user/flagged within the station?
- Water right changes, ROCS would be useful notifications or tab selection for the distribution engineer.



- Suggest having the owner of the measurement be stated somewhere on the site page if not owned / maintained by Water Rights (USGS, Salt Lake County, PRWUA). This would be more for the Water User portal. Additional statements if needed that data reported may be different from the owner reported.

Reviewer Comments on the "Automation and Field Services New Data System Design Requirements" (June 2025)

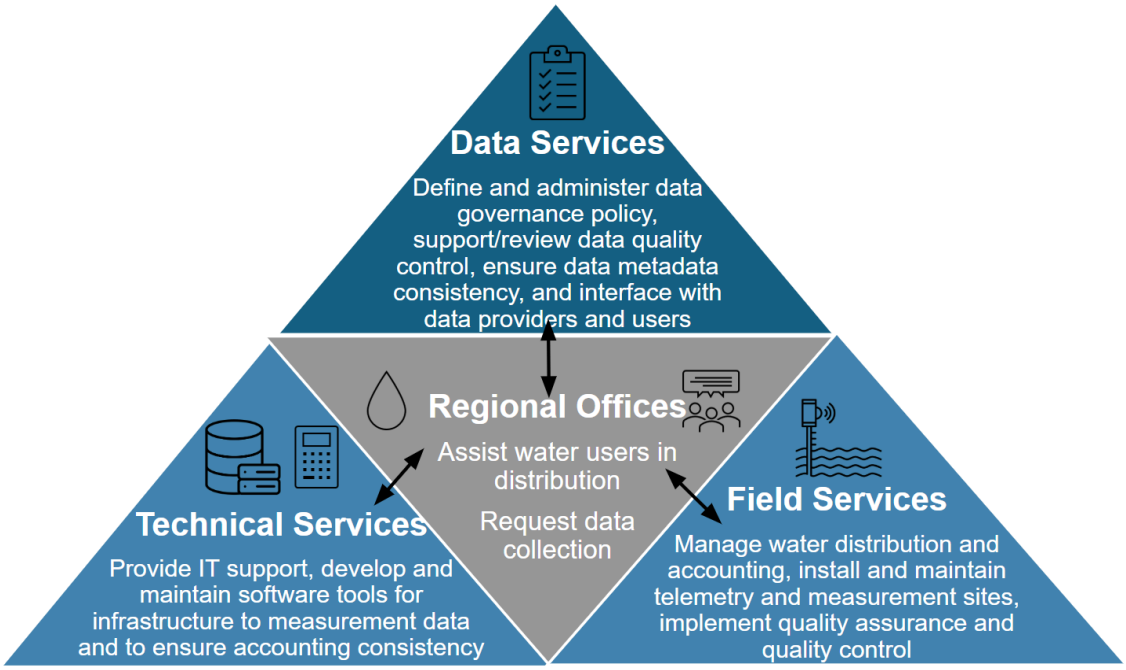
The Automation and Field Services engineers' own design-requirements document (Adel Abdallah and Alex Heilig, Data Services, June 2025) carried in-line reviewer comments and one emailed round of feedback. These are captured here, attributed by name, since they refine several of Fatima's requirements above:

On Site Management & Status — Jacob Olsen:

> What about a category that indicates whether the station is being used in a WCAT model?

On Water Quantity & Storage (reporting total annual diversion) — Tim Heyrend:

> Report the Annual Imported Water.



On Site and Measurement Data Entry (adding manual measurements) — Ian Stockdill:

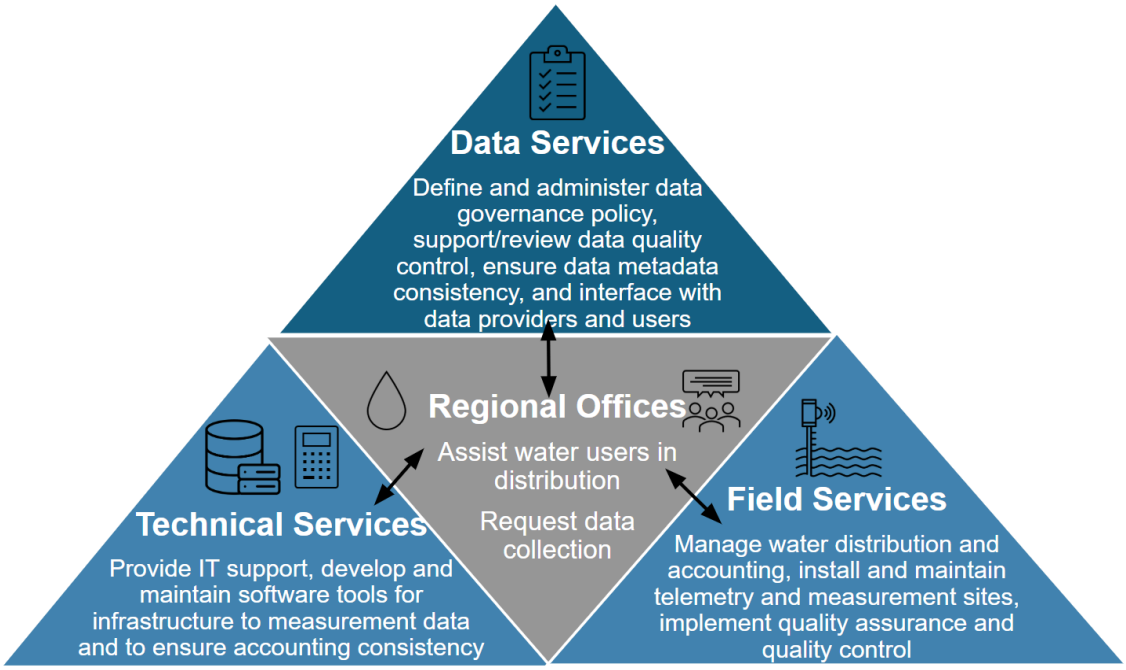
> I don't know where it fits in here, but a way to set custom headers for certain non-standard sites without relying on Jim and David, and set the order for sites where the data comes through without variable headers (current satellite stations). If this doesn't fit into what we can do on the HydroServer end that's okay — just trying to take work off Jim and David from creating custom scripts for the rare sites that need different variable headers or no variable headers.

On Notifications and Data Correction Logging (recording known data issues) — Susan Odekirk:

> For sites without dataloggers, telemetry is integral to data collection. For these sites, the peripheral card address and a link to recent gage server communications might be helpful.

Emailed feedback on the full document — Caleb McCurry:

1. Irrigation seasons are often site- or system-specific; it would be nice to include exact irrigation start and stop dates.
2. For asset verification, remember that assets can be moved from one site to another, which may affect how site visits for asset verification are planned. Tags can also be



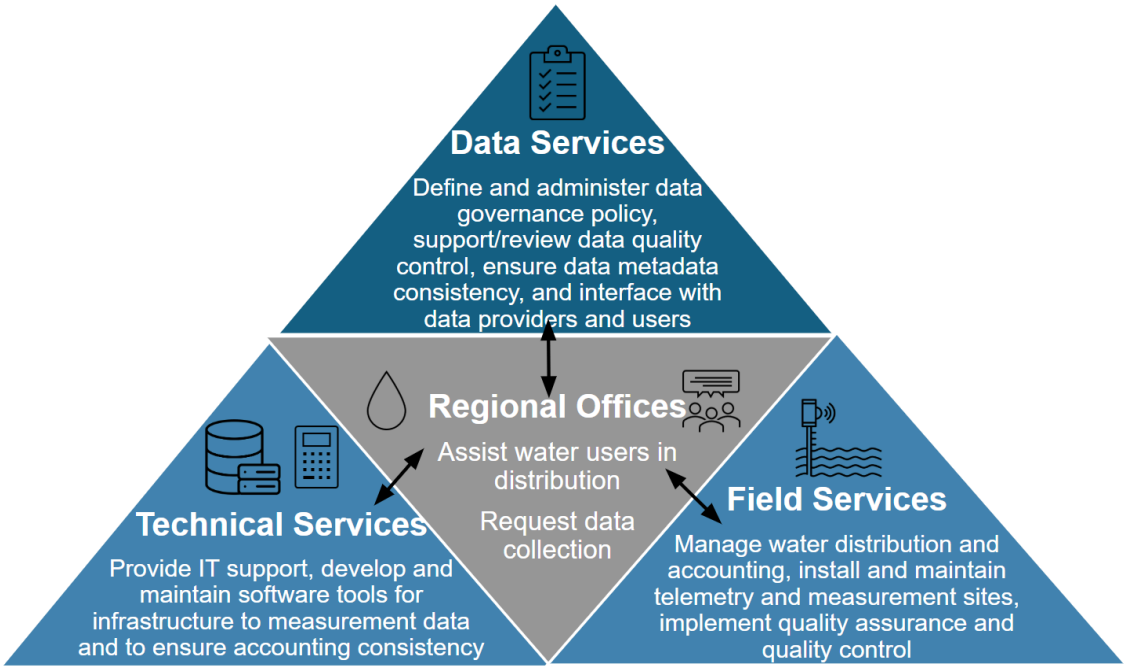
damaged or lost in rough environmental conditions, so a backup to the tag — such as a photo of the serial number — would help.

3. Shifts and other changes might be good to update only once a day or once a week; when repairing a site, the shift may be adjusted several times within a day to dial it in.
4. If possible, adding access points to a satellite map could be helpful, or having the option of multiple points for one site to show important features such as entry point, sensor location, or diversion location — though sometimes it is better to keep it simple.

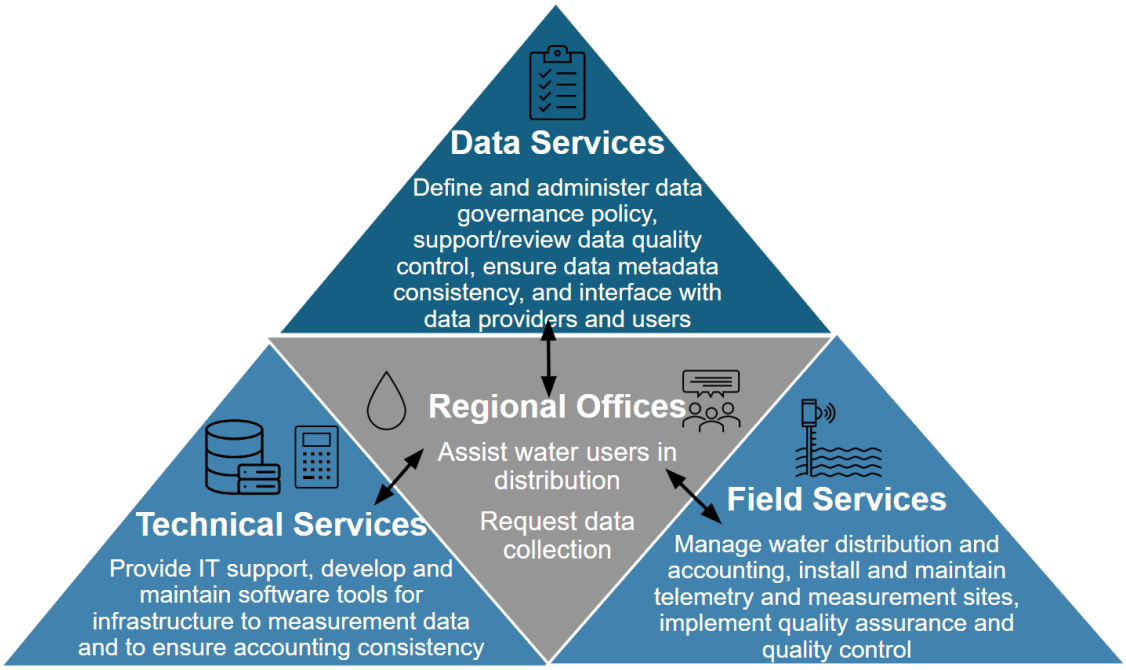
Foundational User Stories (2023): WCAT and Water Right Network

These needs, drawn from the 2023 foundational user-stories document (including the WCAT story list), describe field engineering, modeling, and water-right-network requirements that inform Fatima's requirements above:

- **Teresa:** wants a complete Water Commissioner Accounting Tool (WCAT) model of the Sevier River to help find where measurement is needed, where incorrect assumptions have been made, and to correct the imbalance between paper and real water.
- **Ed Vidmar:** wants a WCAT model of Spanish Fork that replicates the existing Access-based model, so he won't need to seek out a contractor to build one, and so the division doesn't lose the data currently held in Access.



- **The Logan Regional Office:** wants a completed and correct Lower Bear River WCAT model so the Bear River bird refuge has a more realistic picture of what it is using, given wide public interest in Great Salt Lake water usage.
- **Cole Panter:** wants a complete WCAT model of Little Cottonwood so metered and manual measurements can be compared.
- **Cole Panter:** wants a complete WCAT model of the Jordan River, noting that the Audubon Society has become interested in this model's completion.
- **Leah Richardson:** wants active diversion stations for all data on the Olmsted report to be either automated or easy to input, so the commissioner doesn't need to do a massive manual update a few times a year, and so water use can no longer be obscured.
- **Ted and regional office staff:** want a way to match water right records to actual use so division records are more accurate — though the change-application process makes this hard to keep current.
- **Users of water right records:** need a surface-diversion equivalent of a WIN, since duplicate Point of Diversion (POD) records — sometimes listing different locations — make it hard to tell whether records refer to the same surface diversion or different ones.
- Field staff need to know what flow measurement goes with what field.
- **Station data managers:** need the last time a station was checked or rated to be retrievable as structured, queryable data — not only as a text description.

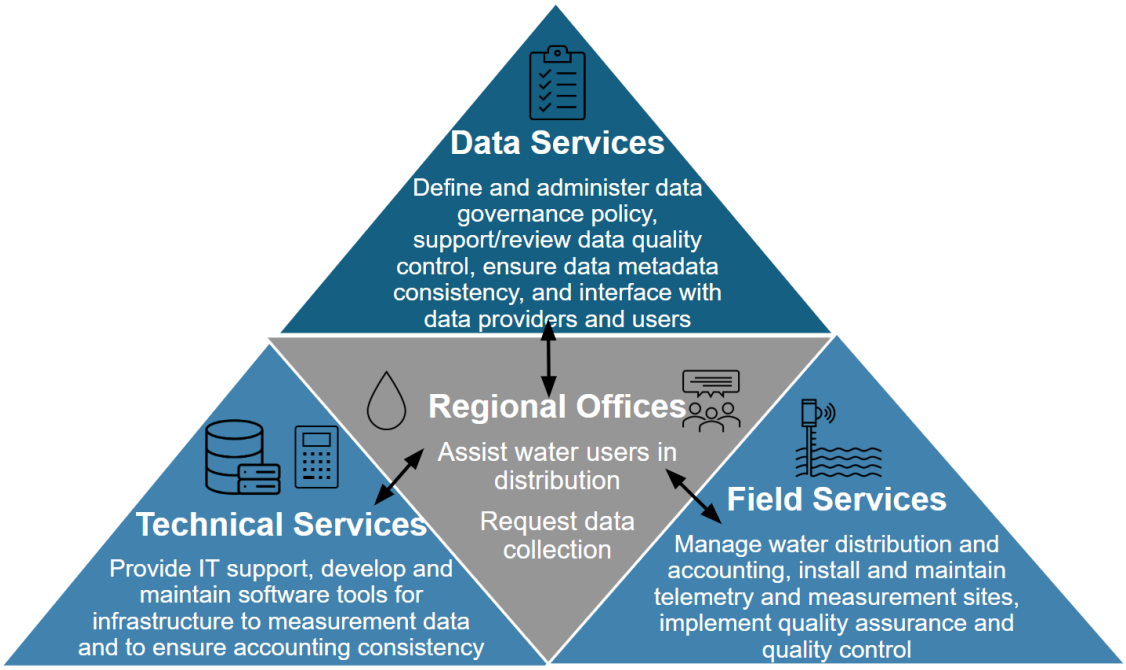


4. Wiley the Water User

Wiley is a water rights owner and user — typically a farmer, rancher, conservation district, canal company, city, or irrigation company, often operating within a distribution system. In principle, a water user can also be a public water supplier, as in the Water Use Program.[14]

To manage water use effectively and comply with their allocation, Wiley relies on timely, accurate water data — such as real-time diversion records, priority regulation decisions, reservoir levels, and measurement reports — to understand how much water is available, verify that distribution is equitable, and plan irrigation or operational needs. Wiley must follow all regulation orders issued by the Commissioner and provide any required water-use reports to the State Engineer, making transparent access to clear, consistent water data an essential part of fulfilling these responsibilities.

[14] Water Use Reporting: https://waterrights.utah.gov/distinfo/wuse.asp



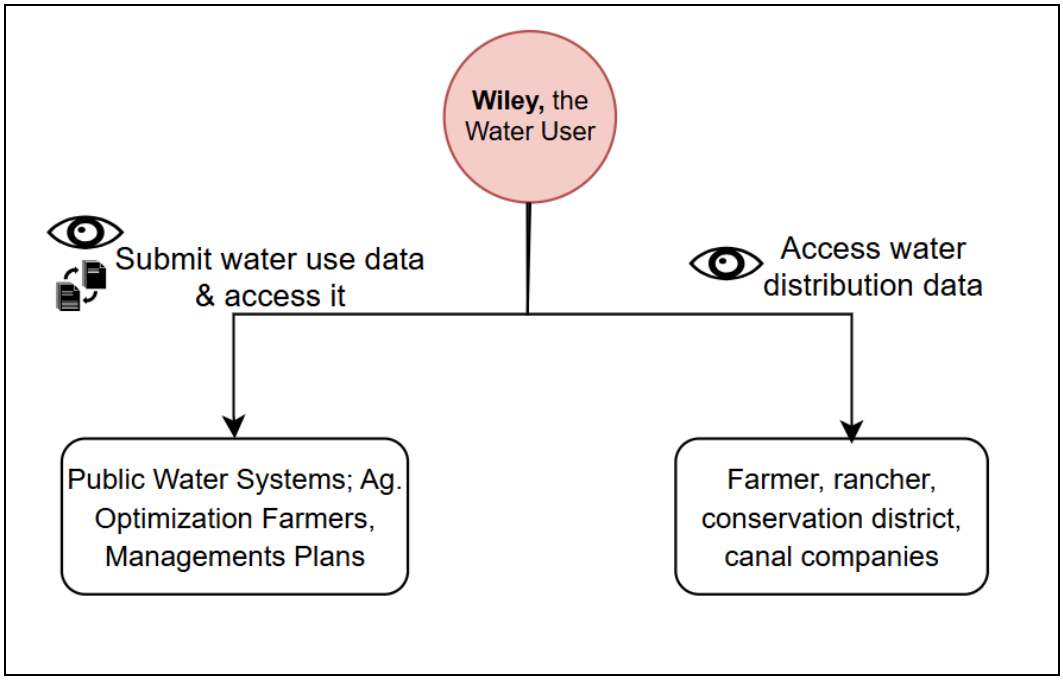
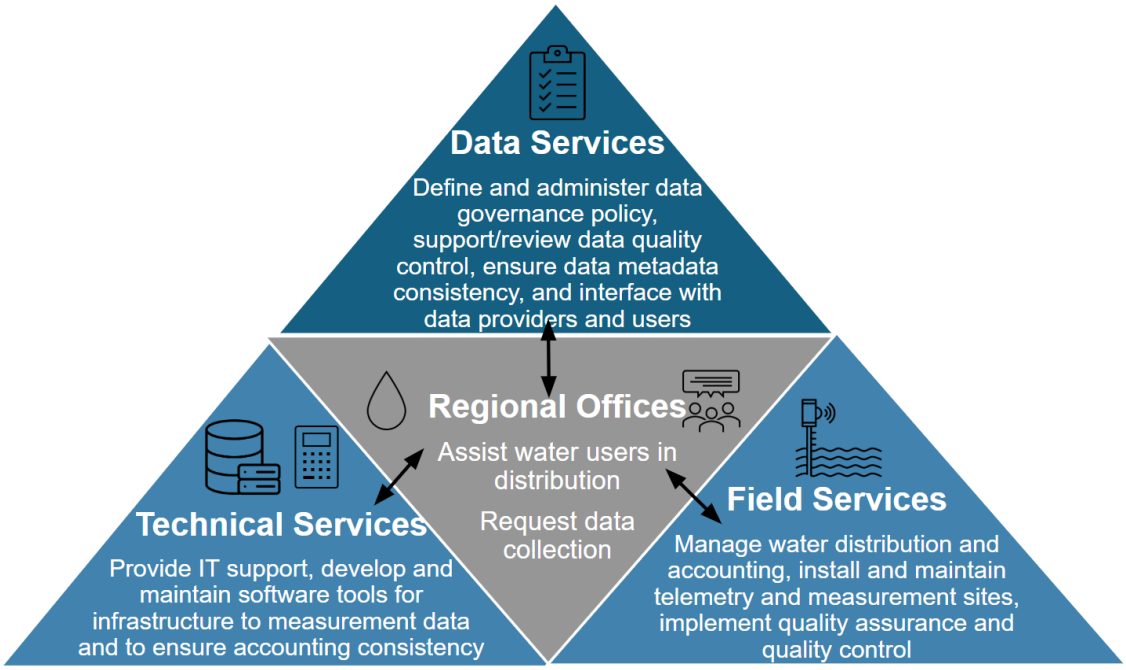


Figure 3: Two roles and responsibilities for Wiley, the water user group — (1) Submit water use data & access it, covering public water systems, agricultural optimization, farmers, and management plans; and (2) Access water distribution data, covering farmers, ranchers, conservation districts, and canal companies.

Key Dashboard Tasks



Wiley needs to use the dashboard to accomplish the following:

My Sites

- Access a "My Sites" tab: search through and favorite any distribution site.
- Set up alerts/notifications for favorite sites.
- Visualize data for any favorite site: view plots of discharge, water level, and battery voltage; view photos and location; view a tea-cup diagram and year-over-year comparison.

Mobile Data Access and Entry (Nice to Have)

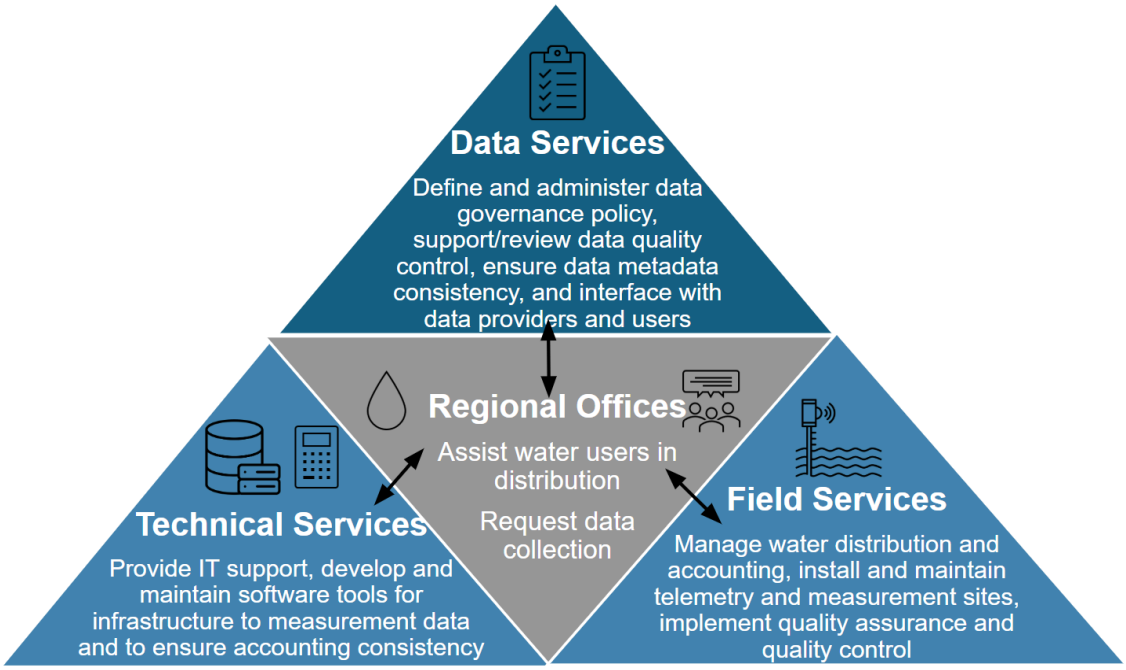
Smartphone-accessible mockup →

The Public Water Supplier (Water Use Program use case)

Public water supplier mockup →

My Sites

- Access a "My Sites" tab: search through and favorite any distribution site.
- Set up notifications for favorite sites.



- Visualize data for any favorite site: view plots of discharge, water level, battery voltage, and pH; view photos and location; view a tea-cup diagram and year-over-year comparison.

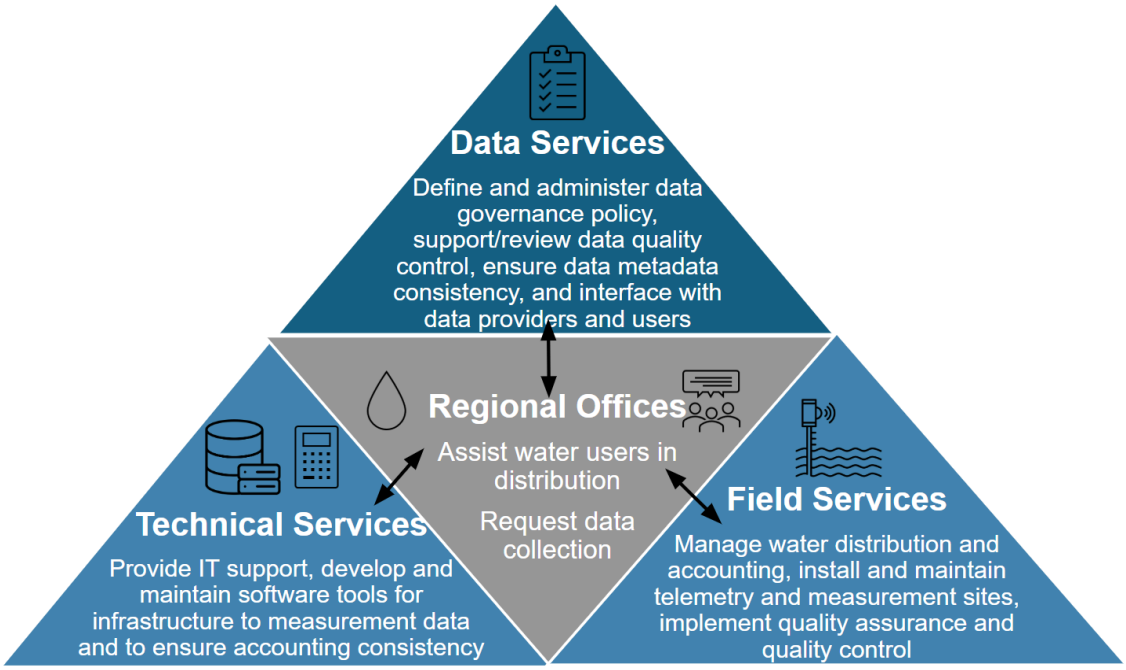
My System

Portal for the user's assigned water area:

- Displays sources, purchases, wholesale, and returns for the area.
- Graph of historical water discharge data, with the ability to view discharge across different years.

Report Water Use

1. The Configure System screen provides basic system information and status updates on boundary approval.
2. Make changes to boundary areas, with the ability to comment on why the change is needed.
3. Download boundary areas as a file.



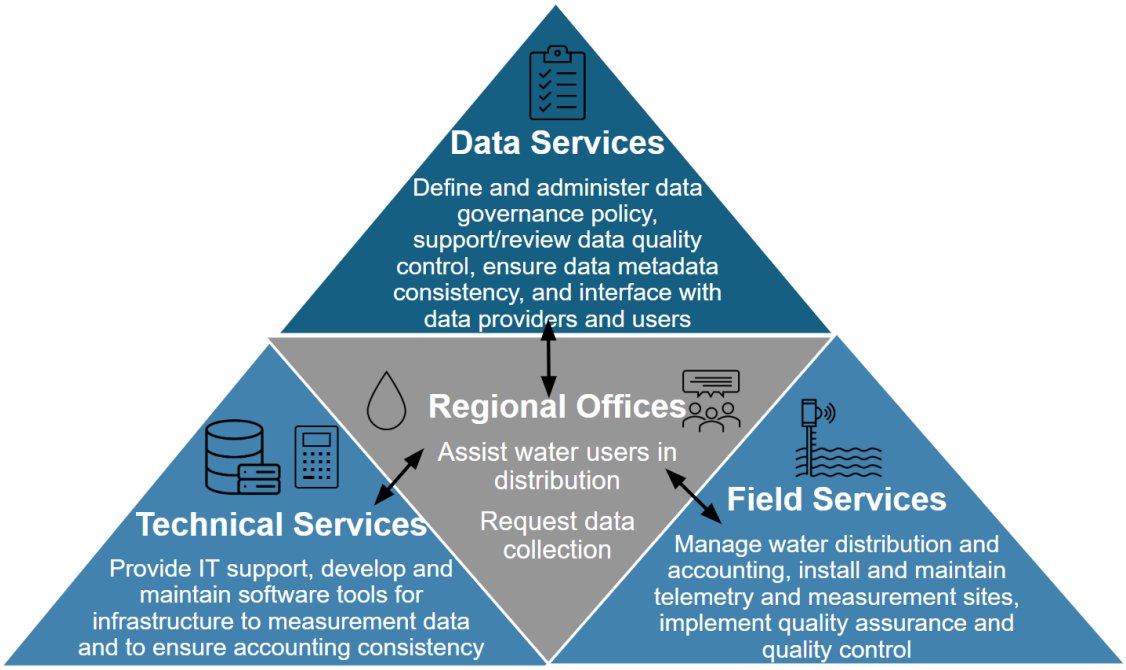
4. Add comments on every section — intended to replace emailing questions or concerns when filing a water use report.
5. Report culinary use, source inventory, purchase inventory, wholesale inventory, returns inventory, source peak demand, and secondary water use.
6. Download the completed Water Use Report as a PDF once it's submitted.
7. Sections are underlined in orange when a value exceeds the predetermined reasonable amount or when a required field is left blank.

Context: Current Reporting Tool

The Water Use Program currently has a reporting tool that water systems (e.g., cities) use to submit an annual report covering the past year's monthly water withdrawals and deliveries: waterrights.utah.gov/distinfo/wuse.asp. Each system has an account holding its historical data and new reports.

Parker (the public water supplier persona) needs a modernized version of this tool — one that is easy to use and navigate, dynamic, and uncluttered. Parker also wants multiple ways to import water use data into the tool (e.g., manual entry, file upload, and other data import options).

Foundational User Stories (2023)

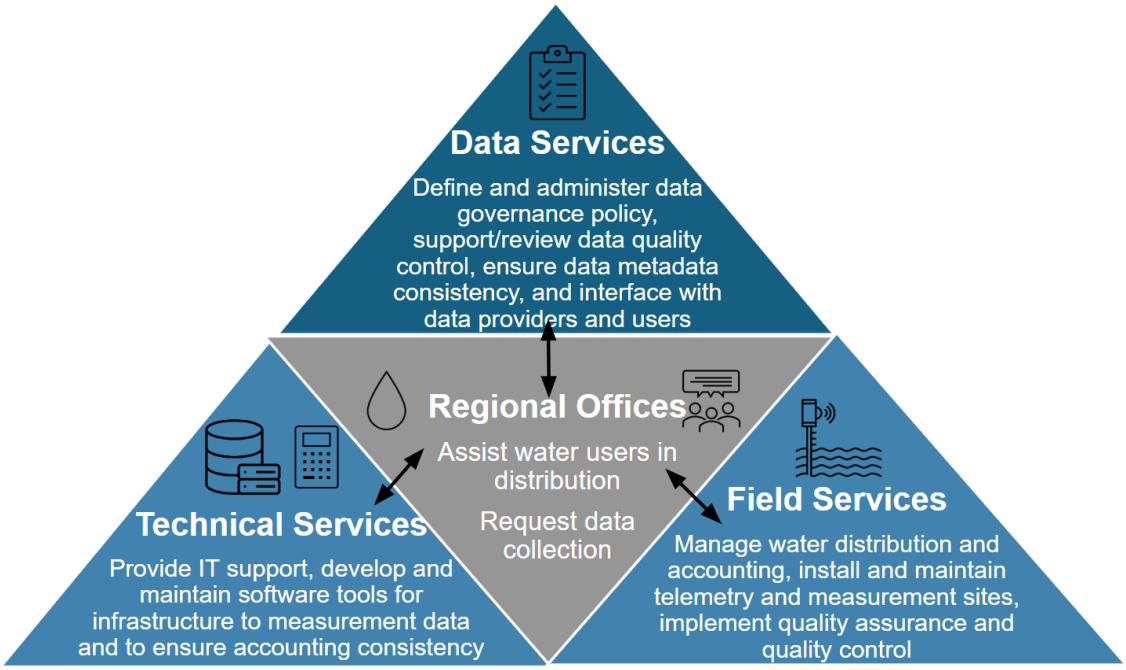


These needs, drawn from the 2023 foundational user-stories document, describe water-user requirements that inform Wiley's requirements above:

- **Thoughtful and proactive water users:** want to understand how priority distribution works in their area, so they can see how future scenarios will affect their water deliveries.
- **Public users:** need to be able to use applications on a mobile phone — that's what they use, not desktops.

5. Donna, the Director, Auditor, Legislator, or Citizen (the public and policymakers)

Donna represents the broadest audience for the system — directors, auditors, legislators, and members of the public — who need to explore and understand water data at a high level without any specialized training or ongoing account management. Unlike Fatima, Sarah, or Wiley, Donna is not managing sites or submitting data; she is looking for clear, trustworthy answers to specific questions, often to inform a report, a policy decision, or general public interest.



Donna's needs are shaped directly by two 2023 reports from the Office of the Legislative Auditor General. The performance audit of DWRi's data management found that water use data was not easily summarized, lacked consistent metadata, and was difficult for outside stakeholders to access and trust.[15] The Best Practice Handbook, released the same year, sets out how government organizations should collect, protect, and communicate data so that non-specialist audiences — like the legislators and members of the public Donna represents — can understand and trust it.[16] Together, these reports inform not just what data Donna needs access to, but how that data should be presented to her.

We could give Donna the option to create an account and gain access to the same permissions as "Wiley the water user."

Key Dashboard Tasks

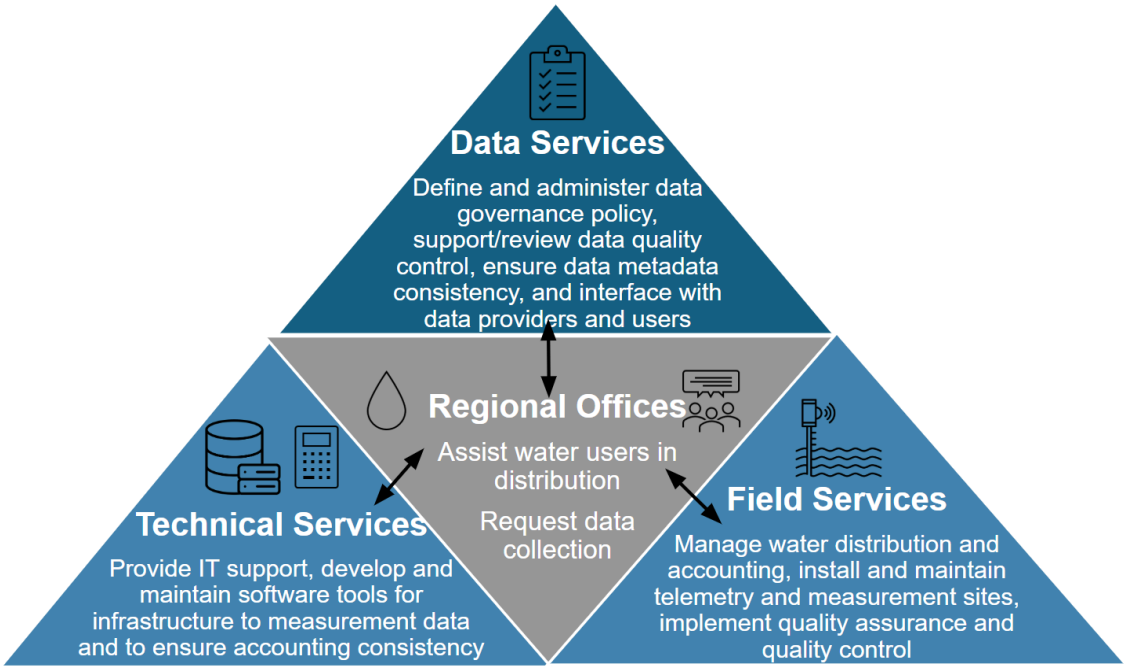
Donna needs to use the dashboard to accomplish the following:

Browse & Filter (No Account Required)

No account creation is needed. Donna can use filters to answer questions of interest, including:

5. Distribution system
6. Site type

[15] A Performance Audit of Ensuring Data Integrity, Program Best Practices, and Comprehensive Water Planning. https://le.utah.gov/interim/2024/pdf/00001105.pdf
[16] The Best Practice Handbook: A Practical Guide to Excellence For Utah Government. https://le.utah.gov/interim/2023/pdf/00002695.pdf

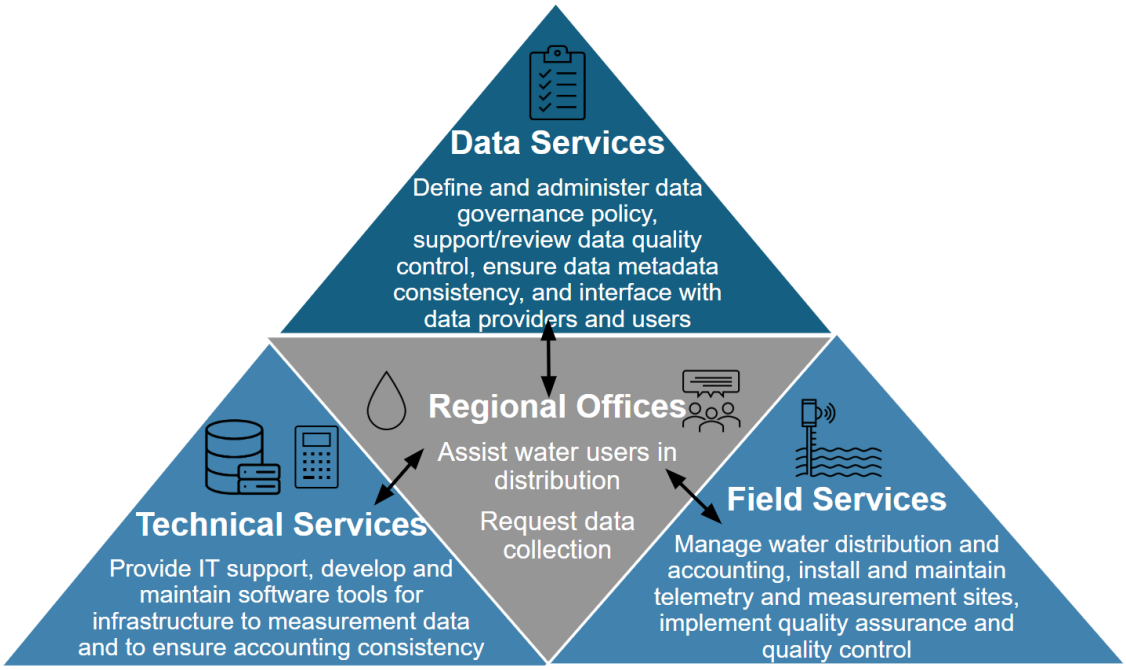


7. Data source / partner organization / owner (e.g., USGS, UDWRi, CUWCD)
8. Water source type (e.g., surface water, groundwater)
9. Date range of data (based on the min/max dates available for each site's properties in the database)
10. Observed properties
11. Sensor / measuring device type
12. Processing level (the highest available processing level is shown by default)
13. Time step (hourly, daily, monthly, annual)
14. Status (active, inactive, proposed)
15. Dataset type (observation, calculated, or modeled)

Per the Handbook's data visualization guidance, Donna's views should lead with an overview or conclusion before she drills into filters, rather than presenting an empty filter panel first.

Statewide Metadata & Performance Reporting

- As an auditor, view metadata about diversion measurement stations in order to produce statewide reports for the legislature and the public.
- Track key performance indicators, distinguishing between **output measures** (e.g., the percentage of stations equipped with telemetry, the percentage of sites whose data has been QA/QC-ed) and **outcome measures** (e.g., how effectively measurement data supports accurate water distribution), consistent with the Handbook's guidance that a



balanced set of measures — not any single metric — gives the clearest picture of program performance.

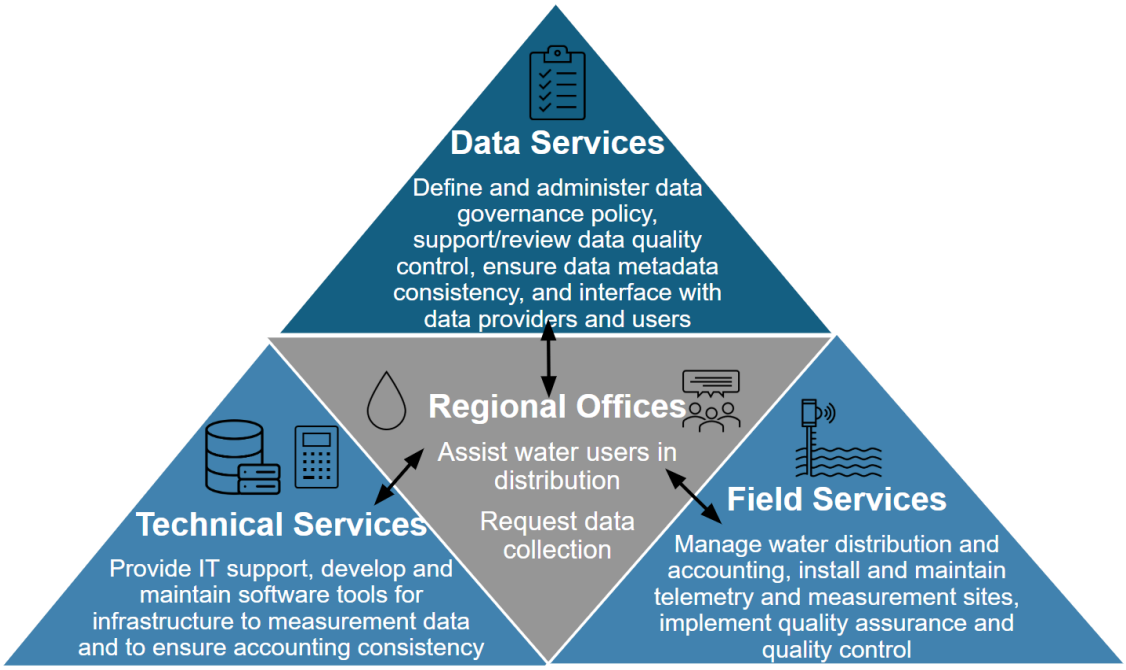
- Track the percentage of sources reporting measured (versus estimated) water use — a gap the 2023 audit flagged, noting that roughly one in ten M&I sources, one in seven secondary sources, and nearly half of agricultural sources report estimated rather than measured data.
- Access documentation equivalent to a data map and data dictionary for the underlying database, consistent with the audit's recommendation that DWRi develop and maintain these tools so that data remains findable, accessible, interoperable, and reusable (FAIR) even as staff turn over.

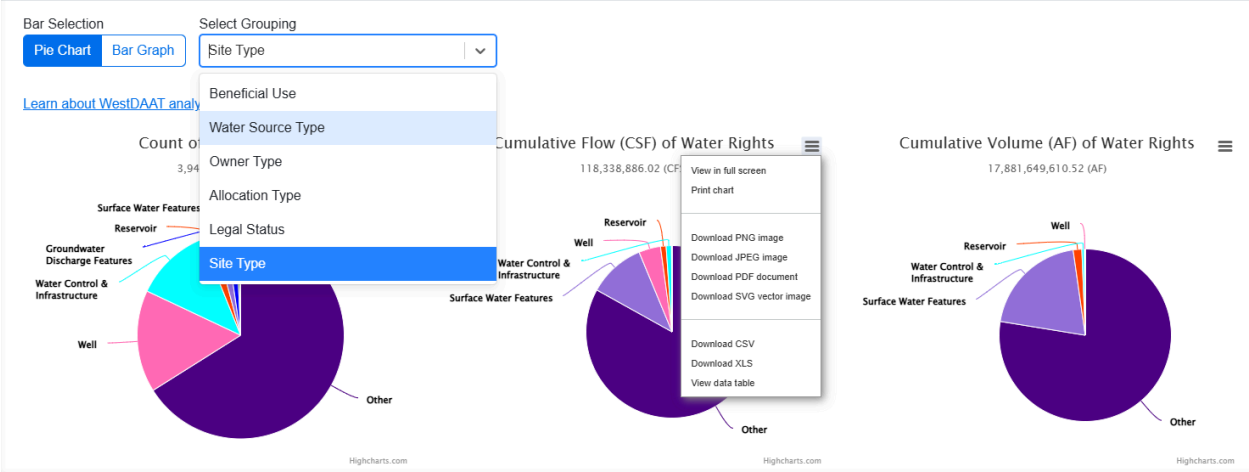
Download & Export

- Download data streams as a CSV file, for one site or many sites at once.

Dynamic Summaries & Aggregation

- Generate dynamic summaries — through charts and tables — that aggregate water use data across a selected space and time (e.g., "How much water did each distribution system use last year?"). This directly addresses an audit finding that the Divert Database, as currently structured, cannot easily produce summarized annual distribution records for the state's major water users.
- Reference example: WestDAAT illustrates the kind of aggregated, exploratory dashboard experience the team wants to support.

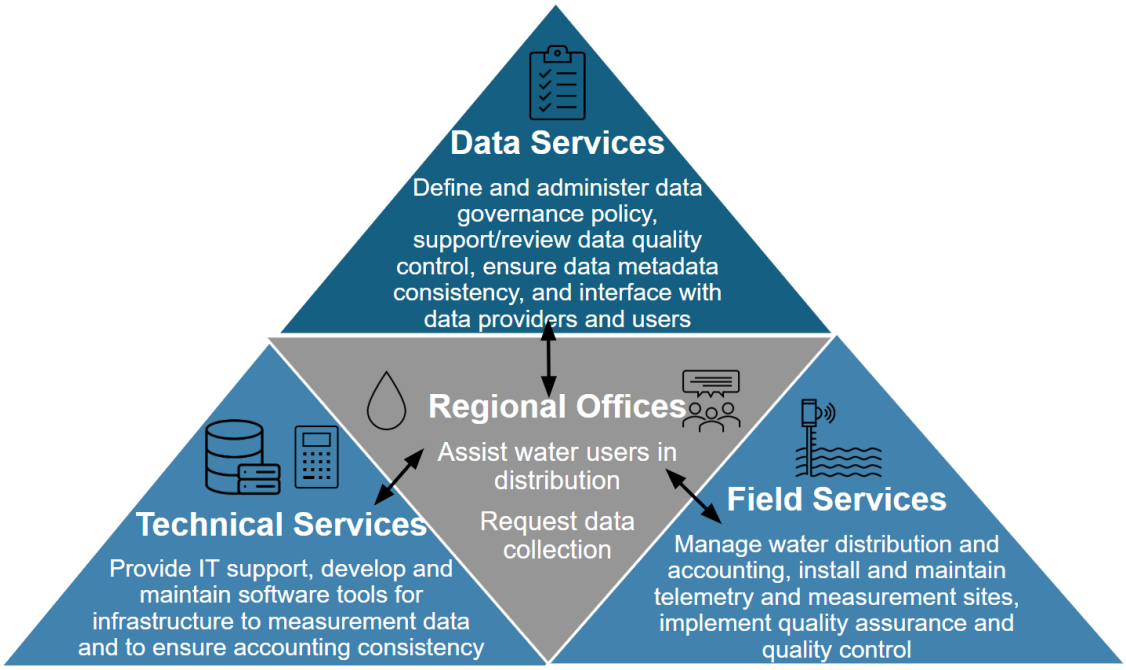




Data Visualization Guidance for Donna's Views

Because Donna and her audience (legislators, auditors, the public) often come to the dashboard without specialized training, the Handbook's data visualization guidance should directly inform how her views are designed:

- Lead with an overview or conclusion before offering filter options, rather than presenting an empty filter panel first.
- Use color and callouts intentionally to highlight the takeaway, not just to decorate the chart.

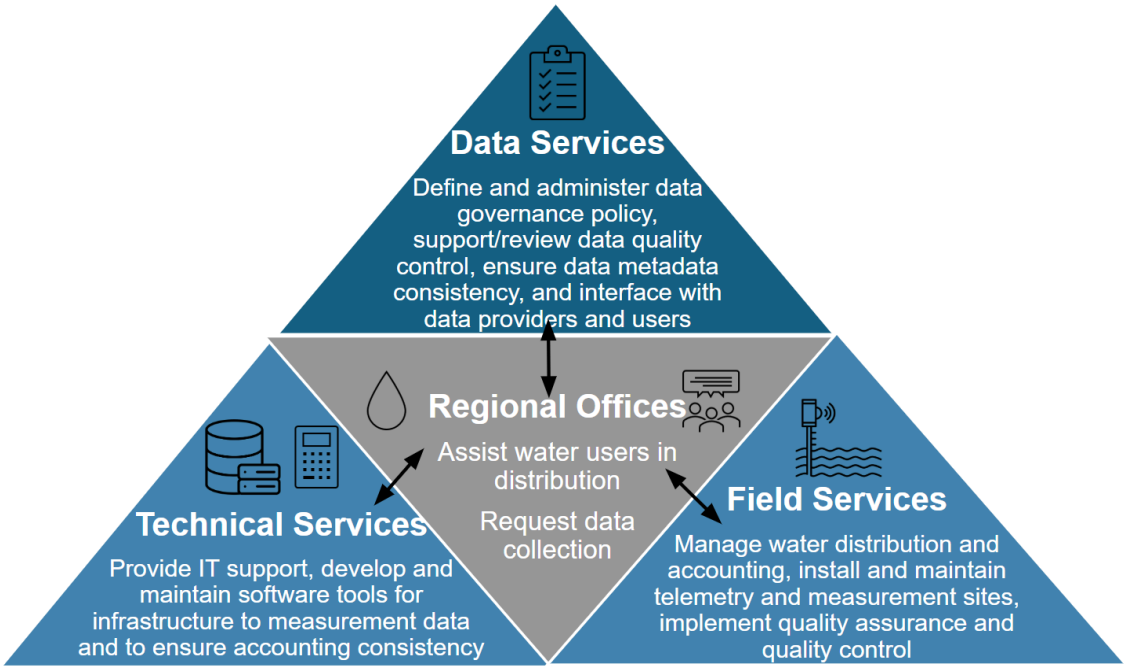


- Choose color scales with color-blind users in mind, and avoid using more than about a dozen colors in one visualization.
- Favor bar and sequential comparisons over pie charts, except when showing a straightforward percentage breakdown.
- Start numeric axes at zero, and avoid tilted axis labels, which are hard to read at a glance.
- Design visualizations to be web-based and interactive rather than static, so non-technical users can explore filters themselves without special software.

Data Integrity and Sharing

The Handbook's guidance on data collection also supports several of Donna's existing needs:

- Adopting consistent data standards, validation rules at entry, and ongoing data audits reduces the kind of inconsistent or unusable data that has hampered comparisons in other state audits — directly reinforcing the QA/QC tracking Donna needs.
- Making data shareable across agencies (USGS, UDWRi, CUWCD, and others already reflected in Donna's "data source / partner organization" filter) increases its value and avoids the siloed-data problems the Handbook and the 2023 audit both flag as a recurring risk in state government.
- Appropriate access controls and de-identification practices should still apply to any sensitive data before it's exposed through Donna's no-account, public-facing views.



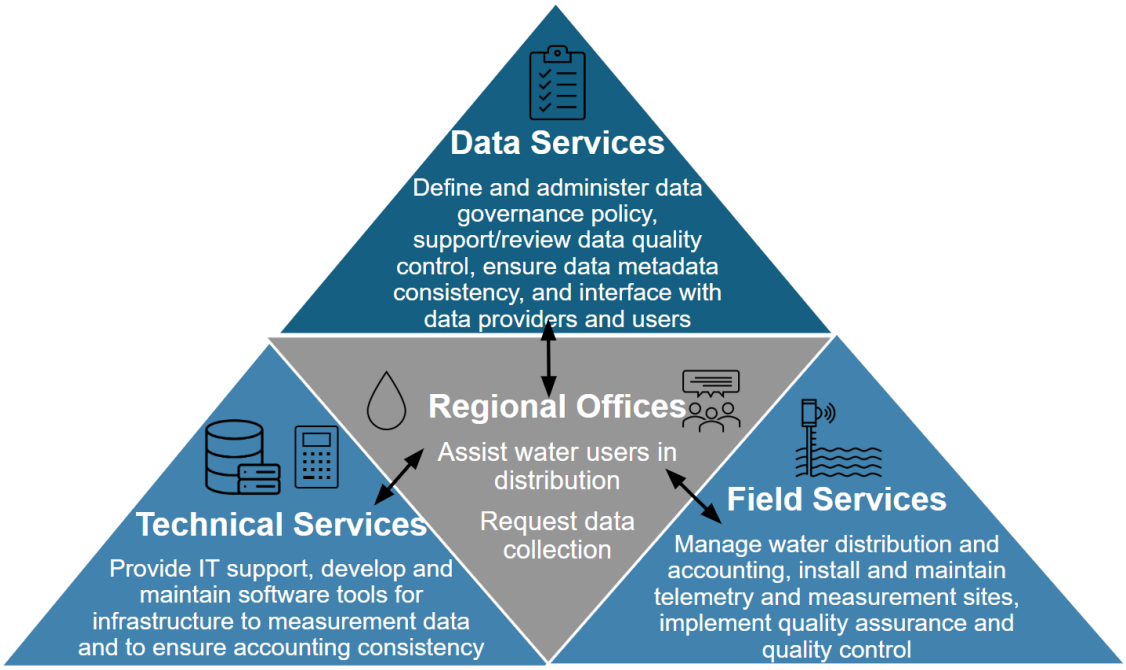
2023 Legislative Audit Findings

These findings, drawn from the Office of the Legislative Auditor General's 2023 performance audit of Utah's water management, further ground Donna's requirements above:

- The audit found that DWRi's Divert Database, while effective for day-to-day water distribution, was not structured to support broader planning, oversight, or accountability needs — restructuring it would let legislators, water planners, and the public access and analyze the data more easily.
- The audit recommended DWRi continue developing the Water Right Network so that data can support a wider range of internal management needs while improving transparency and accountability (Recommendation 1.1).
- The audit recommended DWRi continue its internal data audit to confirm accuracy and completeness, and act on any gaps identified (Recommendation 1.2).
- The audit recommended DWRi strengthen internal data evaluation and management by completing its gap analysis with Utah State University, linking data to the Division's focus areas, and creating and maintaining data tools such as a data map and data dictionary (Recommendation 1.5).

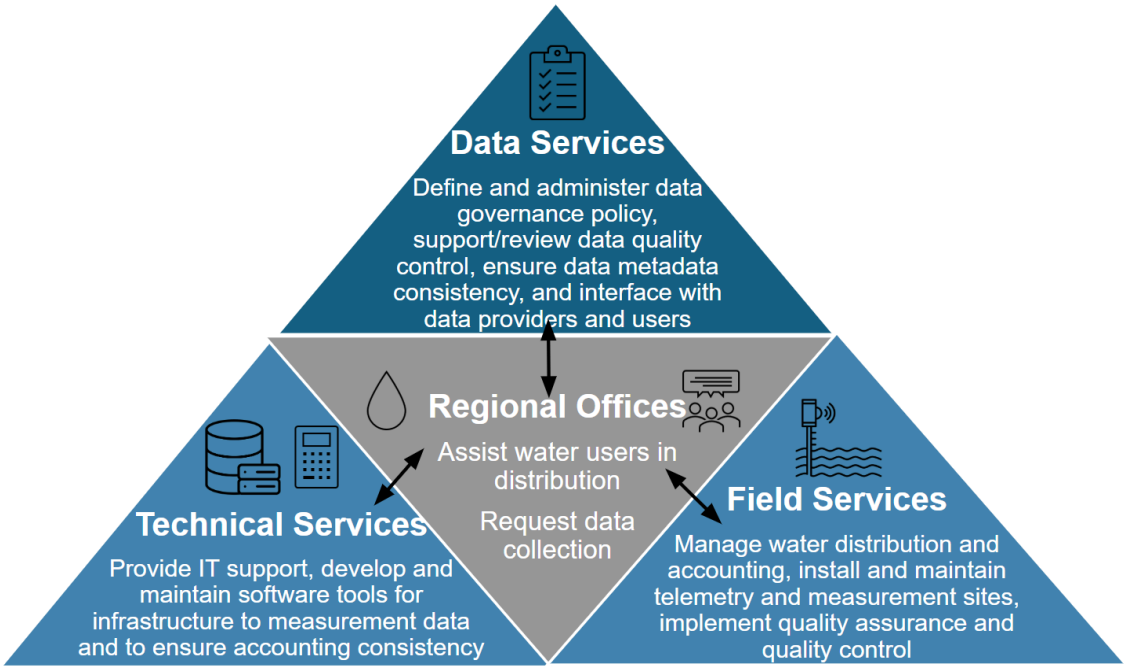
Foundational User Stories (2023)

These needs, drawn from the 2023 foundational user-stories document, describe director-, legislature-, and public-facing requirements that inform Donna's requirements above:



- Teresa: wants to show the legislature and other stakeholders the progress being made on the water right network and water measurements, so the legislature supports the division's efforts rather than legislating solutions that could have unintended consequences for long-standing water law principles.
- Teresa: wants to know depletion by pre- and post-compact water rights to plan for and anticipate how compact curtailment will impact the state; curtailment will require a massive overhaul of division resources and will be controversial, so these depletion amounts need to be defensible and transparent.
- The Legislature: wants to know how well Utah is measuring its water use — you can't manage what you don't measure — and wants to understand what measurements are being made and which are telemetered.
- Legislative auditors: want to generate reports evaluating whether DWRi is fulfilling its legislative mandate.
- Brian: as director of the fund that purchases water rights for the benefit of the Great Salt Lake, wants to know which water rights he should — and should not — buy or lease, so the benefit to the GSL is maximized and the use of these public funds can be defended.

Dynamic Summaries & Aggregation

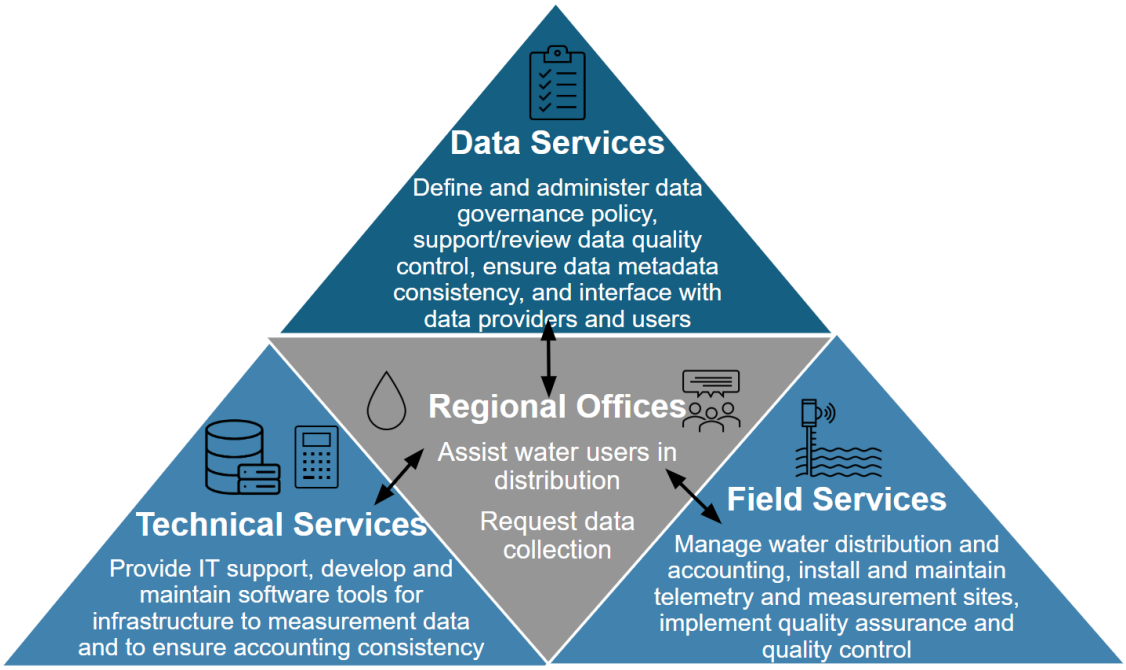


- Generate dynamic summaries — through charts and tables — that aggregate water use data across a selected space and time (e.g., "How much water did each distribution system use last year?").
- Reference example: WestDAAT illustrates the kind of aggregated, exploratory dashboard experience the team wants to support.

Foundational User Stories (2023)

These needs, drawn from the 2023 foundational user-stories document, describe director-, legislature-, and public-facing requirements that inform Donna's requirements above:

- **Teresa:** wants to show the legislature and other stakeholders the progress being made on the water right network and water measurements, so the legislature supports the division's efforts rather than legislating solutions that could have unintended consequences for long-standing water law principles.
- **Teresa:** wants to know depletion by pre- and post-compact water rights to plan for and anticipate how compact curtailment will impact the state; curtailment will require a massive overhaul of division resources and will be controversial, so these depletion amounts need to be defensible and transparent.
- **The Legislature:** wants to know how well Utah is measuring its water use — you can't manage what you don't measure — and wants to understand what measurements are being made and which are telemetered.



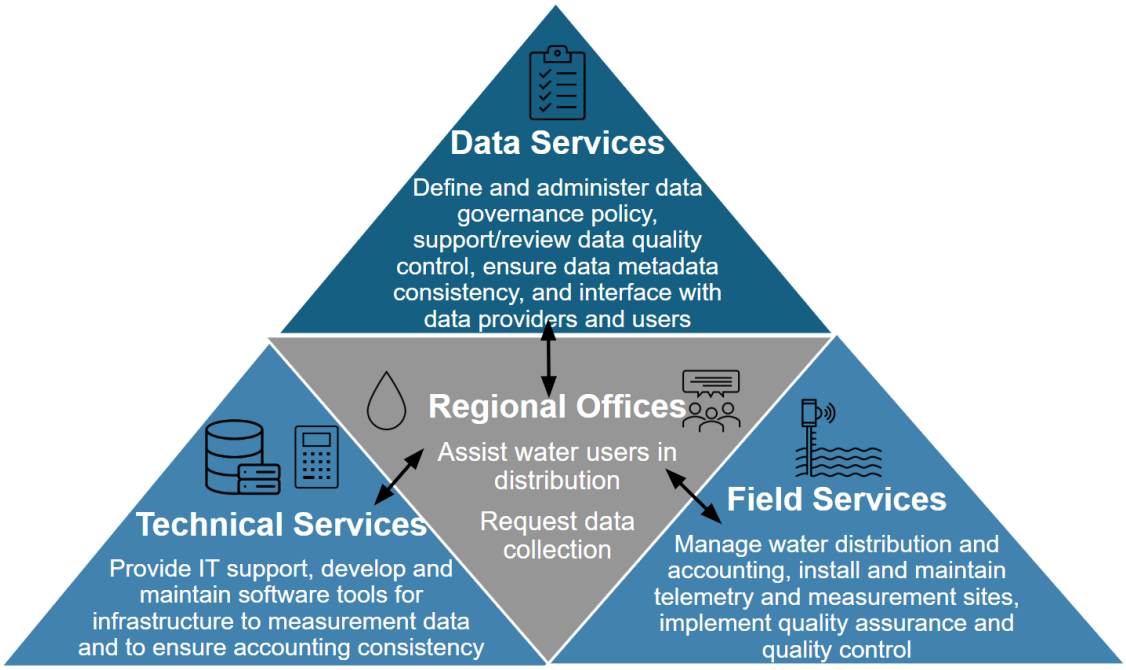
- **Legislative auditors:** want to generate reports evaluating whether DWRi is fulfilling its legislative mandate.
- **Brian:** as director of the fund that purchases water rights for the benefit of the Great Salt Lake, wants to know which water rights he should — and should not — buy or lease, so the benefit to the GSL is maximized and the use of these public funds can be defended.

6. Sarah, the Scientist, Researcher, or Consultant

Sarah is a scientist, researcher, or consultant who is tech-savvy and wants to download data or get programmatic access through the system's Application Programming Interface (API). She is a downstream consumer who needs large volumes of reliable, well-documented data to feed into her own models, analyses, or reports.

Examples of users: Division of Water Resources, Colorado River Authority of Utah, Office of the Great Salt Lake Commissioner, United States Geological Survey (USGS), Bureau of Reclamation, Utah State University, and students.

Sarah wants to access water use data programmatically and frequently, primarily for modeling and analytics purposes. Before building anything, she needs clear, complete API documentation. From there, she may build a software application or pipeline that depends on



the API for ongoing data feeds — or, in a simpler case, she may just want to select one or more sites and export their data streams to a CSV file for one-off analysis.

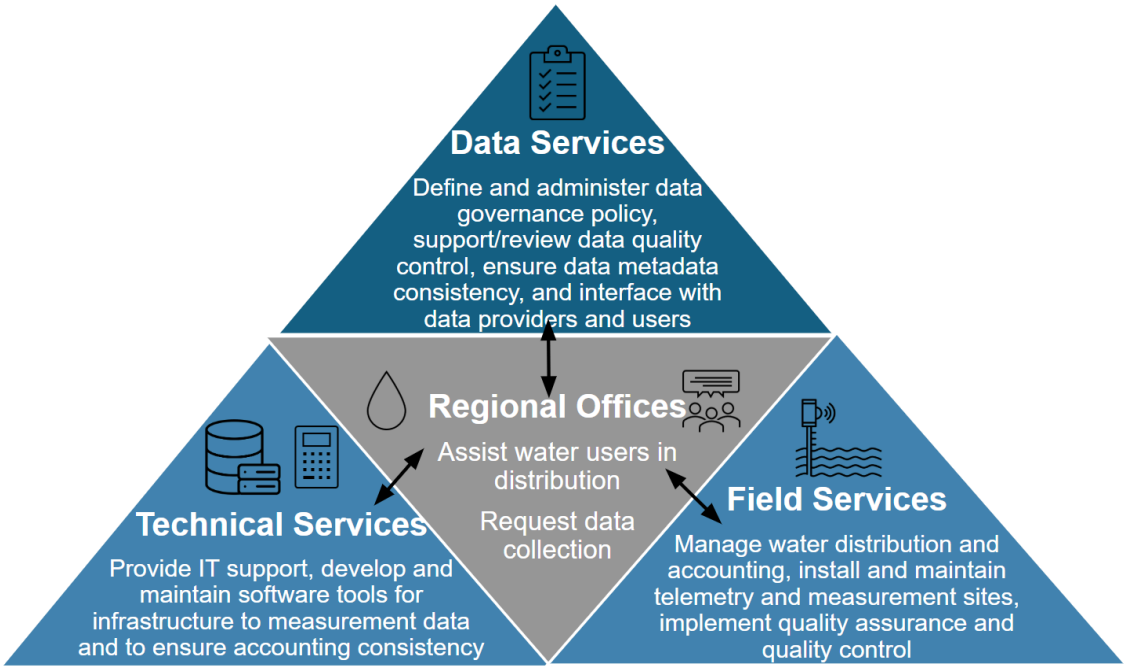
Key Tasks

Account & API Access

- Create a user account and generate an API token (or key) to query the data. It's much easier for users when the token is viewable directly in the account rather than something they have to download and store themselves.
- Regenerate or revoke a token if it's compromised, without having to create a new account.
- Ensure that any API token used within a software application is encrypted and never exposed publicly (e.g., in client-side code or public repositories).

Data Discovery & Quality Assessment

- Distinguish good diversion measurement data from poor-quality measurement data at a glance (e.g., through quality flags, processing level, or confidence indicators), so she can decide whether the data is fit for her purpose without manually investigating every value.



- When data quality is unclear or flagged, see enough context (e.g., known issues, edit history, qualifiers) to assess it herself or know who to contact to get it improved.
- Query data by specific processing level — for example, pulling only provisional data that was available at the time of distribution, which is important for accurately reproducing how a model or decision would have looked in real time, as opposed to best available (revised/finalized) data.

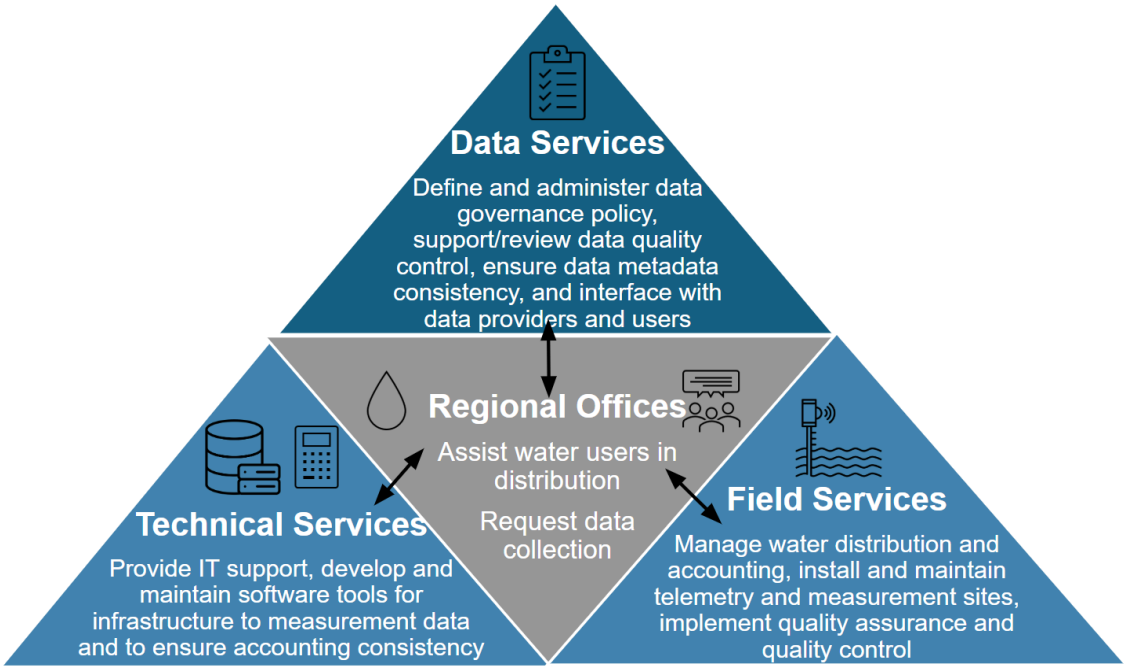
Data Download & Programmatic Access

- Download data for one site, or many sites at once, into a CSV file.
- Query and retrieve data programmatically via the API for use in her own software, models, or automated pipelines.
- Filter/query by parameters such as site, date range, observed property, and processing level, so she can pull only the data relevant to her use case rather than downloading everything and filtering locally.

Foundational User Stories (2023)

These needs, drawn from the 2023 foundational user-stories document, describe researcher, consultant, and modeler requirements that inform Sarah's requirements above:

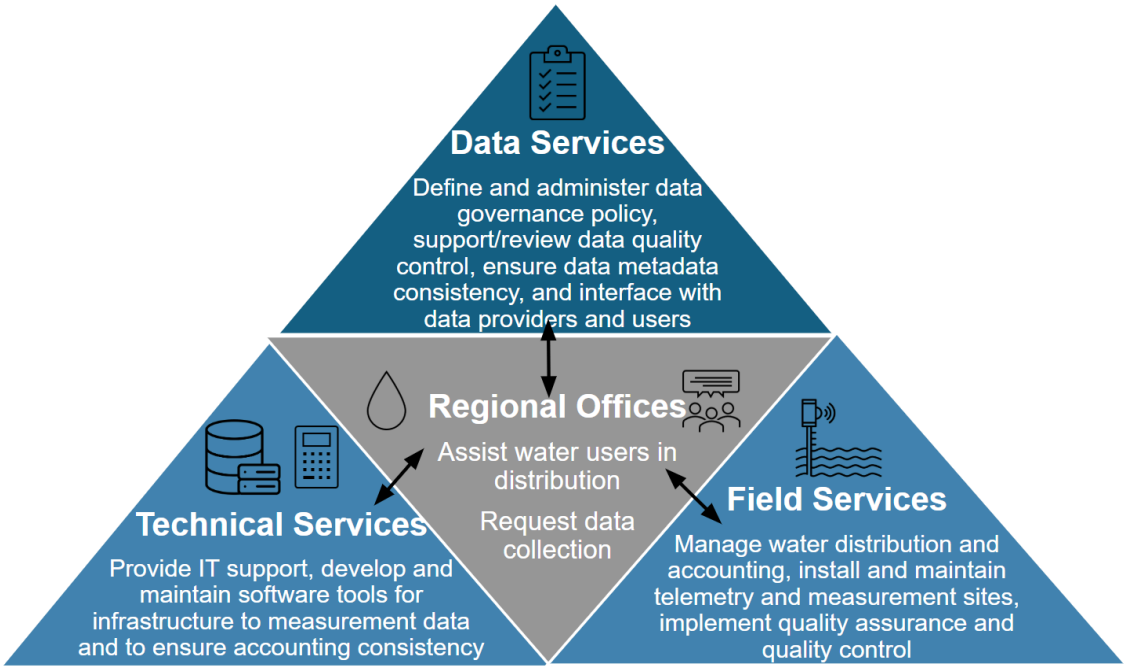
- **Bart, of the Colorado River Authority of Utah (CRAU):** wants to be able to tell good diversion measurement data from poor measurement data — because his work is so



important, if it isn't clear the data is good he has to take action himself to assess and improve its quality (part of the motivation for the division's gap analysis).[17]

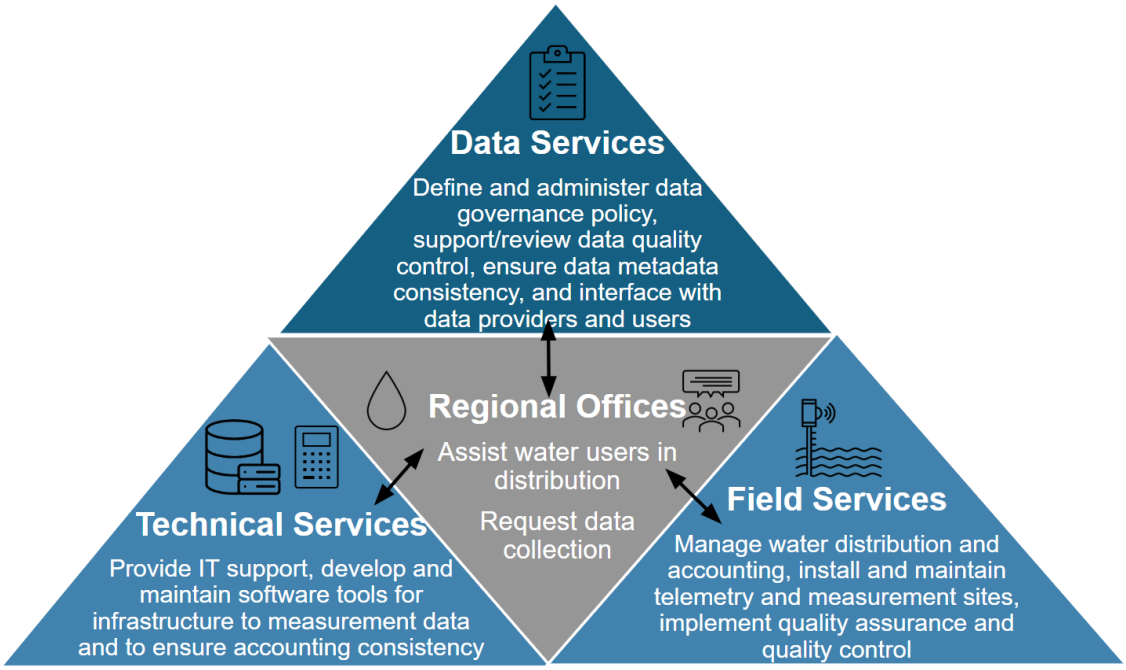
- **A water modeler hired by CRAU:** wants to know which water rights go with which fields so the model being built (CRSS) is accurate and useful for analyzing alternatives like demand management or fallowing; the modeling tool can easily learn and consume a well-documented data service API.
- **Chad, a former Richfield DWRi staff member now consulting and frequently requesting data:** wants an easy way to get and understand data so he doesn't take up so much DWRi staff time.
- **David, or someone developing a generalized accounting model:** wants to display the accounting results the draft model produces so others can critique it and identify issues quickly, enabling faster feedback and a more efficient development process.
- **Researchers working on safe-yield calculations, numeric modeling, or water planning:** want to see all the locations where diversions are measured — and not measured — to proceed with their analysis.
- **Folks evaluating the reliability of OpenET products:** want to compare field depletion from OpenET to water actually diverted to the field.

[17] Utah Division of Water Rights Hydroinformatics and Technology Gap Analysis. Utah Water Research Laboratory, Utah State University, 2024.



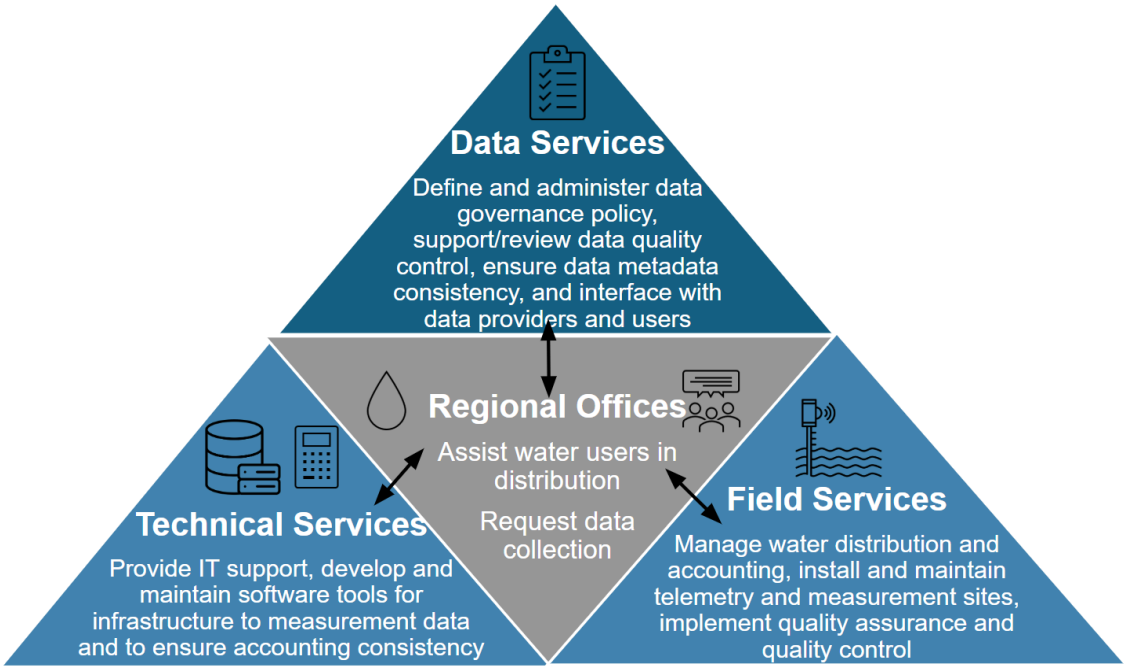
- **Researchers and analysts:** need a dataset showing how a water right with an approved change application is actually being used — according to the “heretofore” or the “hereafter” description.
- There is a need for a transparent, machine-readable transaction schedule — supporting efforts like a general apportionment solver — so the public can understand how a system operates and can support planning and modeling work, since non-trivial water rights currently require expert interpretation from the division.

Note: The 2023 foundational document also included needs specific to the Adjudication Database and PD/LUR Book Processing (e.g., map/search enhancements, UCA 73-4-22(2) summons tracking fields, and individual page printing tools). These pertain to a separate system outside the scope of the six measurement-dashboard personas above and are retained here for future reference.



Appendix: Summary of Key Requirements by Persona

Persona	Primary Goal	Key Dashboard Feature	Key Output
Diego (Data Manager)	System integrity & metadata governance	Centralized user management, orchestrator health monitoring, automated QC rule definitions	Daily connection digests, standardized metadata dictionaries
Cody (Commissioner)	Daily field distribution & manual entry	Mobile entry app, site flow plots, simplified spreadsheet block pasting	Annual distribution reports, verified flow logs
Fatima (Field Engineer)	System compliance, hardware QA, & modeling	Dual-layer time series, point audit log, rating curve history, WCAT model synchronization	QA/QC flagged datasets, site inspection records
Wiley (Water User)	Usage tracking & annual reporting	"My Sites" notification hub, Water Use Program annual reporting portal	Verified annual water use reports (PDF)
Donna	Legislative oversight	Macro summary charts,	Statewide audit



Persona	Primary Goal	Key Dashboard Feature	Key Output
(Director/Auditor)	& policy planning	FAIR data dictionary, legislative KPI performance scorecards	metrics, aggregated basin summaries
Sarah (Researcher)	Technical modeling & data ingestion	REST API with token management, raw/provisional data filtering, bulk CSV exports	Automated data feeds for hydrologic models

